Power, phasors and complex power

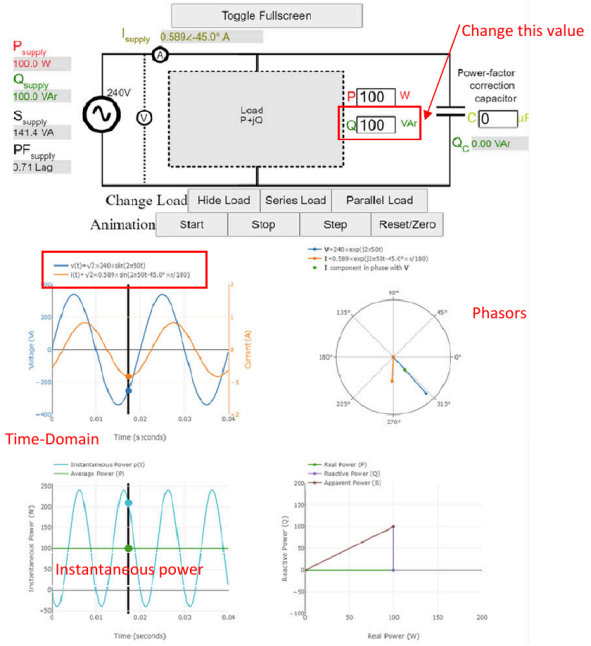
Practice questions · Day 01 · Sat 24 Oct · pack M1

5. Practice

From your course files

Power Circuit Tool WS1 Q1 (workshop). *Time-Domain Waveforms:* Using the default values of load P = 100 W and Q = 100 VAr, examine the equations for $v(t)$ and $i(t)$ which are given just above the voltage and current time waveforms.

a. What are the peak and rms values of the voltage and current waveforms?
b. For each of the values of Q = 100 VAr, 0 VAr and −100 VAr: (i) indicate whether the current is leading, lagging or in-phase; (ii) indicate whether the power factor is leading, lagging or unity.



Power Circuit Tool WS1 Figure 1: the tool interface, with the default load P = 100 W, Q = 100 VAr on 240 V (from the workshop sheet)

Power Circuit Tool WS1 Q2 (workshop). *Phasors:* examine the relationship between the phasor graph and the time-domain graphs.

a. Which direction are the phasors rotating?

b. Does the x-axis or y-axis component of the phasors correspond to the instantaneous voltage and currents? You may find it convenient to use the animation control buttons: start, stop, step and reset/zero.
c. Reset the phasors to stop them rotating. As you change Q = 100 VAr, 0 VAr and −100 VAr, how do you work out whether the current is leading, lagging or in-phase with the voltage?

Power Circuit Tool WS1 Q3 (workshop). *Instantaneous Power:* the instantaneous power waveform $p(t) = v(t) \times i(t)$ is shown in blue in the graph below the time-domain waveforms.

a. What is the frequency of the instantaneous power? Hint: find the period over which it repeats (T) and hence find its frequency ($f = 1/T$). How does this frequency compare to that of the voltages and currents?
b. Under what circumstances is the instantaneous power zero?
c. For the values of Q = 100 VAr, 0 VAr and −100 VAr: (i) What is the change in the average instantaneous power (shown in green)? (ii) What is the (approx.) minimum value of the instantaneous power? Note that negative values of instantaneous power are associated with reactive power. We will cover this in the next workshop.

PQ 1.14 (practice questions). Calculate the following quantities

a. Torque when the mechanical power is 7.5 kW and the speed is 2910 rpm.
b. Mechanical power when the speed is 50,000 rpm and the torque is 4.25 Nm
c. Speed (in rpm) when the power is 30 MW and the torque 191 kNm

PQ 1.21 (practice questions). Consider a single-phase AC machine. The input voltage is 110 V, the input current is 15 A and the power-factor is 0.8 lagging. The output speed is 1750 rpm and the output torque is 6 Nm. Calculate the:

a. input power
b. output power
c. efficiency
d. power loss

PQ 1.22 (practice questions). Consider a three-phase AC machine. The voltage of each phase is 240 V and the power-factor is 0.85 lagging. The output speed is 650 rpm and the output torque is 63 Nm. It is known that the machine efficiency is 88%. Calculate the:

a. output power
b. input power
c. power loss
d. input current magnitude

PQ 2.6 (practice questions). A certain car battery can be modelled using an equivalent circuit of a 12.5 V voltage source with a series resistance of 0.02 Ω. Calculate:

a. The open-circuit battery output voltage.
b. The battery current and internal power loss under short-circuit conditions.
c. The battery output voltage and current when powering a load of 1 ohm.
d. When the starter motor is energised, the battery output voltage is observed to fall to 8 V. Calculate the current drawn by the starter motor and the power supplied to the starter motor.
e. Calculate the maximum output power available from the battery.

PQ 2.15 (practice questions, do (f)–(h)). Calculate the magnitude and phase of the current in the load connected to a single-phase 240 V AC source. You can assume the phase angle of the voltage is zero. The load consists of:

a. a 10 ohm resistor
b. a 24 ohm resistor
c. a 24 ohm resistor in series with a 10 ohm resistor
d. a 24 ohm resistor in parallel with a 10 ohm resistor
e. an inductor with a 24 ohm reactance
f. a 24 ohm resistor in series with an inductor with a 24 ohm reactance
g. a 24 ohm resistor in parallel with an inductor with a 24 ohm reactance
h. the load in part (a) in parallel with the load in part (f). [Hint: you can add the currents from (a) and (f)]

PQ 2.17 (practice questions). A single-phase load consists of a parallel-connected 25 ohm resistor and an inductor with a 125 ohm reactance is connected to the 240 V mains, calculate

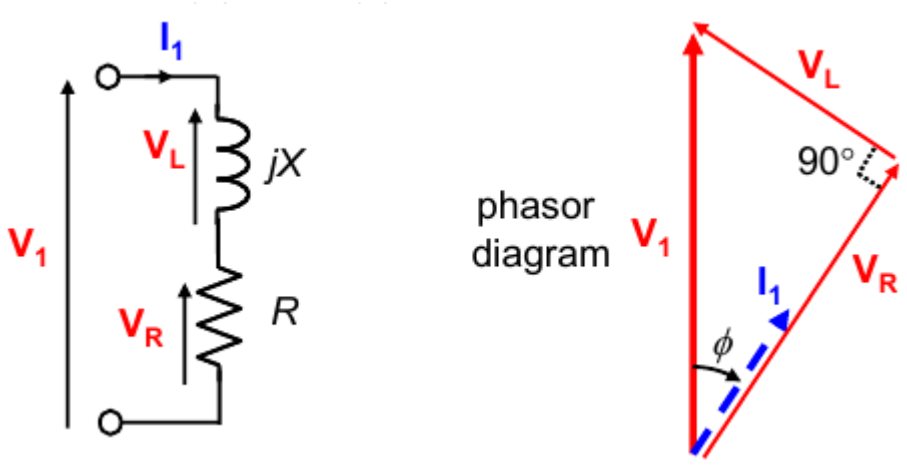
a. the load impedance Z in polar form
b. write the supply voltage $\mathbf{V_1}$ in phasor form (assume it has a phase angle of zero)
c. the resultant phase current phasor $\mathbf{I_1}$
d. the input power factor (remember to show lead or lag)
e. the complex power S_1 in both polar and Cartesian form
f. the input real power P using (i) the real part of S_1; (ii) the input power factor and V_1 and I_1; (iii) the fact that all the power losses occur in the resistor and hence the losses are V^2/R (why not I^2R?)
g. the input reactive power Q
h. the input VA (that is, the apparent power S)
i. sketch a power triangle and verify that $P^2 + Q^2 = S^2$
j. sketch a phasor diagram showing the relationship between the voltage and the current

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PQ 2.18 (practice questions). In the following circuit, it is known that V_1 (magnitude of $\mathbf{V_1}$) = 200 V and V_R = 150 V.

a. Find V_L.
b. Find the power-factor angle and power-factor of the circuit.
c. If the input current magnitude I_1 is 10 A, find the input apparent, real and reactive power. Use this input current magnitude for the following parts of this question.
d. What is the real and reactive power absorbed by the resistor?

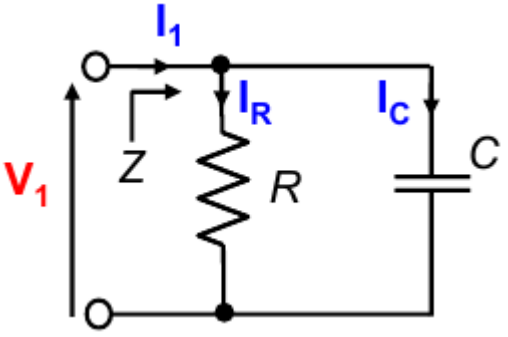
e. What is the real and reactive power absorbed by the inductor?
f. Using the results from (d) and (e), what is the resistance and reactance of the circuit?



PQ 2.18 figure: series R and jX with V_R, V_L and V_1, and the phasor diagram (from the practice questions)

PQ 2.19 (practice questions). In the following circuit, is known that the input voltage V_1 is 140 V, the input power P_1 is 500 W and the input current I_1 is 15 A.

a. Find the value of the resistance R.
b. Find the input apparent and reactive power.
c. Find the power-factor angle ϕ and hence the power-factor.
d. Show that you get the same result for the power-factor using pf = P/S.
e. Using the value of reactive power, find the value of reactance.
f. Sketch a phasor diagram for the circuit showing $\mathbf{V_1}, \mathbf{I_1}, \mathbf{I_R}, \mathbf{I_C}$ and ϕ. It is not necessary to show any values for the above phasors in the diagram.
g. Sketch a power triangle for the circuit showing the values of S, P and Q and ϕ.



PQ 2.19 figure: parallel R and C on V_1, with branch currents I_R and I_C (from the practice questions)

Jun23 Q1a (past exam, 5 marks). *Battery Modelling.* The equivalent circuit on the right is used to model a car battery.

i. The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]
ii. A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5 V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]
iii. Find the battery output voltage when a load drawing a current of 20 A is connected to it. [1 mark]

Jun23 Q1a figure: battery model E in series with R_S, terminal voltage V, load R_L (from the exam paper)

Jun23 Q1c (past exam, 9 marks). *Phasor Calculations.* Consider the single-phase series RLC load as shown in the figure on the right. The current $\mathbf{I_1}$ is known to be $13\angle+20^\circ$ A.

i. Using the known current $\mathbf{I_1}$, find the voltage phasors across the resistance $\mathbf{V_R}$, inductance $\mathbf{V_L}$ and capacitance $\mathbf{V_C}$ in polar form using Ohm's law. [3 marks]
ii. Find the phase impedance Z of the load in polar form and hence the voltage phasor $\mathbf{V_1}$ in polar form using Ohm's law. [3 marks]
iii. The complex power S is given by $S = \mathbf{V_1}\mathbf{I_1}^*$, where * refers to the complex conjugate. Find the complex conjugate of the current $\mathbf{I_1}^*$, thus the complex power S and hence the total real power P and reactive power Q supplied from the power source. [3 marks]

Jun23 Q1c figure: series j30 Ω, 10 Ω and −j50 Ω carrying I_1, with V_L, V_R, V_C and V_1 marked (from the exam paper)

Jun24 Q1a(ii) (past exam, 2 marks). *Fill-In-The Blank.* An electric motor has an input power of 1000 W and a power loss of 300 W. This results in an output power of _____ W and an efficiency of _____ %. Increasing the power loss for the same input power would cause the efficiency to _____ (choose decrease or increase). Increasing the input power for the same power loss would cause the efficiency to _____ (choose decrease or increase).

Jun24 Q1b (past exam, 3 marks). *Electrical and Mechanical Power.* Consider a 3ph, 4 pole, 50 Hz induction motor. Under a certain operating condition, the motor speed is 1420 rpm, input voltage is 415 V, input current is 6.8 A, input electrical power is 4 kW and output torque is 23 Nm.

i. Find its input power factor (you can assume this is lagging). [1 mark]
ii. Find its output power and hence efficiency. [2 marks]

Jun24 Q1c (past exam, 5 marks). *AC Circuit Analysis.* Consider the series resistor-inductor (RL) circuit shown on the right.

i. The input current I_1 has a magnitude of 2 A. Find the value of the real power and reactive power absorbed by the resistor. Find also the real and reactive power absorbed by the inductor. [3 marks]
ii. For a different value of input current I_1, it is known that the resistor voltage V_R has a magnitude of 50 V and the inductor voltage V_L has a magnitude of 20 V. Hence find the magnitude of the input voltage V_1. [2 marks]

Jun24 Q1c figure: series j40 Ω and 100 Ω carrying I_1, with V_L, V_R and V_1 marked (from the exam paper)

New questions

N1. Single-phase pump motor [5 marks]. A single-phase motor runs from 230 V, 50 Hz and draws 6.2 A at a pf of 0.78 lagging. It delivers 0.95 kW to its load at 2,860 rpm.

a. Find the input real power and the efficiency. [2]
b. Find the output torque. [1]
c. Find the input apparent and reactive power. [2]

Circuit for New question 2: 30 Ω in series with 100 µF on 240∠0° V, 50 Hz

N2. Series RC [6 marks]. A 30 Ω resistor in series with a 100 µF capacitor is connected to 240 V, 50 Hz. Take $V_1 = 240\angle 0^\circ$ V.

a. Find the impedance Z in polar form. [2]
b. Find the current I_1 and the voltages V_R and V_C as phasors. Show that $V_R^2 + V_C^2 = V_1^2$. [2]
c. Find S, P, Q and the power factor. [2]

Circuit for New question 3: the 230 V, 5.0 A, 900 W load as series (a) and parallel (b) RL models

N3. One measurement, two models [6 marks]. An inductive load on 230 V, 50 Hz draws 5.0 A and 900 W.

a. Find the apparent power, reactive power and power factor. [2]
b. Find R, X and L if the load is modelled as series RL. [2]
c. Find R, X and L if the load is modelled as parallel RL. [2]

Circuit for New question 4: motor load A in parallel with load B (40 Ω in series with −j30 Ω) on 240 V, 50 Hz

N4. Two loads on one supply (chained) [8 marks]. A 240 V, 50 Hz single-phase supply feeds two loads in parallel. Load A is a motor delivering 1.5 kW of mechanical output, with an efficiency of 80 % and a pf of 0.75 lag. Load B is a 40 Ω resistor in series with a capacitor of reactance 30 Ω.

a. Find the electrical P, S and Q of load A, and the magnitude of its current. [3]
b. Find the current phasor of load B and its complex power. [2]
c. Find the total P, Q and S, the supply pf and the supply current magnitude. [3]

Circuit for New question 5: series R = 8 Ω, L = 40 mH, C = 200 µF on 230∠0° V, 50 Hz

N5. Series RLC with a surprise [7 marks]. A series circuit of $R = 8\ \Omega$, $L = 40$ mH and $C = 200$ µF is connected to $V_1 = 230\angle 0^\circ$ V, 50 Hz.

a. Find X_L, X_C and Z (polar). Is the circuit inductive or capacitive overall? [2]
b. Find I_1, V_R, V_L and V_C. Comment on $|V_C|$. [2]
c. Find S, P, Q and the pf using $S = V_1 I_1^*$, and check P and Q from the element values. [2]
d. At what supply frequency would the pf be unity? [1]

N6. Three-phase motor nameplate (brief preview of M2) [6 marks]. A 3ph induction motor is rated at 415 V, 7.5 kW, 1,455 rpm, with an efficiency of 88 % and a pf of 0.84 lag. Its connection is not stated.

a. Find the rated torque. [1]
b. Find the rated input power and losses. [2]
c. Find the rated line current and the total input reactive power. [2]
d. Find the phase voltage. [1]

6. Quick recall check

1. A UK supply is 230 V rms. What is its peak value?
2. Convert 3000 rpm to rad/s, and 100 rad/s to rpm.
3. A load has $P = 1200$ W and $Q = -500$ VAr. Is its pf leading or lagging, and what is the value?
4. Series RL: $V_R = 80$ V and $V_L = 60$ V. What is V_1?
5. Why can't you add the apparent powers of two parallel loads?

Power-factor correction, 3-phase, equivalent-circuit tests

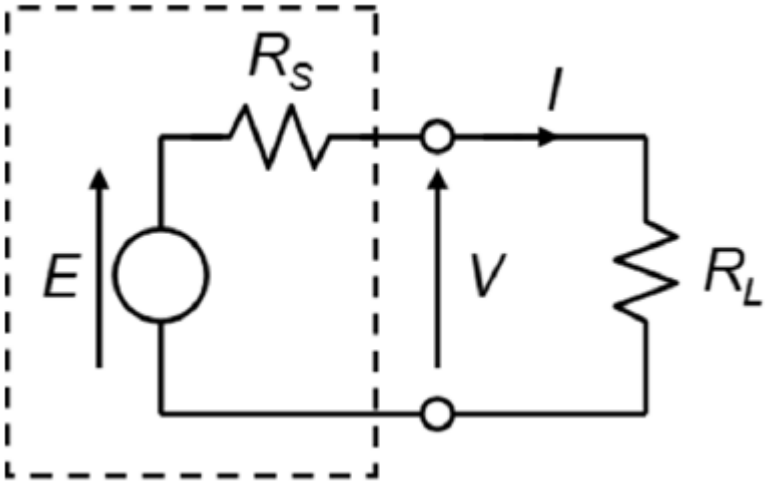
Practice questions · Day 02 · Sun 25 Oct · pack M2

5. Practice

From your course files

Jun23 Q1a (past exam, 5 marks). *Battery Modelling.* The equivalent circuit on the right is used to model a car battery.

i. The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]
ii. A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5 V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]
iii. Find the battery output voltage when a load drawing a current of 20 A is connected to it. [1 mark]



Jun23 Q1a figure: battery model E in series with R_S, terminal voltage V, load R_L (from the exam paper)

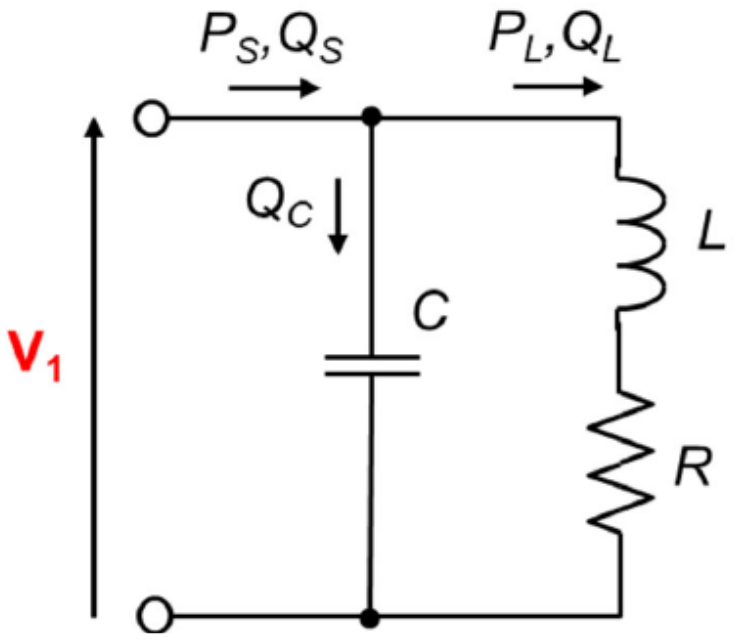
Jun23 Q1b (past exam, 6 marks). *Star-Delta Equivalent Circuits.* Consider a three-phase load consisting of three 70 Ω resistors connected to a 460 V, 3ph supply.

i. If the load is star-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

ii. If the load is delta-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

Jun24 Q1d (past exam, 8 marks). *Power Factor Correction.* Consider the circuit shown on the right which contains a series RL load which is known to absorb a real power of 80 W and a reactive power of 60 VAr. A power-factor correction capacitor C is connected across the input supply.

i. Sketch the power triangle for the RL load and hence calculate its power factor. [2 marks]
ii. By adding the power factor correction capacitor C, it is desired to have a supply (input) power factor of 0.9 lag at the input to the circuit shown. Find the supply real power P_S and the value of supply reactive power Q_S which gives the desired power-factor. Hence find the value of the reactive power absorbed by the capacitor Q_C. [4 marks]
iii. For another desired value of power-factor, the capacitor is found to absorb −30 VAr. What is the required value of capacitance (in μF) for the above capacitor if the input power supply is 115 V, 60 Hz? [2 marks]

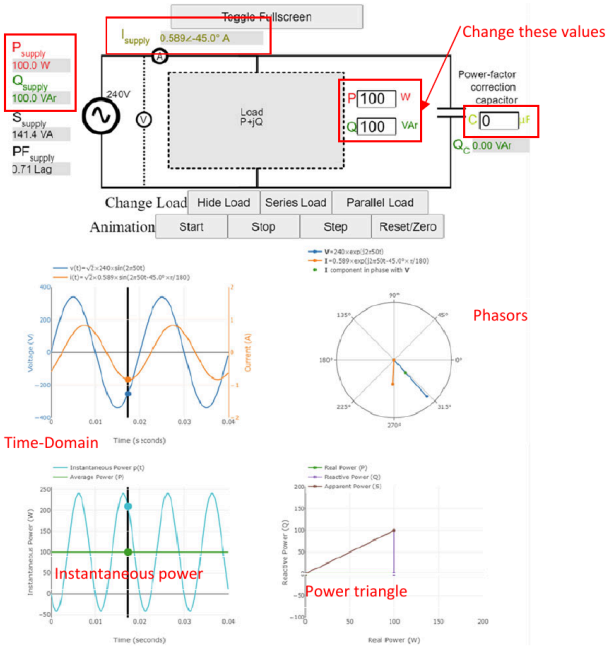


Jun24 Q1d figure: capacitor C across V_1 in parallel with the series L–R load, with P_S, Q_S, Q_C and P_L, Q_L marked (from the exam paper)

Power Circuit Tool WS2 Task 1 (workshop). *Investigating the Load's P and Q Relationship to Supply Current I.* Change the values of P and Q and the record the resultant supply current phasor in the table below.

Case	P (W)	Q (VAr)	$\hat{I}_{supply}$ (A∠°)
1	100	0	
2	0	100	
3	100	100	

1. What is the difference between the supply current phasors for cases 1 and 2?
2. What is the difference between the instantaneous power waveforms between case 1 and 2?
3. Express the supply current phasor for case 3 in terms of the phasors for case 1 and 2? Hint: consider superposition.



Power Circuit Tool WS2 Figure 2: the tool interface, with the P, Q and C inputs and the supply outputs highlighted (from the workshop sheet)

Power Circuit Tool WS2 Task 2 (workshop). *Series and Parallel Equivalent Circuits.* Set the load P = 100 W and Q = 200 VAr. Set "Change Load" to "Series Load" and then "Parallel Load" and record the values in the table below.

Case	$\hat{I}_{supply}$ (A∠°)	R (Ohms)	X (Ohms)
Series			
Parallel			

1. In the series circuit is the voltage or the current the same for both R and X?
2. In the series load, what is the ratio of the values of R and X? Can you explain this using $P = I^2R$ and $Q = I^2X$?
3. In the parallel circuit is the voltage or the current the same for both R and X?
4. In the parallel load, what is the ratio of the values of R and X? Can you explain this using $P = V^2/R$ and $Q = V^2/X$?

Power Circuit Tool WS2 Task 3 (workshop). *Power-Factor Correction.* Set the load P = 100 W and Q = 140 VAr. Set “Change Load” to “Hide Load”. Adjust the magnification of your browser so you can adjust the capacitance and see the supply power triangle at the same time. Record the quantities in the table below with zero capacitance in case 1. Now step the capacitance up in integer steps using the up arrow next to its value. Observe the effect on the current waveform and the power triangle. Find the integer value of capacitance $C_{optimal}$ which results in the smallest supply reactive power and record this as case 3. Then record the values for half this value of capacitance as case 2 and 1.5 times this value of capacitance as case 4.

Case	Capacitance (µF)	P_{supply} (W)	Q_{supply} (VAr)	$\hat{I}_{supply}$ (A∠°)
1: zero capacitance	0			
2: 0.5 $C_{optimal}$				
3: $C_{optimal}$				
4: 1.5×$C_{optimal}$				

1. What is the change in the supply real power as the capacitance is changed? Why is this?
2. Describe the change in the supply current magnitude and angle as the capacitance changes from 0 to $C_{optimal}$ and then beyond $C_{optimal}$.
3. Compare the value of supply current using $C_{optimal}$ versus that for a 240 V source to supply 100 W at unity power-factor.

PQ 2.20 (practice questions). Two single-phase loads $Z_1 = 100 + j0\ \Omega$ and $Z_2 = 10 + j20\ \Omega$ are connected in parallel across a **200 V, 60 Hz** source.

a. Find the magnitude of the two load currents, the total real and reactive power supplied by the source, and the power-factor at the source.
b. Find the capacitance that must be connected in parallel with the loads to improve the power factor to 0.8 lagging.
c. Calculate the total apparent power before and after the capacitor is added.
d. Sketch a power triangle showing apparent, real and reactive power before and after the capacitor is added.

PQ 2.6 (practice questions). A certain car battery can be modelled using an equivalent circuit of a 12.5 V voltage source with a series resistance of 0.02 Ω. Calculate:

a. The open-circuit battery output voltage.
b. The battery current and internal power loss under short-circuit conditions.
c. The battery output voltage and current when powering a load of 1 ohm.
d. When the starter motor is energised, the battery output voltage is observed to fall to 8 V. Calculate the current drawn by the starter motor and the power supplied to the starter motor.
e. Calculate the maximum output power available from the battery.

PQ 2.7 (practice questions). You perform an open-circuit voltage and short-circuit current test on an AA alkaline battery. You measure an open-circuit voltage of 1.61 V and a short-circuit current of 3.1 A.

a. Determine the equivalent circuit parameters E and R.
b. Find its output voltage and power when the load current is 0.5 A.

c. Find the maximum output power from the battery.

PQ 2.21 (practice questions). It is desired to find the equivalent circuit parameters for a single-phase 110 V AC transformer under open-circuit and short-circuit conditions.

a. In the short-circuit condition, the transformer can be modelled by a series resistance and inductive reactance. When 5 V is applied to the transformer, it draws a current of 20 A with a measured input power of 25 W. Find the equivalent parameters for the transformer.
b. In the open-circuit condition, the transformer can be modelled by a parallel resistance and inductive reactance. When 110 V is applied to the transformer, it draws a current of 2.4 A with a measured input power of 40 W. Find the equivalent parameters for the transformer.

PQ 2.26 (practice questions). Each phase of a three-phase load requires a phase voltage of 110 V and a phase current of 10 A. What is the required supply line voltage and current if the load is connected in:

a. star
b. delta

PQ 2.28 (practice questions). A star-connected three-phase resistive load draws 10 A from a 230 V supply.

a. what is the resistance of each phase?
b. The load is reconnected in delta but the supply voltage is unchanged. What are the magnitudes of the new phase voltage, phase current and supply current?
c. By what factors have the phase voltage, phase current and supply current changed?

PQ 2.30 (practice questions). Each phase of a balanced **delta**-connected load consists of a series connected 2 ohm resistor and an inductor with a 3 ohm reactance. It is connected to a 415 V, 3ph supply. Calculate:

a. the phase voltage?
b. the phase current magnitude and phase? (take the phase of the voltage as the reference)
c. the magnitude of the line current?
d. the input power factor?
e. the total input volt-amps (apparent power)?
f. the total input power using the voltage, current and power factor?
g. the total input power as three times the resistive loss $I_1^2 R$ in each phase? (note, this should give you the same result as in the previous part).
h. the total input reactive power?

PQ 2.31 (practice questions). The delta-connected load in the previous question is reconnected in star, however the supply line voltage is unchanged. For each part of the previous question, explain by what factor each quantity changes when connection is changed from delta to star.

Jun23 Q4b (past exam, 7 marks), stretch. *Power Factor Correction.* An industrial plant uses two identical induction motors, each driving a mechanical load of 200 kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i. Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii. Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]
iii. Use the above results to determine the maximum power factor for the plant. [3 marks]

New questions

N1. [4 marks] Three 48 Ω heater elements are connected to a 400 V, three-phase supply.

a. For a delta connection, find the phase current, line current and total power. [2]
b. The elements are reconnected in star. Find the phase voltage, line current and total power, and state the ratio of delta power to star power. [2]

Circuit for New question 2: battery model E and R_S supplying the lights, with the starter motor switched in

N2. [5 marks] A car battery delivers 12.6 V while supplying 5 A to the lights. When the starter motor is engaged the total current is 75 A and the terminal voltage falls to 11.2 V. Use the model E in series with R_S.

a. Find R_S and E. [2]
b. Find the terminal voltage, power to the load and internal loss at 150 A. [2]
c. Find the maximum power the battery could deliver. [1]

Circuit for New question 3: the motor modelled as series R_S and L_S on 230 V, 50 Hz, with the part (d) capacitor across the supply

N3. [6 marks] A single-phase motor on 230 V, 50 Hz draws 6.2 A and 1.1 kW.

a. Find S, Q and the power factor. [2]
b. Find the equivalent series R and L. [2]
c. Find the equivalent parallel R and L. [1]
d. Find the capacitance across the supply needed for unity pf, and the new supply current. [1]

Circuit for New question 4: induction motor, resistive heater and a capacitor bank on the 415 V workshop supply

N4. [8 marks] A workshop on a 415 V, 50 Hz three-phase supply has an induction motor delivering 22 kW of mechanical output at 90% efficiency and 0.82 pf lag, and a 10 kW balanced resistive heater.

a. Find the total P, Q and pf of the workshop, and the line current. [3]
b. Find the reactive power a capacitor bank must supply to raise the pf to 0.98 lag. [2]
c. Find the capacitance per phase if the bank is connected in delta, and if it is connected in star. Which needs the higher voltage rating? [2]
d. The delta-sized capacitors are wired in star by mistake. What pf results? [1]

N5. [7 marks] (Uses your Prac 1 data.) The series-RL prac load absorbs 25.0 W and +31.0 VAr from a 60.5 V, 50 Hz supply.

a. Find the supply current and pf without correction. [1]
b. Find the **two** values of parallel capacitance that give a supply pf of 0.9. State whether each gives a leading or lagging pf. [3]
c. Find the supply current for both values in (b), and the minimum possible supply current. [1]
d. In Table 5 the 36 µF row shows supply Q = +16.6 VAr with load Q = 31.0 VAr and capacitor Q = −45.3 VAr. Explain why the sign must be wrong, and state whether the 0.837 pf is leading or lagging. [2]

6. Quick recall check

1. A 415 V three-phase supply feeds a star load. What is the phase voltage?
2. A balanced load is moved from star to delta on the same supply. By what factor do the line current and the total power change?
3. A load takes 60 W at 0.6 pf lag. What Q_C gives unity pf?
4. A battery reads 12.0 V on open circuit and 11.4 V at 30 A. What is R_S?
5. For the same total Q_C on the same supply, how does the capacitance per phase of a star bank compare with a delta bank?

Magnetic circuits

Practice questions · Day 03 · Mon 26 Oct · pack M3

5. Practice

From your course files

PQ 3.3 (practice questions). Calculate the magnetic reluctance of the following objects:

a. A piece of circular iron wire, 1 mm in diameter and 10 cm long with a relative permeability of 1000.
b. An iron core of mean path length 20 cm, and cross-sectional area 2 cm² and relative permeability of 5000.
c. An airgap of length 1 mm and cross-sectional area 10 cm².

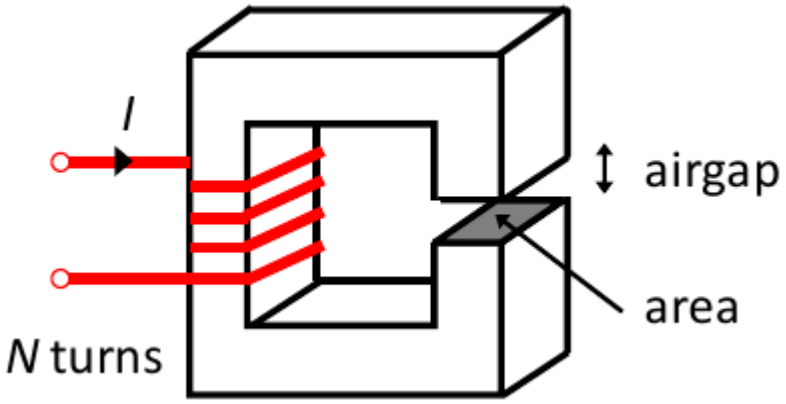
PQ 3.4 (practice questions). In the following cases, calculate the required mmf to produce a 1 T magnetic flux density in the object and hence calculate the required coil current if a coil of 100 turns is used to produce this mmf.

a. An airgap of length of 1 mm.
b. An magnetic core with a mean length of 20 cm and relative permeability 5000

Jun23 Q2a (past exam, 9 marks). *Electromagnetics.* Consider the iron c-core shown, which has a magnetic cross-sectional area of 20 cm² and a mean magnetic path length of 50 cm. The core has a winding with 32 turns and has a relative permeability of 3,000. The airgap length is 1.5 mm.

i. Calculate the reluctances of the airgap $\mathcal{R}_G$ and the core $\mathcal{R}_C$ and hence find the inductance of the coil. [4 marks]
ii. Calculate the maximum energy stored in the airgap only, if the maximum allowable peak core flux density is 1.8 T. [2 marks]
iii. The magnetic core is operating with a peak flux density of 1.5 T at 60 Hz. It has an iron loss of 12 W, of which 30% is hysteresis losses and 70% is eddy-current losses.

Calculate the new hysteresis, eddy-current and total iron loss if the peak flux density is reduced to 1.2 T, whilst the frequency is increased to 100 Hz. [3 marks]



Jun23 Q2a figure: iron C-core with an N-turn coil carrying I, an airgap, and the cross-section area marked (from the exam paper)

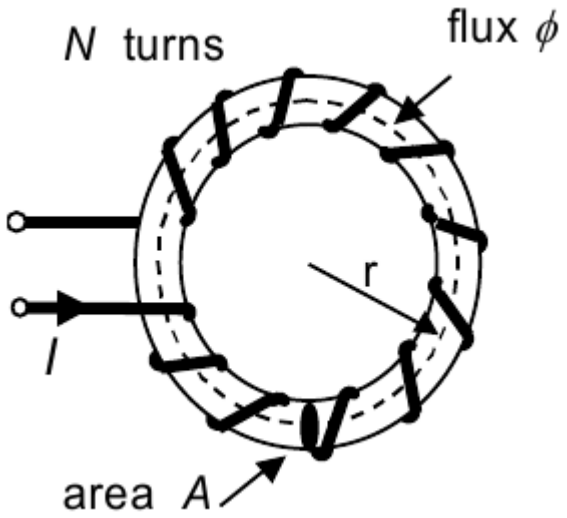
Jun24 Q2a (past exam, 7 marks). *Electromagnetics.* Consider a toroidal magnetic core which has a circular cross-section whose diameter is 2.5 cm. The core has a winding of 120 turns and can be assumed to have an infinite permeability. A fine saw is used to cut a thin slot to produce an airgap of 0.5 mm in one side of the core.

i. Calculate the reluctance of the airgap $\mathcal{R}_g$ and hence the inductance of the winding. [3 marks]
ii. If the maximum allowable peak core flux density is 1.5 T, calculate the maximum 50 Hz voltage which can be applied to the winding. [2 marks]
iii. Using a flux density of 1.5 T, calculate the force between the sides of the slot in the core. [2 marks]

The exam paper has no figure for this part. The sketch in the solution is drawn from the wording.

2023 quiz Q3 (6 marks). *AC Magnetic Circuits.* A winding is placed around a toroid iron core as shown in the figure on the right. The core has a mean diameter of 80 mm, a cross-section area of 20 cm² and a relative permeability μ_R of 1,000. The desired peak operating flux density of the core is 1.5 T.

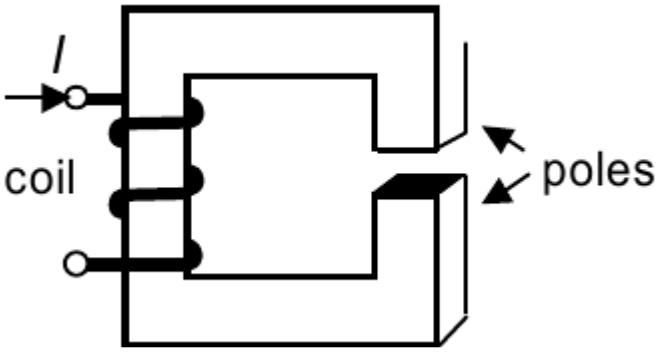
a. Given the winding has a rated voltage of 240 V at 50 Hz, use the back-emf equation to calculate the required number of turns. [2 marks]
b. For a different rated voltage, the number of turns is changed to 200. Calculate the winding inductance and reactance for this number of turns. [3 marks]
c. Based on the results from part (b), calculate the winding current magnitude when 100 V is applied to the winding. [1 mark]



2023 quiz Q3 figure: toroid with N turns carrying I, flux φ, radius r and cross-section area A (from the quiz paper)

2023 quiz Q4 (4 marks). *DC Magnetic Circuits.* An electromagnetic actuator has two iron poles with a 4 mm airgap between them, as shown in the figure on the right. Each pole has an area of 20 cm². A coil of 75 turns is used to create a flux density of 1.1 T in the airgap. Assume the core has infinite permeability.

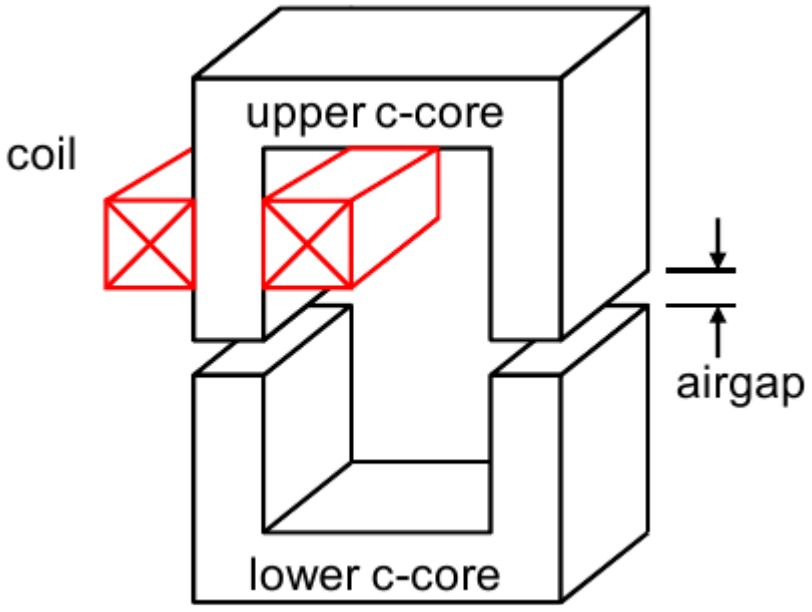
a. Calculate the required DC coil current. [2 marks]
b. Calculate the force of attraction between the poles. [2 marks]



2023 quiz Q4 figure: iron core with a coil carrying I and two poles across one airgap (from the quiz paper)

2024 test Q4 (7 marks). *Electromagnetics.* An electromagnet is constructed using two identical magnetic cores as shown on the right. The upper c-core is held stationary and magnetic attraction is used to hold the lower c-core. Each core has a mass of 0.3 kg, a magnetic cross-section area A of 5 cm² and a relative permeability of 4,000. The mean magnetic iron path in the cores for the flux is 25 cm (this is the total for the two cores). A coil of five turns carrying 3.5 A is placed on the upper core.

a. Assuming initially that the airgap between the cores is zero, what is the flux density in the magnetic circuit? [2 marks]
b. A piece of paper of 0.07 mm thickness is placed between the cores. The relative permeability of paper can assumed to be equal to 1. Calculate the new value of flux density in the magnetic circuit. [3 marks]
c. What is the minimum flux density in the magnetic circuit that would be required to support the weight of the lower core? Assume a value of acceleration due to gravity of 9.81 m/s². [2 marks]



2024 test Q4 figure: upper and lower C-cores with the coil on the upper core and the airgap between them (from the test paper)

2024 test Q5 (5 marks). *Magnetic Circuits.* A single-phase 240 V/48 V, 50 Hz transformer uses a laminated rectangular core with a cross-sectional area of 40 cm², magnetic path length of 0.8 m and relative permeability of 2,000. The peak operating flux density of the core should not exceed 1.6 T.

a. Using the back-emf equation, calculate the required number of primary turns for the transformer. [2 marks]
b. The primary winding consists of copper wire with a length of 40 m and a circular wire of diameter 0.5 mm. Find its resistance using a conductivity of $59 \times 10^6\ (\Omega\mathrm{m})^{-1}$. [2 marks]
c. Choose an appropriate number of turns for the secondary winding. [1 mark]

2023 quiz Q7 (4 marks). *Short-Answer Questions.* Answer the following question using words only (that is, without using sketches or equations).

In designing a transformer using a given magnetic core, what is the process for selecting the number of turns on the primary winding for a given primary voltage? Carefully explain the results of using too few turns. How about if too many turns are used?

PQ 3.10 (practice questions). Ferrite is a magnetic material which has a maximum allowable flux density before saturation of about 0.3 T. A circular ferrite core has a cross-sectional area of 5 cm² and has a small airgap cut in the core. A winding of 50 turns is placed on the core. Assume the ferrite has infinite permeability.

a. What length airgap is required to produce an inductance of 1 mH?
b. Alternatively, if the airgap is 1 mm, how many turns would be required to produce the same inductance? (number of turns needs to be an integer, hence round result to nearest)
c. For case (a), calculate the maximum current the inductor can carry without saturating.

PQ 3.13 (practice questions). A laminated magnetic core is operating at 50 Hz with a peak flux density of 1 T. At this flux density the core has an iron loss of 3 W.

a. The peak flux density is increased to 1.5 T. Estimate the new core loss.
b. The operating frequency is increased to 400 Hz with the original peak flux density of 1 T. Estimate the new iron loss assuming that (i) the core only has hysteresis loss and has no eddy-current loss, that is, the original 3 W of loss was only due to hysteresis. Hint: hysteresis loss is directly proportional to frequency. (ii) the core only has eddy-current loss and has no hysteresis loss (iii) the core has both eddy-current and hysteresis loss and at 50 Hz the eddy-current loss is equal to the hysteresis loss
c. In the situation described in part (b)iii, what is the required peak flux density at 400 Hz to give the same iron loss as operation at 50 Hz with a peak density of 1 T. Hint: remember that iron loss is proportional to B^2

PQ 3.14 (practice questions). Answer the following questions about magnetics.

a. Consider a magnetic core with an airgap. Where is the majority of energy stored?
b. Iron losses are caused by changing magnetic fields in a conductive magnetic material. Are there any iron losses when air is exposed to a changing magnetic field?
c. What are the two types of iron loss?
d. Which type of iron loss is more important when magnetic cores are operated using high frequency currents and why?
e. Why do motors and transformers normally use iron in the form of thin, insulated iron sheets rather than a solid core?
f. What happens to the input current of an inductor as the AC voltage is increased such that the core goes into saturation?

New questions

N1. Gapped rectangular core [5 marks]. A rectangular iron core has a mean path of 60 cm, a cross-section of 12 cm² and $\mu_R = 2500$. It has a single 0.8 mm air gap and a 250-turn coil.

a. Find $\mathcal{R}_C$, $\mathcal{R}_G$ and the coil inductance. [3]

b. Find the coil current for $B = 1.2$ T, and the fraction of the mmf dropped across the gap. [2]

Core for New question 2: shell core with a 200-turn coil and a 1 mm gap on the centre limb; the flux splits into the two outer paths

N2. Shell (E) core with parallel paths [7 marks]. A shell-type core has a 200-turn coil on its centre limb. The centre limb has length 10 cm and area 8 cm², and contains a 1 mm air gap. The flux returns through two identical outer paths. Each outer path has length 30 cm and area 4 cm². All the iron has $\mu_R = 2000$. Neglect fringing.

a. Draw the magnetic equivalent circuit and find the total reluctance seen by the coil. [3]
b. Find the coil inductance. [1]
c. Find the coil current needed for 1.2 T in the gap, and the flux density in each outer limb. [3]

N3. Back-emf design and loss scaling [8 marks]. A 230 V, 50 Hz iron-cored reactor has a core area of 15 cm², and its peak flux density must not exceed 1.5 T.

a. Find the number of turns and the actual peak flux density. [2]
b. At 230 V, 50 Hz the iron loss is 20 W, of which 40% is hysteresis. The reactor is moved to a 230 V, **60 Hz** supply. Find the new peak flux density and the new hysteresis, eddy-current and total iron loss. [3]
c. It is then used on a 40 Hz supply. Explain why 230 V is unsafe, and find the highest voltage that keeps $\hat{B} \leq 1.5$ T. Estimate the iron loss at that voltage. [3]

Lifter for New question 4: U-core with a 400-turn coil pulling an armature across two 1.5 mm gaps

N4. Electromagnetic lifter, two gaps [6 marks]. A U-shaped iron core with a 400-turn coil attracts an iron armature across two air gaps of 1.5 mm each. Each pole face is 6 cm². Treat the iron as infinitely permeable.

a. Find the coil current needed for a total pull of 150 N. [3]
b. Find the coil inductance and the energy stored at that current. [2]
c. The armature moves in until each gap is 0.5 mm, with the same current. Estimate the new force and comment. [1]

N5. Gapped inductor design, chained [8 marks]. You need a 5 mH inductor that carries a peak current of 10 A. The core has a cross-section of 6 cm², and its peak flux density must stay at or below 1.2 T. Assume infinite core permeability and a single air gap.

a. Using $\lambda = N\Phi = LI$, find the minimum number of turns (integer). [2]
b. Find the gap length that gives 5 mH with this number of turns. [2]
c. Find the actual peak flux density at 10 A, and show that $\frac{1}{2}L\hat{I}^2$ equals the gap energy. [2]
d. The inductor is used on a 50 Hz sinusoidal supply at a 10 A peak. Use the back-emf equation to find the rms voltage across it, and check it against $\omega L I_{\mathrm{rms}}$. [2]

6. Quick recall check

1. What are the units of reluctance, and what is the electric analogue of mmf?
2. Same coil and current, but the air gap (which dominates the reluctance) is doubled. What happens to L, to B, and to the force?
3. A transformer's applied voltage is halved at the same frequency. By what factor does the iron loss change?
4. Same voltage, but the supply changes from 50 Hz to 60 Hz. What happens to $\hat{B}$, P_H and P_E?
5. A winding is rated 240 V at 50 Hz. What is the highest voltage you should apply at 25 Hz, and why?

Transformer equivalent circuit

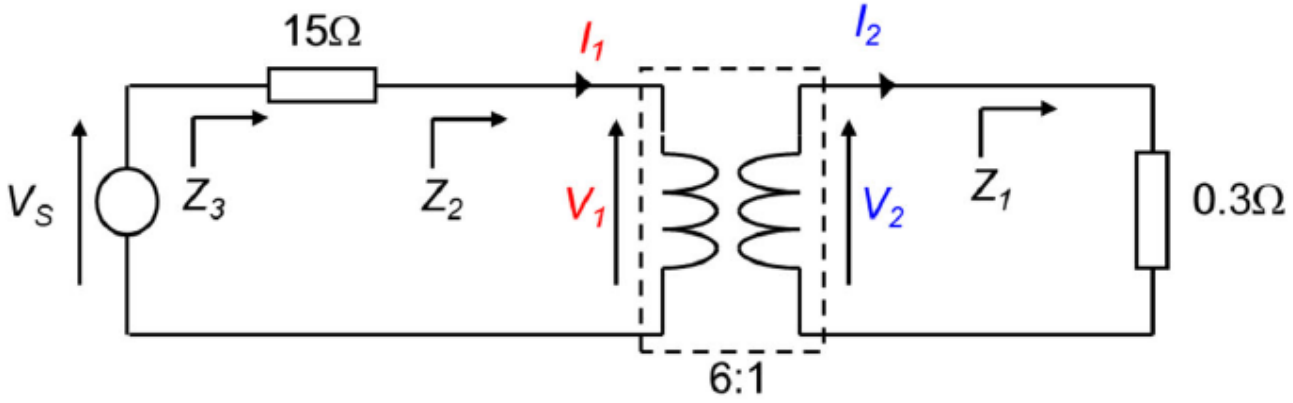
Practice questions · Day 04 · Tue 27 Oct · pack M4

5. Practice

From your course files

Jun24 Q2b (past exam, 5 marks). *Ideal Transformers.* In the diagram on the right the transformer is ideal and has a turns ratio of 6:1.

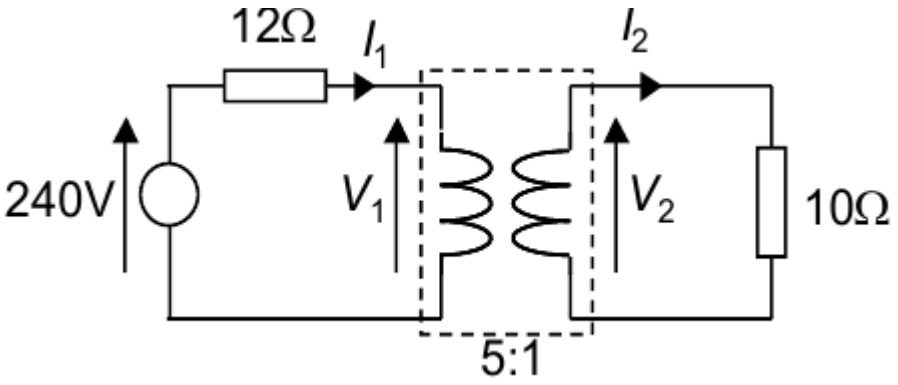
i. Find the values of the impedances Z_1, Z_2 and Z_3 shown in the figure. [2 marks]
ii. Assuming the AC supply voltage V_S is 12 V, find the magnitudes of the primary current I_1 and the actual (not referred) secondary current I_2. Find also the magnitudes of the primary voltage V_1 and the actual secondary voltage V_2. [3 marks]



Jun24 Q2b figure: V_S, 15 Ω series resistor, ideal 6:1 transformer and 0.3 Ω load, with Z_3, Z_2 and Z_1 marked (from the exam paper)

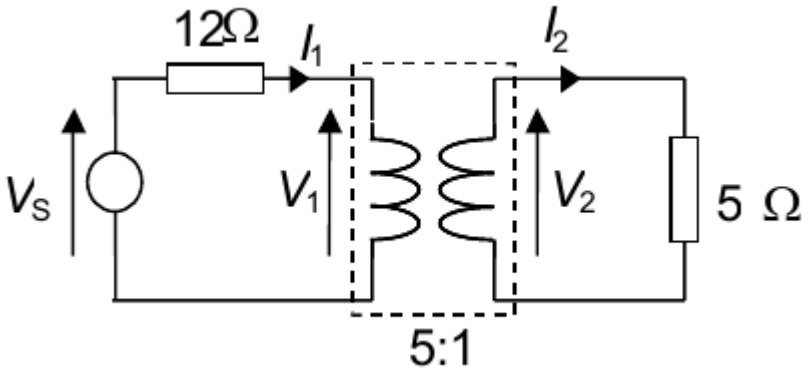
PQ 3.20 (practice questions). Consider the ideal transformer circuit shown below.

a. Find the load resistance referred to the primary.
b. Calculate the primary current I_1. [hint: it may be helpful to draw the equivalent circuit using the load resistance referred to the primary.]
c. Find the primary voltage V_1 [hint: Ohm's law]
d. Find the secondary current and voltage.
e. Calculate the output power delivered to the load
f. Calculate the power loss in the 12 Ω resistor.
g. Calculate the efficiency of power transfer from the voltage source to the load.



PQ 3.20 figure: 240 V source, 12 Ω series resistor, ideal 5:1 transformer and 10 Ω load (from the practice questions)

PQ 3.21 (practice questions). Consider the ideal transformer circuit shown below. The input voltage is 240 V. Calculate the secondary voltage.



PQ 3.21 figure: source V_S, 12 Ω series resistor, ideal 5:1 transformer and 5 Ω load (from the practice questions)

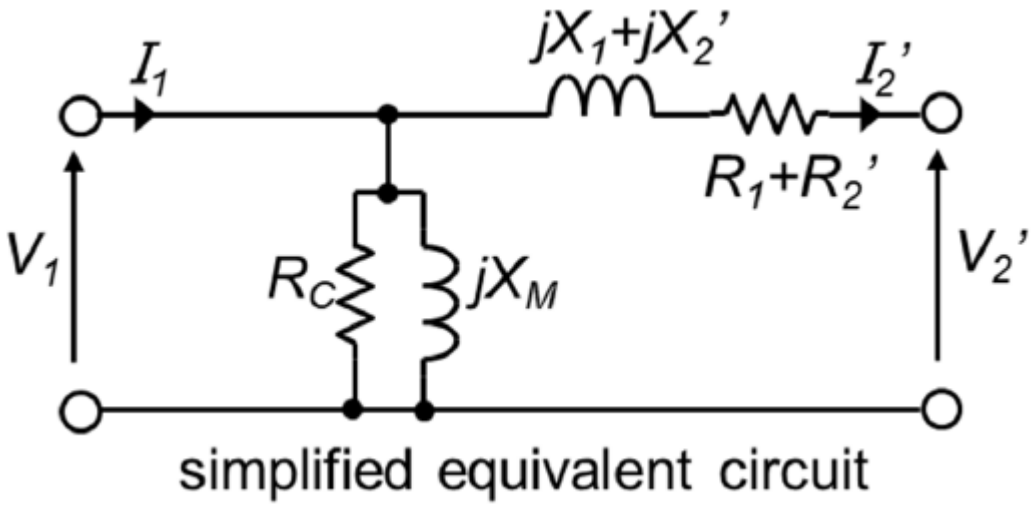
Jun23 Q2b (past exam, 5 marks). *Transformer Parameters.* The following tests are performed on a single-phase 240 V:32 V, 2400 VA transformer.

- Short-circuit test. Primary measurements are 27 V, 9.7 A, and 55 W.
- Open-circuit test. Primary measurements are 240 V, 1.3 A, and 45 W.

Find all the simplified equivalent circuit parameters.

Jun23 Q2c (past exam, 6 marks). *Transformer Analysis.* Consider the transformer simplified equivalent circuit shown on the right. The values of the primary voltage $\mathbf{V_1}$ and current $\mathbf{I_1}$ phasors are known and it is desired to find the value of the secondary voltage $\mathbf{V_2'}$ phasor.

i. Using Kirchhoff's voltage law (KVL), write an expression for $\mathbf{V_2'}$ in terms of $\mathbf{V_1}$, $\mathbf{I_2'}$ and the relevant equivalent circuit parameters shown. [2 marks]
ii. Assuming the load is purely resistive with a value of R_L', use the voltage divider rule to write an expression for $\mathbf{V_2'}$ in terms of $\mathbf{V_1}$ and the relevant equivalent circuit parameters shown. [2 marks]
iii. For a purely resistive load, would you expect that $\mathbf{V_2'}$ will be leading, in phase or lagging with respect to $\mathbf{V_1}$? Carefully explain why. [2 marks]



Jun23 Q2c figure: simplified equivalent circuit with R_C and jX_M across V_1, then jX_1+jX_2′ and R_1+R_2′ in series to V_2′ (from the exam paper)

PQ 3.27 (practice questions). A single-phase 1000 VA, 240 V/48 V transformer has the following parameters: primary resistance 0.6 ohm, secondary resistance 0.02 ohms, primary leakage reactance 2 ohms, secondary leakage reactance 0.06 ohms, magnetising reactance 550 ohms, core-loss resistance 3200 ohms. Note the secondary parameters are NOT referred to the primary. It is connected to a 240 V, 50 Hz supply for parts (a) to (c).

a. Draw a simplified equivalent circuit for the transformer referred to the primary and calculate all the circuit parameters.
b. Find the no-load (that is, with the secondary winding open-circuited) input current magnitude and input power at rated voltage.
c. Find the input current magnitude and input power at rated input voltage with the secondary winding short-circuited. You can neglect the core-loss resistance and the magnetising reactance in this part and for part (d).
d. Find the rated input current. Then find the required supply voltage magnitude to produce rated input current with the secondary winding short-circuited.

PQ 3.28 (practice questions). Using the input voltage and current magnitudes and input power results from parts (b) and (c) of the above question, calculate the reactances of the simplified equivalent circuit. Verify the calculated parameters are identical to the actual parameters.

PQ 3.29 (practice questions, do (a)–(c)). The following tests are performed on a single-phase 240 V:24 V, 2400 VA transformer. Short-circuit test. Secondary winding is short-circuited. Primary measurements are 27 V, 9.5 A, and 55 W. Open-circuit test. Primary winding voltage is 240 V, current is 1.2 A, power is 45 W.

a. Find the simplified equivalent circuit parameters.
b. The transformer is delivering rated output at a power-factor of 0.87 lagging at rated output voltage. Calculate the secondary voltage and current referred to the primary.
c. Find the required primary voltage corresponding to the case in part (b).
d. Find the output power, copper loss, iron loss and efficiency of the transformer corresponding to the case in part (b).

2024 test Q6 (4 marks). *Transformer Parameters.* Consider a star-connected, three-phase transformer. The following test results were obtained:

- open-circuit test: 5.2 kV, 45 A, 49 kW
- short-circuit test: 710 V, 390 A, 140 kW

Find the winding resistance and the leakage reactance for the simplified transformer equivalent circuit. It is not necessary to find the other equivalent circuit parameters.

2024 test Q7 (4 marks). *Transformer Analysis.* A three-phase, 14 kV:135 kV 250 MVA transformer is supplying a load. The load has an apparent power of 150 MVA and is generating 35 MVAr of reactive power. The output voltage is equal to its rated value. Find the secondary phase voltage $\mathbf{V}_2'$ and phase current $\mathbf{I}_2'$ phasors referred to the primary. Assume the phase angle of $\mathbf{V}_2'$ is zero. Hint: remember the power triangle!

2024 test Q8 (4 marks). *Short-Answer Question.* Answer the following question using words only (that is, without using sketches or equations).

Consider an ideal transformer driving a resistive load of resistance R. Carefully explain what would be the effect on the output voltage and current if the load resistance was doubled. Carefully explain why and how this would be different for a real transformer.

2023 quiz Q6 (9 marks). *Transformer Analysis.* A three-phase 50 kVA, 3.3 kV/415 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance 35 ohms, total winding resistance 4 ohms, magnetising reactance 2,300 ohms, core loss resistance 18,000 ohms. The transformer is delivering 27 kW at a power-factor of 0.78 lagging at rated output voltage.

a. Calculate the magnitude and phase of the secondary phase voltage $\mathbf{V}_2'$ and current $\mathbf{I}_2'$ referred to the primary. [4 marks]
b. Under a different load condition, it is known that the secondary phase voltage $\mathbf{V}_2' = 2100\angle 0^\circ$ V and that the secondary phase current $\mathbf{I}_2' = 7.7\angle -25^\circ$ A (both referred to the primary). Draw a simplified equivalent circuit for the transformer referred to the primary and hence calculate the required primary (line) voltage magnitude. [5 marks]

PQ 3.30 (practice questions, overflow: do (a), (b)). A three-phase 3000 VA, 415 V/48 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance X 8.5 ohms, total winding resistance R 0.95 ohms, magnetising reactance X_M 510 ohms and core loss resistance R_C 3300 ohms. The transformer is delivering rated output at unity power-factor at rated output voltage.

a. Calculate the secondary phase voltage and current referred to the primary.
b. Calculate the required supply voltage magnitude.
c. Calculate both the total copper winding loss and total core loss.
d. Hence calculate the transformer efficiency.

PQ 3.31 (practice questions, overflow: do (a), (b)). The following tests are performed on a three-phase 15 kV:135 kV, 100 MVA star-connected transformer. Short-circuit test with secondary winding short-circuited. Primary measurements are 95 V, 500 A, and 12 kW. In the open-circuit test the primary winding voltage is 415 V, current is 15 A, power is 800 W.

a. Find the simplified equivalent circuit parameters referred to the primary winding.
b. Assuming the transformer is delivering 50 MW at a power-factor of 0.8 lagging at rated output voltage, calculate the secondary phase voltage and current referred to the primary and hence the required input voltage.
c. The transformer efficiency corresponding to the operating case described in part (b).

New questions

New Question 1 (easy) [6 marks]. A 240 V, 50 Hz source with an internal series resistance of 8 Ω feeds the primary of an ideal 8:1 transformer. The secondary supplies a load $Z_L = 0.4 + j0.3\,\Omega$.

a. Find the load impedance referred to the primary. [1]
b. Find I_1 and V_1 in polar form, taking the source at 0°. [3]
c. Find the actual secondary current and voltage, and the real power delivered to the load. [2]

New Question 2 OC test: meters on the 110 V winding, 415 V winding open

New Question 2 SC test: meters on the 415 V winding, 110 V winding shorted

New Question 2 (medium) [5 marks]. A single-phase 5 kVA, 415 V:110 V transformer is tested. The 415 V winding is the primary.

- Open-circuit test, done on the **110 V side** (415 V side open): 110 V, 1.8 A, 55 W.
- Short-circuit test, done on the **415 V side** (110 V side shorted): 22 V, 12.05 A, 120 W.

Find all the simplified equivalent circuit parameters referred to the 415 V primary. Draw the circuit.

New Question 3 (medium) [5 marks]. A 3-phase, star-connected 11 kV/415 V, 500 kVA transformer gives these primary-side test results:

- OC test: 11 kV, 1.2 A, 1.9 kW

- SC test: 480 V, 26.24 A, 5.2 kW

a. Find the per-phase simplified equivalent circuit parameters referred to the primary. [4]
b. Show that the SC test was carried out at rated current. [1]

Circuit for New Question 4: the New Question 2 transformer on 415∠0° V with the 2.42 Ω load referred to the primary

New Question 4 (hard, chained) [8 marks]. Use the transformer from New Question 2 (referred to the 415 V primary: $R = 0.8264\,\Omega$, $X = 1.628\,\Omega$, $R_C = 3131\,\Omega$, $X_M = 905.5\,\Omega$). Rated voltage, $415\angle 0^\circ$ V, is applied to the primary. A resistive load of **2.42 Ω** (actual value) is connected to the 110 V secondary.

a. Find the referred load resistance R_L'. [1]
b. Write the KVL expression for V_2' in terms of V_1, I_2' and the circuit parameters. [1]
c. Use the voltage divider rule to find V_2' in polar form, and the actual secondary voltage V_2. [3]
d. Find the primary current I_1 in polar form. [2]
e. A student claims that replacing the load with one of the same $|Z_L'|$ at 0.8 pf leading would make V_2' lead V_1. Check this numerically and comment. [1]

New Question 5 (hard, saturation) [5 marks]. A single-phase 240 V, 50 Hz transformer has two OC tests on its primary:

- at 120 V: 0.050 A, 2.2 W
- at 240 V: 0.42 A, 8.6 W

a. Find R_C and X_M at both voltages. [2]
b. A designer uses only the 120 V values to predict the no-load current and iron loss at 240 V. What do they predict, and how does that compare with the measurement? [2]
c. Explain the difference in one or two sentences, and say which values belong in the equivalent circuit for rated operation. [1]

6. Quick recall check

1. In the SC test, why can R_C and X_M be neglected?
2. A 10:1 ideal transformer has a 2 Ω load on its secondary. What impedance does the primary see?
3. Which formula gives X_M from the OC test? Which gives X from the SC test?
4. For a resistive load, does V_2' lead, lag or stay in phase with V_1? Why, in one line?
5. Star-connected 3-phase OC test: 6.6 kV, 3 A, 6 kW. What are V_1 and P_1 per phase?

Regulation, efficiency, 3-phase transformers

Practice questions · Day 05 · Wed 28 Oct · pack M5

5. Practice

From your course files

Jun24 Q2c (past exam, 8 marks). Three-Phase Transformers. A three-phase transformer is rated at 16 kV/220 kV, 150 MVA. It has the following equivalent circuit parameters (referred to the primary winding where appropriate): primary resistance $R_1 = 0.08$ ohms, secondary resistance $R_2' = 0.06$ ohms, primary leakage reactance $X_1 = 0.14$ ohms, secondary leakage reactance $X_2' = 0.14$ ohms, magnetising reactance $X_M = 14$ ohms, core-loss resistance $R_C = 250$ ohms.

i. Draw a simplified equivalent circuit for the transformer showing the values for each of the equivalent circuit parameters as impedances. [2 marks]
ii. Consider the transformer in the open-circuit condition with rated input voltage. Find the input current magnitude and (real) power. [4 marks]
iii. Consider the open-circuit condition as discussed above, if the input voltage was halved, by what factor (e.g. x0.25, x0.5, x1, x2 or x4) would you expect the following quantities to change by? Input current magnitude; input real power; input reactive power; input power-factor. [2 marks]

2023 quiz Q5 (quiz, 5 marks). Transformer Losses. Consider a three-phase transformer with the following simplified single-phase equivalent circuit parameters referred to the primary: magnetising reactance X_M: 9,100 ohms; core loss resistance R_C: 100,000 ohms; total winding resistance R: 8.5 ohms; total leakage reactance X: 50 ohms. Under a particular operating condition it is known that the: primary voltage is 11.7 kV; secondary current referred to the primary is 6.8 A; input power is 115 kW. Using the simplified equivalent circuit, calculate the copper losses, the iron losses and hence the transformer efficiency.

2023 quiz Q6 (quiz, 9 marks). Transformer Analysis. A three-phase 50 kVA, 3.3 kV/415 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance 35 ohms, total winding resistance 4 ohms, magnetising reactance 2,300 ohms, core loss resistance 18,000 ohms. The transformer is delivering 27 kW at a power-factor of 0.78 lagging at rated output voltage.

a. Calculate the magnitude and phase of the secondary phase voltage V_2' and current I_2' referred to the primary. [4 marks]
b. Under a different load condition, it is known that the secondary phase voltage $V_2' = 2100\angle 0^\circ$ V and that the secondary phase current $I_2' = 7.7\angle -25^\circ$ A (both referred to the primary). Draw a simplified equivalent circuit for the transformer referred to the primary and hence calculate the required primary (line) voltage magnitude. [5 marks]

2023 quiz Q7 (quiz, 4 marks). Short-Answer Questions. Answer the following question using words only (that is, without using sketches or equations). In designing a transformer using a given magnetic core, what is the process for selecting the number of turns on the primary winding for a given primary voltage? Carefully explain the results of using too few turns. How about if too many turns are used?

PQ 3.29 (practice questions). The following tests are performed on a single-phase 240V:24V, 2400VA transformer. Short-circuit test. Secondary winding is short-circuited. Primary measurements are 27 V, 9.5 A, and 55 W. Open-circuit test. Primary winding voltage is 240 V, current is 1.2 A, power is 45 W.

a. Find the simplified equivalent circuit parameters.
b. The transformer is delivering rated output at a power-factor of 0.87 lagging at rated output voltage. Calculate the secondary voltage and current referred to the primary.
c. Find the required primary voltage corresponding to the case in part (b).
d. Find the output power, copper loss, iron loss and efficiency of the transformer corresponding to the case in part (b).

PQ 3.30 (practice questions). A three-phase 3000 VA, 415V/48V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance X 8.5 ohms, total winding resistance R 0.95 ohms, magnetising reactance X_M 510 ohms and core loss resistance R_C 3300 ohms. The transformer is delivering rated output at unity power-factor at rated output voltage.

a. Calculate the secondary phase voltage and current referred to the primary.
b. Calculate the required supply voltage magnitude.
c. Calculate both the total copper winding loss and total core loss.
d. Hence calculate the transformer efficiency.

PQ 3.31 (practice questions). The following tests are performed on a three-phase 15kV:135kV, 100 MVA star-connected transformer. Short-circuit test with secondary winding short-circuited. Primary measurements are 95 V, 500 A, and 12 kW. In the open-circuit test the primary winding voltage is 415 V, current is 15 A, power is 800 W.

a. Find the simplified equivalent circuit parameters referred to the primary winding.
b. Assuming the transformer is delivering 50 MW at a power-factor of 0.8 lagging at rated output voltage, calculate the secondary phase voltage and current referred to the primary and hence the required input voltage.
c. The transformer efficiency corresponding to the operating case described in part (b).

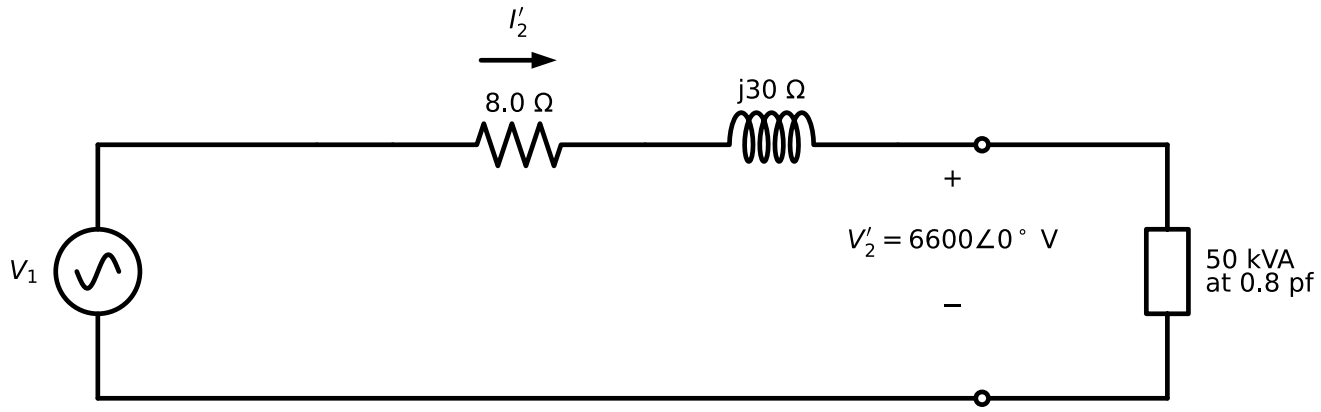
Jun24 Q4c(ii) (past exam, 3 marks). Answer the following questions using words only (that is, without using any sketches, drawings or equations).

ii. Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power. [3 marks]

New questions

N1. Loss scaling and maximum efficiency [5 marks]. A single-phase 10 kVA, 2400/240 V transformer has an iron loss of 120 W at rated voltage and a full-load copper loss of 180 W. Assume rated output voltage throughout.

a. Find the efficiency at full load, 0.8 pf lagging. [2]
b. Find the efficiency at half load, 0.8 pf lagging. [1]
c. Find the load (kVA) at which efficiency is maximum, and the maximum efficiency at unity pf. [2]

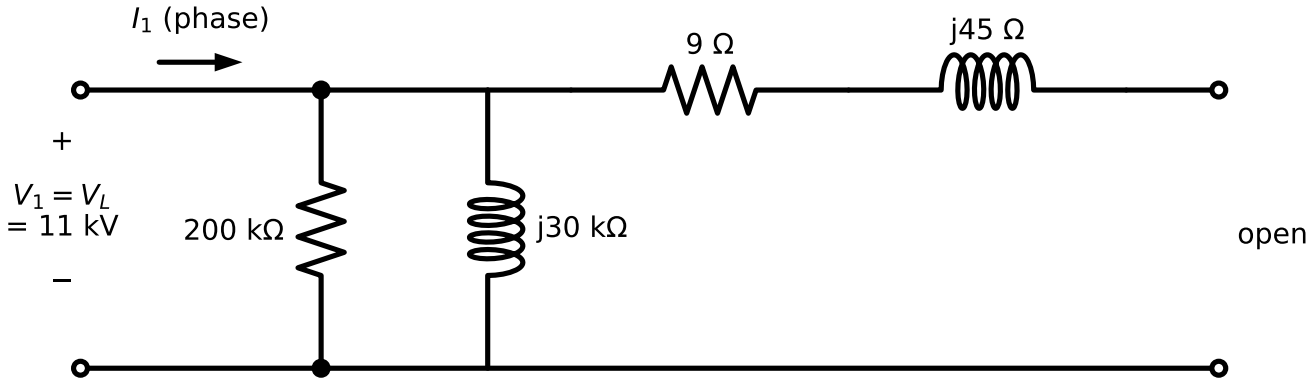


referred to the 6600 V primary; shunt branch neglected

Circuit for New question 2: series 8.0 Ω + j30 Ω (shunt branch neglected) delivering rated kVA at V_2′ = 6600∠0° V

N2. Regulation with lagging and leading loads [6 marks]. A single-phase 50 kVA, 6600/415 V transformer has $R = 8.0\ \Omega$ and $X = 30\ \Omega$ referred to the primary. The shunt branch can be neglected for the voltage calculation. The transformer delivers rated kVA at rated output voltage.

a. At 0.8 pf lagging, find the required V_1 and the regulation. [3]
b. Repeat for 0.8 pf leading. Explain the sign of the answer in one sentence. [2]
c. Using the approximate formula, find the leading pf at which the regulation is zero. [1]



one delta phase, referred to the primary

Circuit for New question 3: one delta phase on open circuit, 200 kΩ ‖ j30 kΩ across 11 kV, series 9 Ω + j45 Ω carrying no current

N3. Delta-connected primary on open circuit [6 marks]. A three-phase 500 kVA, 11 kV/415 V transformer has a **delta**-connected primary. The per-phase parameters, referred to the primary delta phase, are $R = 9\ \Omega$, $X = 45\ \Omega$, $R_C = 200\ \text{k}\Omega$ and $X_M = 30\ \text{k}\Omega$.

a. With rated voltage and the secondary open, find the phase current, the line current, the total input power and the pf. [3]
b. Express the OC line current as a % of rated line current. [1]
c. The supply falls to 5.5 kV. By what factors do the line current, P, Q and pf change? Would you expect the real transformer to follow these factors exactly? [2]

N4. Full chained analysis from test data [10 marks]. A three-phase, star-connected 1 MVA, 11 kV/433 V transformer gives these primary test results:

- OC: 11 kV, 3.0 A, 2.4 kW.
- SC: 550 V, 52.5 A, 9.0 kW.

a. Find R, X, R_C and X_M per phase. [3]
b. It delivers rated kVA at 0.85 pf lagging at rated output voltage. Find V_2', I_2', the required primary line voltage and the regulation. [3]
c. Find the efficiency for case (b). [2]
d. Find the kVA at which efficiency is maximum (at 0.85 pf) and the value of η_{max}. [2]

N5. Short answer: no-load current and the Prac 2 evidence [4 marks]. In words only:

a. Why is the input current of a transformer not zero when its secondary is open-circuited? [2]
b. In Prac 2 the open-circuit current rose from 0.04 A at 100 V to 0.66 A at 240 V. Explain why it rose so much more than the voltage. [2]

6. Quick recall check

1. A transformer runs at half its rated load. By what factor does the copper loss change, and by what factor does the iron loss change?
2. In the simplified circuit, which voltage do you use for the iron loss, and which current for the copper loss?
3. State the condition for maximum efficiency and the formula for the per-unit load at which it occurs.
4. On open circuit, the supply voltage is halved. Give the factors for current, P, Q and pf.
5. A star-connected transformer has a 33 kV line rating. What V_2' do you use for "rated output voltage", and how do you get the required supply line voltage from V_1?

Machine basics, DC machines, rotating fields

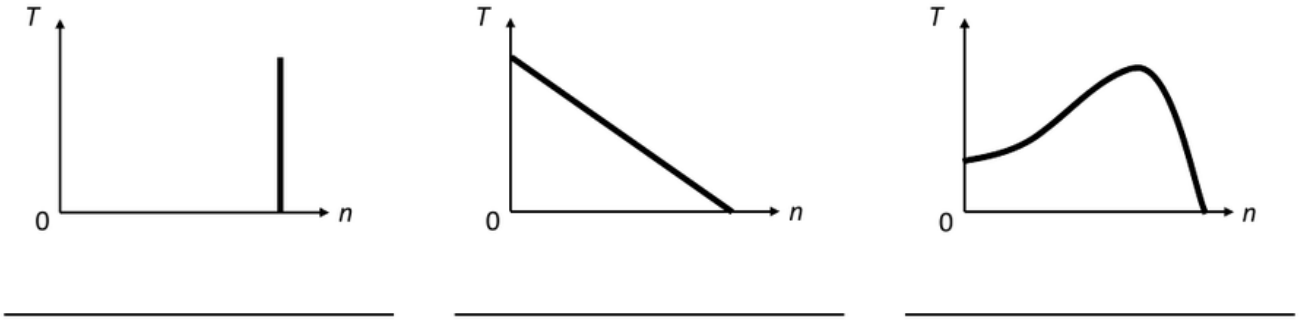
Practice questions · Day 07 · Fri 30 Oct · pack M6

5. Practice

From your course files

Jun24 Q3a (past exam, 6 marks). Torque vs. Speed Characteristics. Consider the torque versus speed curves shown in the figure below.

i. For each curve, write the type of machine in the space below the curve and draw the resulting torque curve if the no-load/synchronous speed was reduced to half of its original value. [3 marks]
ii. Alongside each curve, indicate how you could halve the no-load/synchronous speed while ensuring safe machine operation. [3 marks]



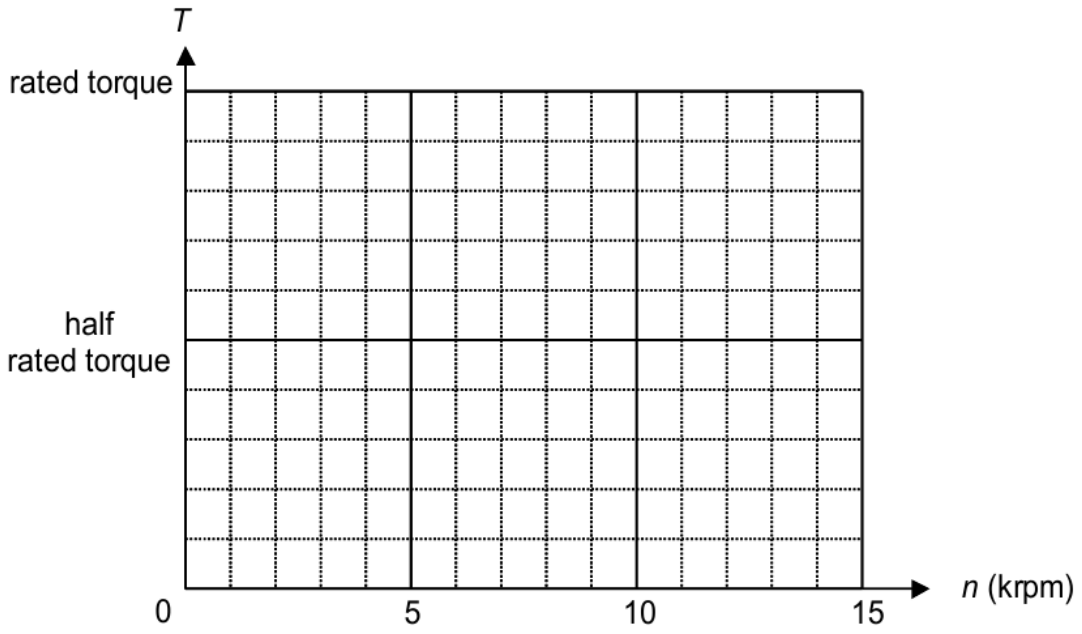
Jun24 Q3a figure (from the exam paper): three torque–speed axes (torque is labelled T on the paper) showing, left to right, a vertical line, a straight falling line, and the induction-motor curve

PQ 4.3 (practice questions). A 500W DC machine has a no-load speed of 12,000 rpm and a rated speed of 10,000 rpm when operated at rated voltage.

a. What is the rated output power of the DC machine?
b. Calculate the rated output torque.
c. Calculate the motor speed (in rpm) corresponding to half rated torque.
d. Calculate the motor output power with half rated torque. Why is this higher than half rated power (250W)?

PQ 4.4 (practice questions). Consider the DC machine described in the question above except this time it is operated with half rated voltage.

a. What is the new no-load speed?
b. What is the new full-load speed?
c. What is the speed corresponding to half rated output torque?
d. Using the template provided below: (i) Draw the torque versus speed curves corresponding to operation with rated and half rated voltage. (ii) Draw circles representing the full-load torque, half-load torque and no-load operating points for the both torque versus speed curves.



PQ 4.4 template (from the practice questions): torque axis marked rated torque and half rated torque (labelled T on the sheet), speed axis n from 0 to 15 krpm

PQ 4.7 (practice questions). A 0.75 kW induction machine is operating from a 110 V, 60 Hz AC power supply. Its synchronous speed is 1200 rpm and its rated speed is 1100 rpm.

a. Calculate the required supply voltage and frequency to give a synchronous speed of 900 rpm.
b. With the conditions in part (a), calculate the speed corresponding to: (i) rated torque (ii) half rated torque

PQ 4.11 (practice questions). You are required to select an AC machine to drive a mechanical load. The AC machine must operate from the 50 Hz AC supply. It is desirable that the machine have a synchronous speed as close as possible to 630 rpm as possible. Select an appropriate number of poles for the motor and calculate the exact synchronous speed.

PQ 4.13 (practice questions). A 11 kW, eight pole synchronous motor is operating from the 50 Hz mains.

a. What is its synchronous speed (in rpm)?
b. What is its rated speed (in rpm)?
c. What is its rated torque?
d. What is its speed at half rated torque?
e. What is its output power at half rated torque?

PQ 5.8 (practice questions). For the machine in the previous question, calculate

a. its no-load speed
b. its rated efficiency $= \omega/\omega_{NL} = n/n_{NL}$
c. the following when the speed is zero (stall condition): (i) induced voltage (ii) input current (iii) output torque (iv) input power (v) output power (vi) efficiency
d. the speed (in rpm) when the torque is 0.2 Nm

The previous question (PQ 5.7) reads: A DC machine has a back-emf constant of 0.24 V/(rad/s) and a resistance of 0.5 ohms. It is operating from a 12V DC power supply. At its rated speed of 400 rpm, calculate its (a) rated back-emf voltage (b) rated current (c) rated torque (d) rated output power $P_{OUT} = \tau\omega$ (e) rated output power $P_{OUT} = EI$ (f) rated input power (g) rated efficiency $= P_{OUT}/P_{IN}$ (h) rated efficiency $= E/V$.

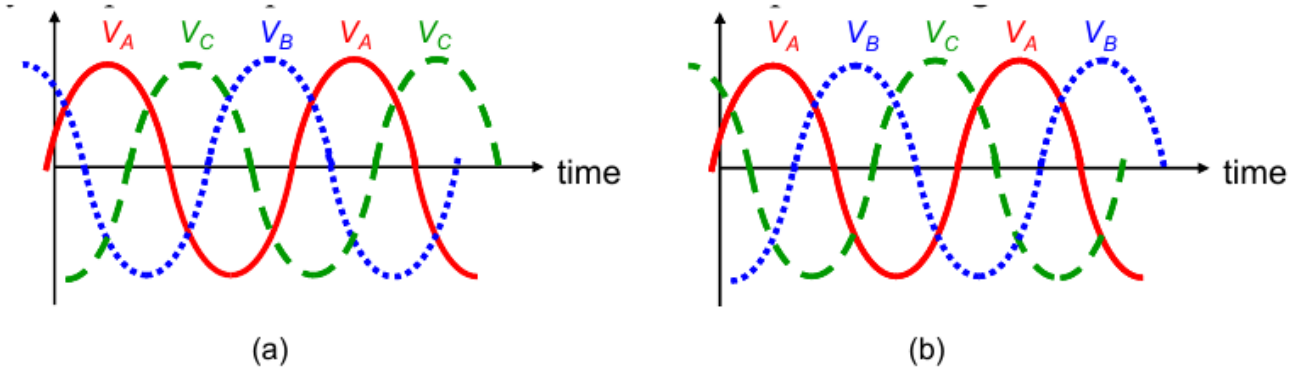
PQ 5.10 (practice questions). A DC machine is connected to a 24 V supply. When the machine is held at zero speed (stall), a current of 35 A flows and the machine produces a torque of 15 Nm. Find:

a. the no-load speed (in rpm)
b. the machine resistance
c. the machine output power at 400 rpm

PQ 5.19 (practice questions). A DC machine is driven by a small wind turbine. When the wind turbine is spinning at 150 rpm the measured open-circuit voltage is 10.5 V. The DC machine has an armature resistance of 0.4 ohms. The machine is then connected to a battery which acts like a constant voltage source of 14.6 V.

a. What is the back-emf constant of the machine?
b. At what turbine speed in rpm does the induced voltage just equal the battery voltage? What is the machine output current under this condition?
c. At what turbine speed in rpm, does the machine output current equals 10A?
d. For the condition in part (c), what is the efficiency of the machine?
e. In a practical application, a diode would be placed in series with the output of the wind generator. This diode allows power to flow from the generator into the battery but not vice-versa. Why is this diode required?

PQ 6.4 (practice questions). Identify the phase sequence for each of the three-phase voltage waveforms shown below:



PQ 6.4 figure (from the practice questions): two sets of three-phase voltage waveforms, (a) peaking in the order VA, VC, VB and (b) VA, VB, VC

PQ 6.5 (practice questions). You connect a three-phase AC induction machine to the mains supply. On turning it on you find that it rotates in the wrong direction. How do you reverse the direction of rotation of the machine?

PQ 6.7 (practice questions). Are the following a valid number of poles for an AC machine?

a. 2 poles (b) 15 poles (c) 4.5 poles (d) 16 poles

PQ 6.9 (practice questions). Synchronous generators are designed to feed power into the power system. In order to do so, they need to operate at the same frequency as the power grid.

a. In Australia, what is the highest speed a synchronous generator can operate at, and how many poles does this correspond to?
b. You are designing a power-station generator for a hydro-electric plant in New South Wales. It is designed to operate at 100 rpm. How many poles should this be designed

for?

PQ 5.11 (practice questions, extra). A DC machine has a no-load speed of 10,000 rpm and a rated speed of 8,000 rpm when operating with its rated supply voltage of 20 V. It has a resistance of 0.5 Ω.

a. Calculate the input current and output torque when the machine is at zero speed.
b. Calculate the input current and output torque when the machine is at rated speed.
c. Sketch the machine voltage-current and torque-speed characteristics on the template provided. Label the no-load, full-load and stall operating points on both graphs and indicate the corresponding induced voltage, current, torque and speed on the appropriate axes.
d. By observation from the graphs (without doing any calculations), what is the current and speed corresponding to operation at half rated torque?

PQ 5.11 template (from the practice questions): I against E axes and torque (labelled T on the sheet) against n axes

PQ 5.12 (practice questions, extra). Consider the machine in the previous question. On the axes provided sketch the torque versus speed characteristics when operating at rated supply voltage and half rated supply voltage. For both curves show the values of the stall current and torque, and the no-load induced voltage and speed.

PQ 5.12 template (from the practice questions): I against E axes and torque (labelled T on the sheet) against n axes

New questions

N1. [4 marks] A three-phase induction motor is rated at 1140 rpm on a 60 Hz supply.

a. Find the number of poles and the synchronous speed. [2]
b. Find the ideal (maximum) efficiency at rated speed. [1]
c. Assuming a linear τ–n curve, find the speed at 60 % of rated torque. [1]

Circuit for New question 2: PM DC motor with R = 0.25 Ω on 48 V, rated current 12 A

N2. [6 marks] A PM DC motor has a rated voltage of 48 V, an armature resistance of 0.25 Ω, a no-load speed of 3000 rpm at rated voltage, and a rated current of 12 A.

a. Find the back-emf constant k. [1]
b. Find the rated torque, rated speed (rpm) and rated efficiency. [3]
c. Find the stall current and stall torque, and express the stall torque as a multiple of rated torque. [1]
d. Find the maximum possible output power and the speed at which it occurs. Why is this not a sensible operating point? [1]

N3. [6 marks] (chains from N2) The motor in N2 drives a constant-torque load equal to its rated torque.

a. The supply is reduced to 24 V. Find the new no-load speed, the operating speed and the efficiency. [3]
b. What supply voltage gives an operating speed of 1000 rpm with this load? [2]
c. At 24 V, the load torque falls to half of rated. Find the speed. [1]

Circuit for New question 4: wind-driven DC machine (k = 0.5 V/(rad/s), R = 0.8 Ω) connected directly to a 24 V battery

N4. [6 marks] A DC machine with k = 0.5 V/(rad/s) and R = 0.8 Ω is driven by a small wind turbine and connected directly to a 24 V battery.

a. Find the turbine speed (rpm) at which the charging current just becomes zero. [1]
b. Find the speed needed to charge at 5 A. For this condition find the shaft torque, the mechanical input power, the electrical power into the battery and the efficiency. [3]
c. With no diode fitted, the wind drops and the turbine slows to 400 rpm. Find the current and its direction, and the torque. Explain what is happening. [2]

N5. [5 marks] A 4-pole, 415 V, 50 Hz induction motor has a rated speed of 1440 rpm. It is fed from a V/f inverter.

a. Find the inverter frequency and voltage so that the motor runs at 900 rpm at rated torque. [3]
b. With those settings, find the speed at half rated torque. [1]
c. A 6-pole synchronous motor with the same 415 V, 50 Hz rating must also run at 900 rpm. Find the frequency and voltage. [1]

N6. [4 marks] (words only)

a. A three-phase motor is connected with supply phases a, b, c to terminals A, B, C, and runs clockwise. An electrician reconnects it as b to A, c to B, a to C. Which way does it now run? Explain. [2]
b. A hydro generator for the Australian grid is designed to turn at 100 rpm. How many poles does it need? How many would it need for the same speed on a 60 Hz grid? [1]
c. In a 4-pole machine on 50 Hz, how many electrical cycles occur per mechanical revolution of the field? [1]

6. Quick recall check

1. What is the synchronous speed of an 8-pole machine on 60 Hz?
2. A DC motor has $n_{NL} = 3000$ rpm and $n_R = 2700$ rpm at 24 V. What are n_{NL} and n_R at 12 V?
3. At stall, what are E and I for a DC motor?
4. Why must V be reduced with f in an AC drive?
5. An induction motor runs at 1470 rpm on 50 Hz. What is its ideal maximum efficiency?

Induction machine fundamentals

Practice questions · Day 10 · Mon 2 Nov · pack M7

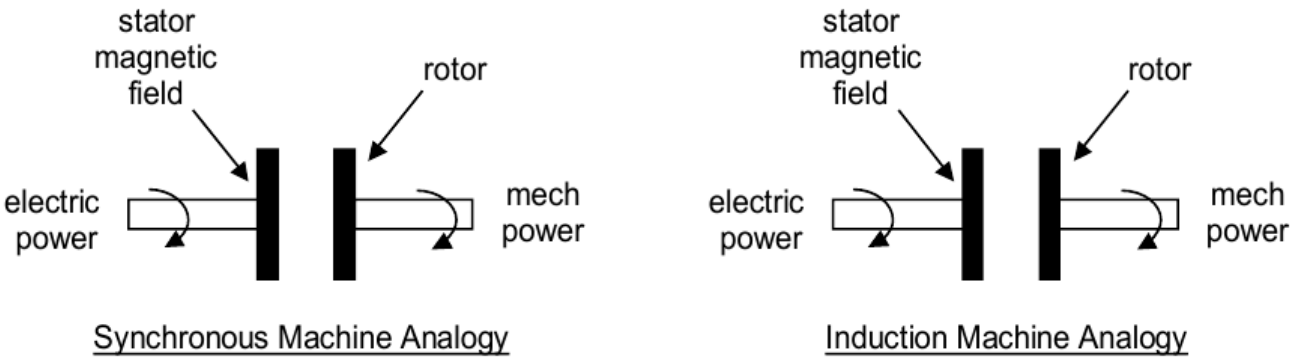
5. Practice

From your course files

PQ 7.1 (practice questions). Answer the following questions about induction machines.

a. The rotor winding of an induction machine consists of _____.
b. List three advantages of induction motors.
c. The most common type of induction motor rotor is called the _____ rotor. The rotor winding in this rotor normally consists of _____ bars.
d. To a first approximation the torque of an induction machine is directly proportional to _____.
e. An induction machine acts a generator when the rotor speed is _____ than the synchronous speed. It acts as a motor when the rotor speed is _____ than the synchronous speed.
f. The equivalent circuit of an induction machine is similar to that of a _____.
g. In an induction machine equivalent circuit, the power conversion between electrical and mechanical power is modelled by a _____.
h. What is the starting and pull-out torque?

PQ 7.2 (practice questions). There are two common mechanical analogies used to describe the operation of synchronous and induction machines. They are both based on modelling the stator magnetic field and the rotor as rotating disks where the stator magnetic field is rotating at synchronous speed. Using the templates below, illustrate these analogies. Label any additional elements added to the figures. No explanation of the analogies are required.



PQ 7.2 templates (from the practice questions): stator-field disk and rotor disk on a shaft, electric power in on the left and mechanical power out on the right, for the synchronous and the induction machine

PQ 7.3 (practice questions). Calculate the synchronous speed (in rpm) for the following AC machines:

a. 4 pole, 50 Hz (b) 8 pole, 60 Hz (c) 2 pole, 50 Hz (d) 4 pole, 800 Hz (e) 24 pole, 50 Hz

PQ 7.4 (practice questions). Calculate the most likely number of poles and supply frequency for motors with the following synchronous speeds.

a. 1,500 rpm (b) 1800 rpm (c) 750 rpm (d) 1,200 rpm (e) 360 rpm (f) 150 rpm

PQ 7.5 (practice questions). Calculate the number of poles, the synchronous speed and the rated slip speed in rpm for three-phase induction motors which have nameplate data as follows:

a. 415 V, 50 Hz, 1440 rpm (b) 3.3 kV, 60 Hz, 3520 rpm (c) 135 V, 50 Hz, 680 rpm (d) 230 V, 50 Hz, 950 rpm

PQ 7.8 (practice questions). Calculate the rated slip and rated slip frequency for the four motors in the question three questions prior to this one.

PQ 7.6 (practice questions). Consider a 7.5 kW, 2910 rpm three-phase 50Hz induction machine operating at rated voltage. Refer to Section 7.2 of your notes if you have difficulty with this.

a. Calculate its synchronous speed in rpm and hence the rated slip speed in rpm.
b. The machine is running at no-load, calculate its operating speed in rpm.
c. At what speed does rated output torque occur at? Calculate the rated output torque.
d. Calculate the rated operating speed as a percentage of the no load operating speed. Is the description of induction machines as almost fixed speed machines valid?
e. Calculate the speed (in rpm) at which the motor operates with half rated output torque.
f. Calculate the output power corresponding to part (e). Express this as a percentage of rated output power. Why is this not exactly half the rated output power?
g. Calculate the operating speed corresponding to a load of 20 Nm.
h. The machine is driven at 3070 rpm. Is the machine acting as a motor or generator? Calculate the output torque.

PQ 7.7 (practice questions). Consider a large induction machine rated at 1470 rpm, 50 Hz operating at rated voltage. Refer to Section 7.2 of your notes if you have difficulty with this.

a. If the machine is delivering half rated output torque, calculate the operating speed in rpm.
b. The machine is driven as a generator with an input torque equal to rated torque. Calculate its operating speed.

PQ 7.9 (practice questions). A 50 Hz induction machine has a rated speed of 960 rpm and a rotor resistance $R_2' = 1$ ohm. For each of the following speeds, calculate the slip, the frequency of the rotor currents and the value of R_2'/s in the equivalent circuit.

a. locked-rotor (that is, at standstill) (b) no-load (c) rated speed (d) operation as a generator at 1050 rpm

PQ 7.10 (practice questions). Answer the following questions about induction machines.

a. An induction machine acts a generator when its rotor speed is _____ than synchronous speed.
b. What is the slip speed?
c. What are the main differences between the equivalent circuit of an induction machine and a transformer?
d. A 50 Hz induction motor is operating at a speed which is 90% of synchronous speed. What is the rotor frequency and slip?
e. The power crossing the airgap between the stator and rotor is called the _____.
f. This power crossing the airgap is less than the power entering the stator because of what losses?
g. The power crossing the airgap is less than the mechanical output power due to what loss?
h. Why are the rotor iron losses often neglected?
i. Briefly explain how the stator resistance R_1 and rotor resistance referred to the primary R_2' can be obtained?
j. What is the relationship between torque and slip at (i) low values of slip? (ii) high values of slip?
k. As far as the rotor resistance R_2' is concerned, what is the difference between a squirrel-cage rotor and a wound-rotor?
l. Changing the rotor resistance, changes the _____ at which maximum torque occurs but does not change its _____.
m. Constant torque loads require motors with high _____ torque.
n. An induction motor is operating at 85% of synchronous speed. What is its ideal efficiency?
o. Why is the power-factor of induction motors low under no-load conditions?

p. The ratio of pullout torque to rated torque for an induction motor is typically about _____.

q. The effective rotor resistance R_2'/s of an induction machine can be thought of as consisting of two separate resistances. Explain the physical significance of these two resistances.

PQ 7.11 (practice questions). A 2.2 kW three-phase induction machine has the following ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

a. Calculate the synchronous speed in rpm
b. Calculate the rated output torque in Nm
c. Calculate the rated input power to the machine: i) Using the voltage, current and power-factor ii) Using the output and efficiency
d. Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

PQ 4.7 (practice questions, only if you skipped M6). A 0.75 kW induction machine is operating from a 110 V, 60 Hz AC power supply. Its synchronous speed is 1200 rpm and its rated speed is 1100 rpm.

a. Calculate the required supply voltage and frequency to give a synchronous speed of 900 rpm.
b. With the conditions in part (a), calculate the speed corresponding to: (i) rated torque (ii) half rated torque

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New questions

N1. [3 marks] A 6-pole, 60 Hz induction motor runs at 1164 rpm. Its referred rotor resistance is $R_2' = 0.45\,\Omega$. Find the slip, the frequency of the rotor currents, and the two parts of R_2'/s.

N2. [3 marks] A 6-pole, 50 Hz motor runs at 970 rpm. Its rotor copper loss is 400 W. Find the airgap power, the mechanical power and the torque.

N3. [5 marks] A 400 V, 50 Hz, 4-pole, **delta-connected** induction motor runs at 1455 rpm. Per phase, $R_1 = 1.2\,\Omega$, $R_2' = 1.0\,\Omega$ and $X_1 + X_2' = 4\,\Omega$. Neglect the magnetising branch.

a. Find the phase current and the line current. [2]
b. Find the airgap power, the torque and the efficiency. [3]

Circuit for N3: delta phase voltage across R1 = 1.2 ohm, j4 ohm and R2′/s, magnetising branch neglected

N4. [10 marks] A 415 V, 50 Hz, 1440 rpm, star-connected, three-phase induction motor has (referred to the stator) $R_1 = 0.8\,\Omega$, $R_2' = 0.6\,\Omega$, $X_1 = X_2' = 1.5\,\Omega$, $X_M = 40\,\Omega$ and $R_C = 500\,\Omega$. It runs at rated voltage and speed.

a. Find the slip and draw the simplified per-phase equivalent circuit with values. [2]
b. Find the rotor current I_2', the input current I_1 (magnitude and phase) and the power factor. [3]
c. Find the input power, the airgap power and the output torque. [3]
d. Find the output power and efficiency, and compare it with the ideal efficiency. Account for the difference with a loss breakdown. [2]

N5. [5 marks] For the motor in N4, the supply voltage sags by 10 % (to 373.5 V line). The load torque stays at 63.59 Nm.

a. What torque would the motor produce at the original slip of 0.04? [1]
b. Estimate the new operating speed using the low-slip approximation $\tau \propto V_1^2 s$. [2]
c. Find the exact new slip from the torque formula, and the new rotor copper loss. Comment. [2]

6. Quick recall check

1. A 4-pole, 50 Hz motor runs at 1470 rpm. What is the rotor frequency?
2. Write R_2'/s as two resistors and state what the power in each one represents.
3. $P_{AG} = 10$ kW at $s = 0.05$. What are the rotor copper loss and the mechanical power?
4. In $\tau = P_{AG}/\omega$, which ω do you use: ω_S or the rotor speed? And for P_M?
5. A 690 V delta-connected motor: what is V_1 in the per-phase circuit?

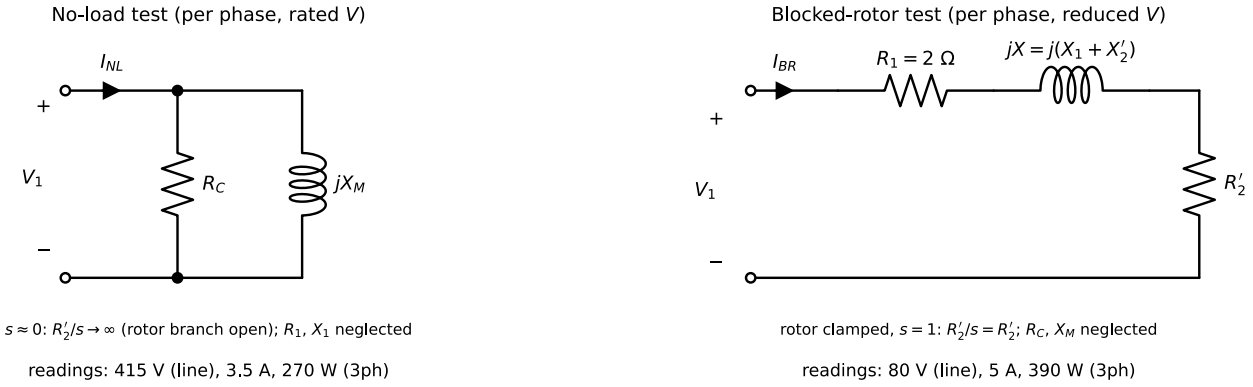
Machine tests, maximum torque, generator mode

Practice questions · Day 11 · Tue 3 Nov · pack M8

5. Practice

From your course files

Jun24 Q3b (past exam, 4 marks). Induction Machine Parameters. The following tests are performed on a 415V, 50Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 2 ohms per phase
No-load test: input voltage 415V, input current 3.5A, input power 270W
Blocked-rotor test: input voltage 80V, input current 5A, input power 390W
Find the following simplified equivalent circuit series parameters: R_1, R_2' and X.



Jun24 Q3b test circuits: no-load 415 V, 3.5 A, 270 W; blocked rotor 80 V, 5 A, 390 W; R1 = 2 ohm

Jun23 Q3b (past exam, 5 marks). Induction Machine Torque vs Speed. The equivalent-circuit parameters of a wound-rotor induction machine are listed below.
stator resistance $R_1 = 0.4\ \Omega$, rotor resistance $R_2' = 0.5\ \Omega$
stator leakage reactance $X_1 = 1.3\ \Omega$, rotor leakage reactance $X_2' = 1.5\ \Omega$
magnetising reactance $X_M = 95\ \Omega$, core loss resistance $R_C = 270\ \Omega$

i. Calculate the external resistance, R_{EXT}, required such that maximum torque occurs at standstill. [2 marks]
ii. On the axes below, draw a typical torque vs. speed characteristic curves for an induction machine – assume that this curve represents the above induction machine with zero

external resistance. In addition, draw the torque curves for external resistance equal to 0.5 and 1pu, where 1pu represents the R_{EXT} calculated in part i) above. [3 marks]



Jun23 Q3b(ii) answer axes (from the exam paper): torque (labelled T on the paper) against speed n

PQ 7.11 (practice questions). A 2.2 kW three-phase induction machine has the following ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

a. Calculate the synchronous speed in rpm
b. Calculate the rated output torque in Nm
c. Calculate the rated input power to the machine: i) Using the voltage, current and power-factor ii) Using the output and efficiency
d. Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

PQ 7.12 (practice questions). The following tests are performed on a 415 V, 1455rpm, 50 Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 3 ohms per phase

No-load test: input voltage 415 V, input current 3.0 A, input power 205 W
Blocked-rotor test: input voltage 65 V, input current 4.8 A, input power 350 W

a. Find the simplified equivalent circuit parameters.
b. Draw the simplified equivalent circuit showing the calculated parameter values.
c. The motor is running at rated speed. Calculate the magnitude of the input current, the power-factor, the input power, output torque and output power, and hence the efficiency of the motor.
d. Calculate the rated copper losses, iron loss and total loss. Using the output power calculated in part (c), hence calculate the efficiency of the motor.

PQ 7.13 (practice questions). The following tests are performed on a 415V, 1410rpm, 50Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 1 ohm per phase
No-load test: input voltage 415V, input current 5.4A, input power 300W
Blocked-rotor test: input voltage 90V, input current 8A, input power 270W
Find the simplified equivalent circuit parameters.

PQ 7.15 (practice questions). A three-phase, 460 V, 60 Hz, 1760 rpm, wound-rotor induction motor has the following equivalent circuit parameters
$R_1 = 0.25$ ohm : $R_2' = 0.2$ ohm : $X_1 = 1.2$ ohm : $X_2' = 1.1$ ohm : $X_M = 35$ ohm
The motor is connected to a 3ph, 460V, 60Hz supply and is star-connected. Use the simplified equivalent circuit. Initially the external rotor resistance is set to zero.

a. Calculate the speed (in rpm) at which maximum torque occurs.
b. Using the torque equation, calculate the output torque at the following speeds (i) synchronous, (ii) rated speed, (iii) speed corresponding to maximum torque, (iv) locked rotor (0 rpm)
c. Using the calculated torques in part (b), sketch out the torque versus speed characteristic.
d. The total effective rotor resistance R_2' is adjusted by adding external rotor resistance. What is the total effective rotor resistance required to give maximum torque at standstill? What is the starting torque under this condition?

Jun24 Q3c (past exam, 10 marks). Induction Machine Calculation. A 415 V, 50 Hz, 960 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):
stator resistance $R_1 = 2.5$ ohms, rotor resistance $R_2' = 3$ ohms
stator leakage reactance $X_1 = 6$ ohms, rotor leakage reactance $X_2' = 6$ ohms
The magnetising reactance and core loss resistance can be neglected.

i. The above induction machine is run as a generator at 1025 rpm. Find the synchronous speed, the value of slip s and hence the effective rotor resistance R_2'/s. [3 marks]
ii. Draw a single-phase simplified equivalent circuit for the machine when operating at 1025 rpm, showing the parameter values as impedances. Assuming that the machine is operating at rated voltage, find the magnitude and phase of the input phase current. Hence find the input power. You should assume that the phase angle of the phase voltage is zero. [5 marks]
iii. Using the results from part (ii) find the corresponding airgap power P_{AG} using the fact that in this case $I_2' = I_1$, and hence find the output torque τ. [2 marks]

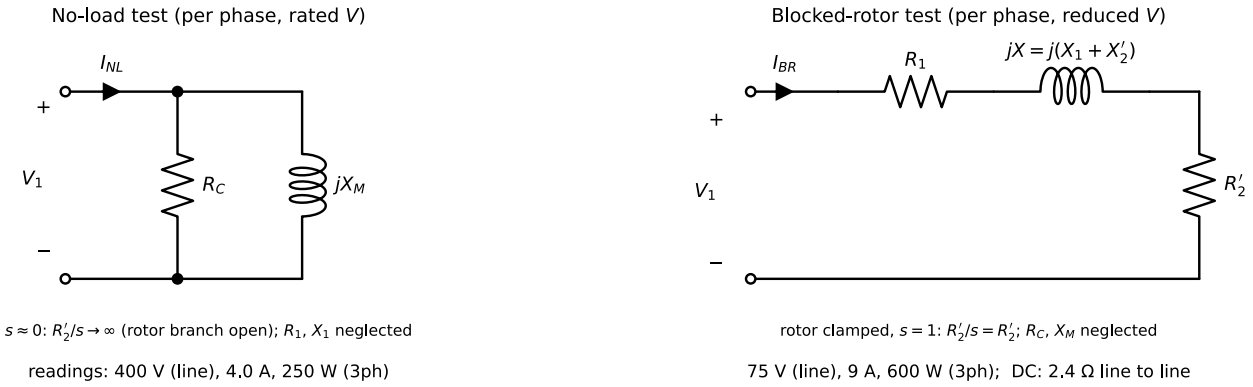
PQ 7.14 (practice questions, if time: parts (d)–(f)). A 415 V, 50 Hz, 2790 rpm, 3 phase, star-connected induction motor has the following equivalent circuit parameters (all referred to the stator winding): stator leakage reactance $X_1 = 0.3$ ohm, stator resistance $R_1 = 0.2$ ohms, rotor leakage reactance $X_2' = 0.3$ ohm, rotor resistance $R_2' = 0.16$ ohm and magnetising reactance $X_M = 12$ ohm. You can ignore the core-loss resistance. Use the simplified equivalent circuit for the following calculations. The motor is operated at rated voltage.

a. A locked-rotor test (slip = 1) is performed at rated voltage. Ignoring the magnetising reactance for this part only, calculate the input current magnitude, input power and output torque
b. At rated speed find the input current magnitude, input power factor and input power (do not neglect the magnetizing reactance!).
c. At rated speed find the rated output torque and output power and hence the efficiency.
d. What are the starting torque and starting current as a fraction of the rated torque and rated current respectively?
e. Determine the maximum torque available and the speed at which it occurs. What is the maximum torque as a fraction of rated torque?

f. Calculate the magnitude of the input current at no-load. What is this as a fraction of rated current?
g. Using the calculated torque at starting, maximum torque and rated torque, and your knowledge of the general shape of the induction motor torque versus speed characteristics, draw a rough sketch showing the output torque as a fraction of rated torque as a function of speed in rpm.

New questions

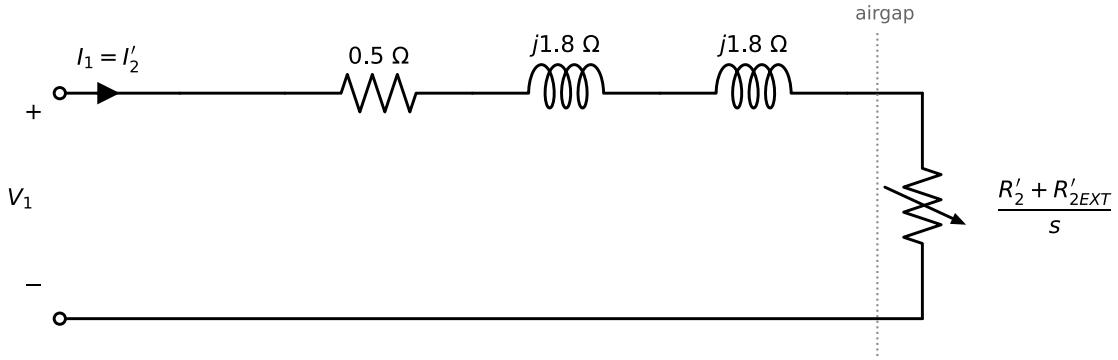
New Q1 (easy) [5 marks]. A 400 V, 50 Hz, star-connected induction motor is tested. DC test: 2.4 Ω measured between two lines. No load: 400 V, 4.0 A, 250 W. Blocked rotor: 75 V, 9 A, 600 W. Find R_1, R_2', X_1, X_2', R_C and X_M.



New Q1 test circuits: no-load 400 V, 4.0 A, 250 W; blocked rotor 75 V, 9 A, 600 W

New Q2 (easy–medium) [6 marks]. A 415 V, 50 Hz, 6-pole, star-connected wound-rotor induction motor has $R_1 = 0.5\ \Omega$, $R_2' = 0.6\ \Omega$ and $X_1 = X_2' = 1.8\ \Omega$ (simplified circuit).

a. Find the speed at which maximum torque occurs, and the maximum torque. [3]
b. Find the starting torque and starting current (neglect the magnetising branch). [1]
c. Find the external rotor resistance (referred) that gives maximum torque at standstill, and the new starting torque and current. [2]



415 V (line), star, 6-pole; $R_2' = 0.6$ Ω; $R_{2EXT}' = 0$ except in (c)

New Q2 circuit: 0.5 ohm, j1.8 ohm, j1.8 ohm and (R2′ + R2EXT′)/s

New Q3 (medium) [5 marks]. The New Q2 motor (no external resistance) is fed from an inverter with constant V/f.

a. What line voltage is needed at 25 Hz? [1]
b. Find the speed at maximum torque and the maximum torque at 25 Hz. Compare with 50 Hz and explain the difference. [3]
c. At 50 Hz the motor drives a 60 Nm load at 976.8 rpm. At 25 Hz it drives the same load at 475.9 rpm. What does this show about the slip speed? [1]

New Q4 (medium) [6 marks]. The New Q2 motor (no external resistance) is driven by a wind turbine at 1040 rpm, connected to the 415 V, 50 Hz grid. Neglect the magnetising branch.

a. Motor or generator? Find s and R_2'/s. [2]
b. Find the phase current (magnitude and angle), the real power and the reactive power at the terminals. [2]
c. Find the airgap power, the torque and the mechanical power the turbine must supply. [2]

New Q5 (hard, chained) [10 marks]. Use the New Q1 parameters. The motor is 4-pole and rated at 1440 rpm. Use the simplified equivalent circuit at 400 V.

a. At rated speed find I_1 (magnitude and phase), the power factor, the input power, the output torque, the output power and the efficiency. [5]
b. Find the maximum torque, the speed at which it occurs, and τ_{max}/τ_R. Find also τ_{start}/τ_R. [3]
c. The shaft is now driven at 1530 rpm. Find the torque and say what it means. [1]
d. Find the no-load current as a fraction of the rated current. Is this typical? [1]

New Q6 (short answer, words only) [6 marks].

a. Carefully explain why an induction motor draws a significant current at no load, and why its power factor is low there. [3]
b. A squirrel-cage rotor is made with zero resistance. How would this change the machine's behaviour? [3]

6. Quick recall check

1. In the blocked-rotor test, which branch is neglected and why? Which quantity do you need from another test?
2. Write $s_{\tau max}$ in terms of the circuit parameters. What happens to τ_{max} if R_2' doubles?
3. For a wound rotor, what is R_{2EXT}' for maximum torque at standstill?
4. A 4-pole, 50 Hz machine runs at 1560 rpm. Give the sign of s, R_2'/s, τ, P_{IN} and Q_{IN}.
5. Under V/f control at half frequency, which three quantities in the torque formula change, and by what factor?

Exam-style induction problems

Practice questions · Day 12 · Wed 4 Nov · pack M9

5. Practice

From your course files

PQ 7.11 (practice questions). A 2.2 kW three-phase induction machine has the following ratings : 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

(a) Calculate the synchronous speed in rpm
(b) Calculate the rated output torque in Nm
(c) Calculate the rated input power to the machine
  i) Using the voltage, current and power-factor
  ii) Using the output and efficiency
(d) Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

PQ 7.16 (practice questions). Constant volts-per-Hertz control is applied to an induction motor which is rated at 460 V and 60 Hz in order to keep the flux in the motor at its rated value. Calculate the required stator voltage corresponding to a stator frequencies of 30 Hz and 50 Hz.

PQ 7.12 (practice questions; do (c)–(d) here). The following tests are performed on a 415 V, 1455rpm, 50 Hz three-phase induction motor which is star-connected.

Stator resistance measurement : 3 ohms per phase
No-load test : input voltage 415 V, input current 3.0 A, input power 205 W
Blocked-rotor test : input voltage 65 V, input current 4.8 A, input power 350 W

(a) Find the simplified equivalent circuit parameters.
(b) Draw the simplified equivalent circuit showing the calculated parameter values.
(c) The motor is running at rated speed. Calculate the magnitude of the input current, the power-factor, the input power, output torque and output power, and hence the efficiency of the

motor.
(d) Calculate the rated copper losses, iron loss and total loss. Using the output power calculated in part (c), hence calculate the efficiency of the motor.

PQ 7.14 (practice questions; do (a)–(d) and (f) here). A 415 V, 50 Hz, 2790 rpm, 3 phase, star-connected induction motor has the following equivalent circuit parameters (all referred to the stator winding) : stator leakage reactance X_1 = 0.3 ohm, stator resistance R_1 = 0.2 ohms, rotor leakage reactance X_2' = 0.3 ohm, rotor resistance R_2' = 0.16 ohm and magnetising reactance X_M = 12 ohm. You can ignore the core-loss resistance. Use the simplified equivalent circuit for the following calculations. The motor is operated at rated voltage.

(a) A locked-rotor test (slip = 1) is performed at rated voltage. Ignoring the magnetising reactance for this part only, calculate the input current magnitude, input power and output torque
(b) At rated speed find the input current magnitude, input power factor and input power (do not neglect the magnetising reactance!).
(c) At rated speed find the rated output torque and output power and hence the efficiency.
(d) What are the starting torque and starting current as a fraction of the rated torque and rated current respectively?
(e) Determine the maximum torque available and the speed at which it occurs. What is the maximum torque as a fraction of rated torque?
(f) Calculate the magnitude of the input current at no-load. What is this as a fraction of rated current?
(g) Using the calculated torque at starting, maximum torque and rated torque, and your knowledge of the general shape of the induction motor torque versus speed characteristics, draw a rough sketch showing the output torque as a fraction of rated torque as a function of speed in rpm.

PQ 7.15 (practice questions; do (b)(i), (ii), (iv) here). A three-phase, 460 V, 60 Hz, 1760 rpm, wound-rotor induction motor has the following equivalent circuit parameters

R_1 = 0.25 ohm : R_2' = 0.2 ohm : X_1 = 1.2 ohm : X_2' = 1.1 ohm : X_M = 35 ohm

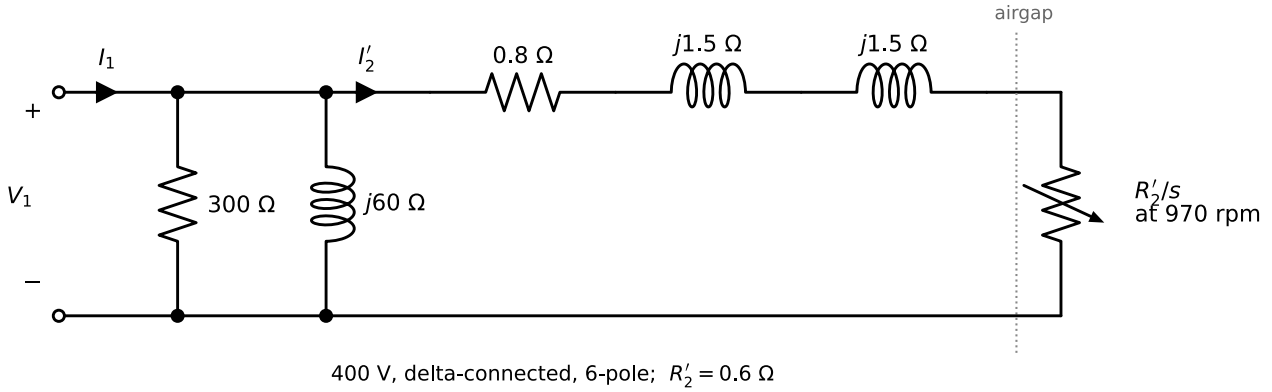
The motor is connected to a 3ph, 460V, 60Hz supply and is star-connected. Use the simplified equivalent circuit. Initially the external rotor resistance is set to zero.

(a) Calculate the speed (in rpm) at which maximum torque occurs.
(b) Using the torque equation, calculate the output torque at the following speeds (i) synchronous, (ii) rated speed, (iii) speed corresponding to maximum torque, (iv) locked rotor (0 rpm)
(c) Using the calculated torques in part (b), sketch out the torque versus speed characteristic.
(d) The total effective rotor resistance R_2' is adjusted by adding external rotor resistance. What is the total effective rotor resistance required to give maximum torque at standstill? What is the starting torque under this condition?

New questions

New Q1 (15 marks, easy–medium). A 400 V, 50 Hz, 6-pole, **delta-connected** induction motor runs at 970 rpm. Simplified-circuit parameters per phase: $R_1 = 0.8\ \Omega$, $R_2' = 0.6\ \Omega$, $X_1 = X_2' = 1.5\ \Omega$, $X_M = 60\ \Omega$, $R_C = 300\ \Omega$.

a. Find n_S, the slip, the rotor current frequency and R_2'/s. [3]
b. Find the phase current I_1 (magnitude and phase), the line current and the power factor. [5]
c. Find P_{IN}, P_{AG}, the output torque, the output power and the efficiency. [5]
d. Find the stator copper, rotor copper and iron losses, and show that they account for the difference between P_{IN} and P_{OUT}. [2]



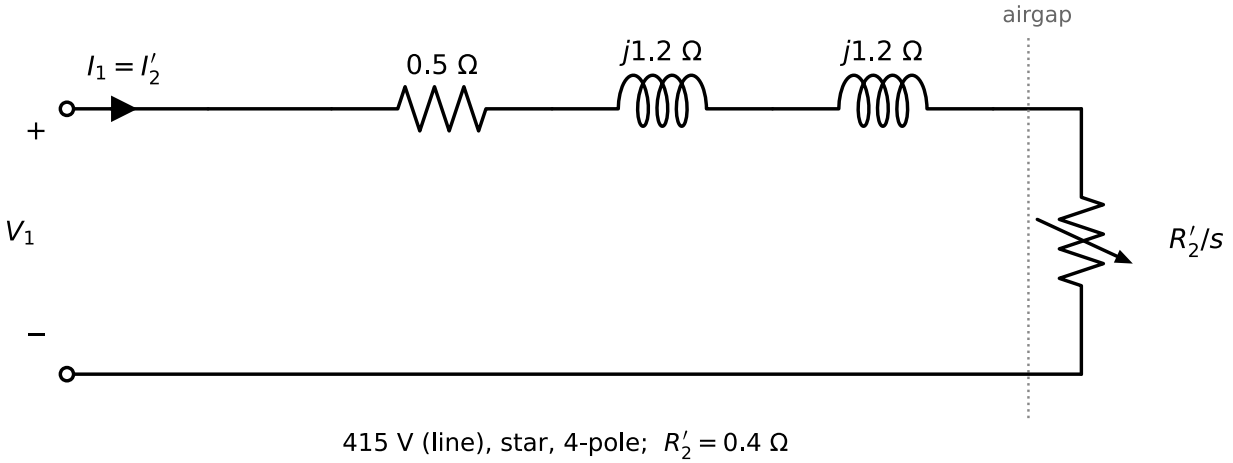
New Q1 circuit: 300 ohm parallel j60 ohm, then 0.8 ohm, j1.5 ohm, j1.5 ohm and R2′/s, delta phase voltage

New Q2 (17 marks, medium). A 415 V, 50 Hz, 4-pole, star-connected induction machine has $R_1 = 0.5\ \Omega$, $R_2' = 0.4\ \Omega$, $X_1 = X_2' = 1.5\ \Omega$, $X_M = 40\ \Omega$. Neglect R_C. It is driven by a wind turbine at 1530 rpm. Use the simplified equivalent circuit.

a. Is the machine motoring or generating? Find s and R_2'/s. [3]
b. Find I_2', I_1 (magnitude and phase), the real power and the reactive power drawn from the supply. Comment on the signs. [6]
c. Find P_{AG}, the shaft torque, the mechanical input power and the generator efficiency. [5]
d. Explain briefly why an induction generator connected to the grid cannot supply its own magnetising current, and what this means for the grid. [3]

New Q3 (18 marks, medium–hard). A 415 V, 50 Hz, 4-pole, star-connected induction motor has $R_1 = 0.5\ \Omega$, $R_2' = 0.4\ \Omega$, $X_1 = X_2' = 1.2\ \Omega$. Neglect X_M and R_C.

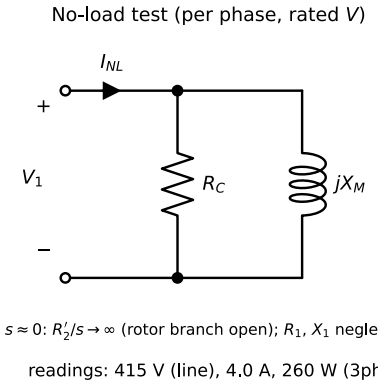
a. Find the torque and the input current magnitude at 1470 rpm. [3]
b. The motor drives a constant-torque load of 40 Nm. Find the exact operating slip and speed. Explain which root you choose. [6]
c. Estimate the speed in (b) using the linear torque–speed approximation and the result of (a). Comment on the accuracy. [3]
d. The supply voltage falls by 10% with the same 40 Nm load. Find the new speed (exact), and compare with the estimate $s \propto 1/V^2$. [4]
e. Find the starting torque. Can the motor start the 40 Nm load? Find the ratio of starting current to the running current in (b). [2]

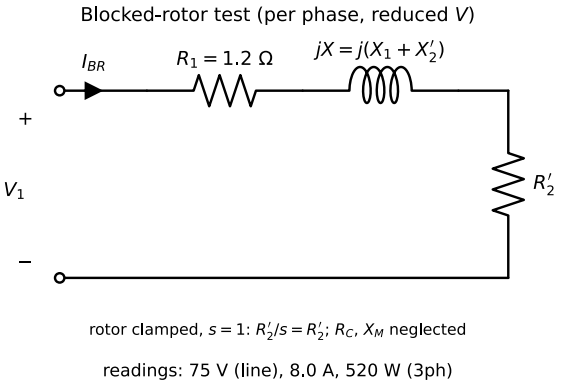


New Q3 circuit: 0.5 ohm, j1.2 ohm, j1.2 ohm and R2′/s, magnetising branch neglected

New Q4 (20 marks, hard: a full Q3-style chain). A 415 V, 50 Hz, 1440 rpm, star-connected induction motor gives these test results:

- DC stator resistance: 1.2 Ω per phase
- No-load test: 415 V, 4.0 A, 260 W
- Blocked-rotor test: 75 V, 8.0 A, 520 W





New Q4 test circuits: no-load 415 V, 4.0 A, 260 W; blocked rotor 75 V, 8.0 A, 520 W; R1 = 1.2 ohm

a. Find R_2', X, R_C and X_M for the simplified equivalent circuit. [4]
b. Find the number of poles, n_S and the rated slip. State the ideal efficiency at rated speed. [3]
c. At rated voltage and speed, find I_1 (magnitude and phase), pf, P_{IN}, τ, P_{OUT} and η. [7]

d. The same machine is driven at 1560 rpm on the same supply. Find the torque that must be applied, the electrical power delivered to the supply, and the generator efficiency. [4]
e. Explain why the actual efficiency in (c) is lower than the ideal efficiency in (b). [2]

6. Quick recall check

1. A 415 V star machine. What is V_1 in the per-phase circuit, and what is it for delta?
2. Which speed do you divide P_{AG} by to get torque, and which speed do you multiply τ by to get P_{OUT}?
3. In generator mode, which of s, R_2'/s, P_{IN}, Q_{IN}, τ are negative?
4. Write the rotor copper loss in terms of P_{AG}, and the mechanical power in terms of P_{AG}.
5. A quadratic for R_2'/s gives two roots. Which one is the normal running point?

Per-unit and synchronous reactance

Practice questions · Day 13 · Thu 5 Nov · pack M10

5. Practice

From your course files

PQ 8.1 (practice questions). For a single-phase system calculate the following assuming a base voltage of 240V and base power of 2000VA.

(a) base current
(b) base impedance
(c) base resistance
(d) base reactance

PQ 8.2 (practice questions). Using the results from the previous question, solve Question 2.17 using the per-unit method, for parts (a) -(c) and (e) give the answers in pu, for parts (f)-(h) convert the results back to real units

Question 2.17 (for reference): A single-phase load consists of a parallel-connected 25 ohm resistor and an inductor with a 125 ohm reactance is connected to the 240 V mains, calculate
(a) the load impedance Z in polar form
(b) write the supply voltage $\mathbf{V}_1$ in phasor form (assume it has a phase angle of zero)
(c) the resultant phase current phasor $\mathbf{I}_1$
(d) the input power factor (remember to show lead or lag)
(e) the complex power $\mathbf{S}_1$ in both polar and Cartesian form
(f) the input real power P using
  i) the real part of $\mathbf{S}_1$
  ii) the input power factor and V_1 and I_1
  iii) the fact that all the power losses occur in the resistor and hence the losses are V^2/R (why not I^2R?)
(g) the input reactive power Q
(h) the input VA (that is, the apparent power S)
(i) sketch a power triangle and verify that $P^2 + Q^2 = S^2$
(j) sketch a phasor diagram showing the relationship between the voltage and the current

PQ 8.3 (practice questions). Repeat the calculations in Question 2.20 Part (a) only to find the magnitude of the two currents using the per-unit system. Use a base VA of 1000VA and base voltage of 200V.

Question 2.20 (for reference): Two single-phase loads $Z_1 = 100 + j0\ \Omega$ and $Z_2 = 10 + j20\ \Omega$ are connected in parallel across a 200 V, 60 Hz source.
(a) Find the magnitude of the two load currents, the total real and reactive power supplied by the source, and the power-factor at the source.

PQ 8.4 (practice questions). A calculation is to be carried out using the per-unit system to a base of 10 MVA. It relates to a section of a (three-phase) power supply system whose nominal line voltage is 11 kV. Calculate the following quantities:

(a) The base line voltage
(b) The base phase voltage
(c) The base phase current
(d) The base phase impedance
(e) The base phase reactance
(f) The base phase resistance
(g) The base real power
(h) The base reactive power
(i) The base VA

PQ 8.5 (practice questions). Express the following as impedances in a three-phase system in per unit form to a base of 750 kVA :

(a) 5 ohm inductive reactance on a 3ph 3300 volt distribution line.
(b) 0.05 ohm resistance on a 3ph 415V supply

PQ 8.6 (practice questions). A three-phase synchronous generator has a rated output voltage of 415 V. Under a particular operating condition it can be represented by an internal voltage source of (line) voltage 440 V and series phase reactance of 7.5 ohms. Carry out the following calculations using a base of 10 kVA.

(a) Calculate the base current and base impedance.
(b) Draw the equivalent circuit for the machine using per-unit values for the internal voltage and impedance.
(c) Calculate the magnitude of the short-circuit output current in per-unit. Then convert this value to amps.
(d) Check this answer by performing the calculations using conventional units.

PQ 8.7 (practice questions). Calculate the magnitude and phase of the per-unit stator voltage $\mathbf{V}$ and current $\mathbf{I}$ for the following situations. You can assume the phase of the voltage is taken as the reference. Use the generator rating as the power base. [Hint : use the relations $I = S/V$ pu and the power triangle relating S, P and Q, and also that the angle of $\mathbf{I}$ equals the power factor angle $= \cos^{-1}(\mathrm{pf})$. Watch the sign of the current angle depending on whether the power-factor is leading or lagging.]

(a) A synchronous generator is operating at full load (rated VA, i.e. S=1pu) at a power-factor of 0.9 lagging. The terminal voltage is at its rated value.
(b) A synchronous generator is operating at half load (that is, half rated output VA, S=0.5pu) at a power-factor of 0.9 leading. The terminal voltage is 5% above its rated value.
(c) A 150 MVA generator is producing 100 MW at a power-factor of 0.85 lagging. The terminal voltage is at its rated value.
(d) A 200 MVA generator is producing 150 MW and 50 MVAr lagging. The terminal voltage is 5% below its rated value

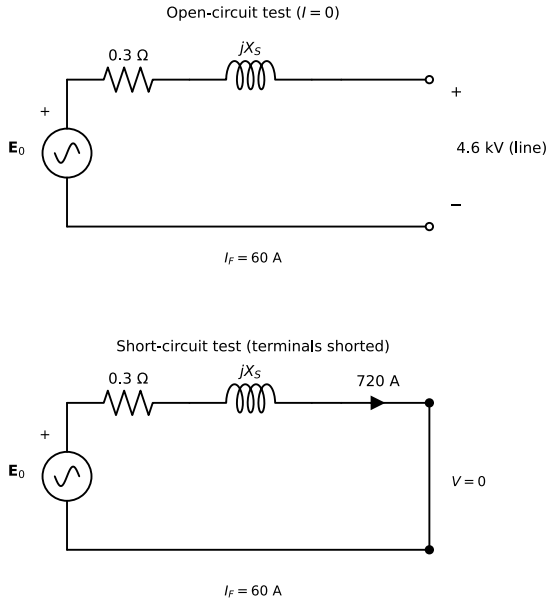
Jun23 Q4a (past exam, 5 marks). *4a) Synchronous Machines [5 marks]* A 150 MVA, 50Hz, 3,000rpm, 14 kV power station generator gives the following test results:

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
60	4.6	720

The stator resistance is measured at 0.3 ohm.

i) Find the value of synchronous reactance in ohms (not per-unit). [2 marks]

ii) The generator is delivering a real power of 100MW and reactive power of 50 MVAr at rated voltage. Calculate the magnitude and phase of the stator voltage and current in per-unit. [3 marks]

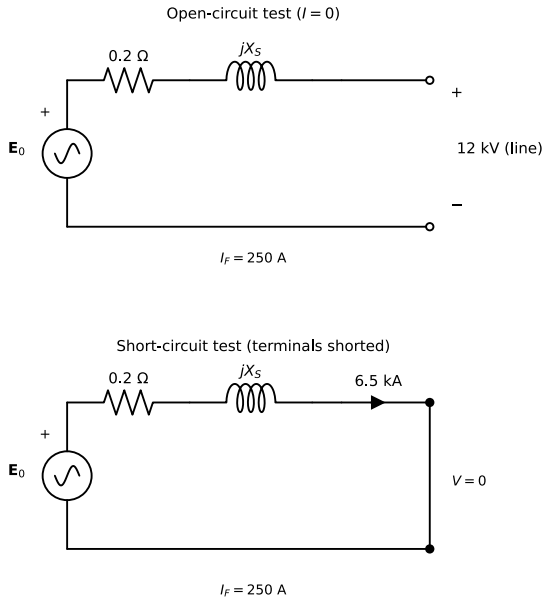


Jun23 Q4a test circuits: OC 4.6 kV line and SC 720 A, both at a field current of 60 A, R = 0.3 ohm

Jun24 Q4a (past exam, 5 marks). *4a) Synchronous Machines [5 marks]* A 14 kV, 400 MVA synchronous generator is tested at a field current of 250 A. The open-circuit voltage was found to be 12 kV and the short-circuit current was 6.5 kA. The stator resistance was measured at 0.2 ohms per phase.

i) Find the value of the synchronous reactance in per-unit. [3 marks]

ii) The synchronous generator is delivering 200MW at a terminal voltage of 3% below rated voltage and a power-factor of 0.8 lagging. Find generator output voltage and current phasors in per-unit. [2 marks]



Jun24 Q4a test circuits: OC 12 kV line and SC 6.5 kA, both at a field current of 250 A, R = 0.2 ohm

PQ 8.11 (practice questions; parts (a)–(b) here). A power station generator is rated at 250 MVA, 15 kV and has a star-connected stator winding. After manufacture, an open-circuit test is performed and a line voltage of 5.75 kV is measured with an excitation of 50 A. Under the same value of excitation, a current of 2050 A is measured with the short-circuit test. The resistance of each stator phase is 0.15 ohm. The generator is supplying its full-load rating at nominal voltage and a power factor of 0.85 lag. Ignoring the effects of magnetic saturation, calculate :

(a) The unsaturated synchronous reactance in per-unit
(b) The per-unit stator current magnitude and phase relative to the terminal voltage
(c) The induced voltage E_0 in per-unit showing both its magnitude and phase angle [Hint : draw the synchronous machine equivalent circuit and hence see that $E_0 = V + I(R + jX_S)$]
(d) The corresponding value of E_0 as a line voltage. [Hint : multiply the magnitude of the per-unit value by the base line voltage]
(e) The excitation required in amps.
(f) The rotor angle?

PQ 8.12 (practice questions; parts (a)–(b) here). A 120 MVA, 50Hz, 3,000rpm, 16 kV power station generator gives the following test results :

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
50	4.4	600

The stator resistance is measured as 0.1 ohm. Magnetic saturation can be ignored.

(a) From the test results, calculate the values of the unsaturated synchronous reactance in per unit
(b) Calculate the stator current magnitude and phase in per-unit when the generator is supplying a load of 102 MW at a power factor 0.95 lag when the terminal voltage is at its rated value. Use the terminal voltage as the reference phasor.
(c) Calculate the required excitation (in amps) for the situation in part (b). What is the ratio of this value of excitation current compared to the excitation current required for synchronization?

New questions

New Question 1 (easy) [4 marks]. A 3ph system has a base of 50 MVA and 11 kV. (a) Find I_B and Z_B. [2] (b) Express a 0.8 Ω per-phase reactance in pu. [1] (c) A bus voltage is 0.95 pu. Give it as a line voltage and as a phase voltage in kV. [1]

New Question 2 (easy–medium) [5 marks]. A 300 MVA, 18 kV, star-connected generator is tested. The OC slope on the linear part is 400 V (line) per field amp, and the SC slope is 80 A per field amp. $R = 0.05\ \Omega$. (a) Find X_S in ohms and in pu. [3] (b) What field current gives rated voltage on open circuit? [1] (c) What field current gives E_0 = 1.8 pu, assuming no saturation? [1]

New Question 3 (medium) [4 marks]. A 250 MVA, 15 kV generator delivers 180 MW and **absorbs** 60 MVAr from the grid, with the terminal voltage 3% above rated. Find $\mathbf{V}$ and $\mathbf{I}$ in pu, and state the pf.

New Question 4 (exam-style, chained) [9 marks]. A 180 MVA, 15 kV, star-connected generator gives these test results at I_F = 80 A: OC line voltage 6.0 kV, SC current 2400 A. $R = 0.12\ \Omega$. Ignore saturation.

a. Find X_S and R in pu. [3]
b. The generator delivers 135 MW at 0.9 pf lagging with the terminal voltage 2% above rated. Find $\mathbf{V}$ and $\mathbf{I}$ in pu, and I in amps. [3]
c. Find $\mathbf{E}_0$ in pu, E_0 as a line voltage, and the field current required. [3]

New Question 5 (motor convention) [5 marks]. A 500 kVA, 415 V synchronous motor has $X_S = 0.9$ pu, with R neglected. It draws 400 kW at 0.8 pf **lagging** at rated voltage. (a) Find the input current in pu and in amps. [2] (b) Using the motor convention (current into the machine), find $\mathbf{E}_0$ in pu and as a line voltage. [2] (c) Is the rotor angle positive or negative, and what does that tell you? [1]

6. Quick recall check

1. A 3ph machine is rated 120 MVA, 16 kV. What is Z_B?

2. The OC test gives 4.4 kV (line) at I_F = 50 A, and the SC test gives 600 A at the same I_F. What is $|Z_S|$ in ohms?
3. A generator supplies 0.6 pu real power at 0.8 pf leading, at 1.0 pu voltage. What is $\mathbf{I}$?
4. Write the course's equivalent-circuit equation for a generator, and the motor version using the input current $\mathbf{I}_m$.
5. Why does the short-circuit characteristic stay linear when the open-circuit characteristic saturates?

Synchronous phasors and the power angle

Practice questions · Day 14 · Fri 6 Nov · pack M11

5. Practice

From your course files

PQ 8.8 (practice questions). A round-rotor synchronous generator has a synchronous reactance of 1.8 per unit. Its excitation system is controlled so that its terminal voltage is always 1 per unit. Resistance and saturation may be neglected.

(a) What excitation (in per unit) is required to produce the generator's rated volt-amp output at unity power factor?
(b) What is the rotor angle at this loading?
(c) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power that can be delivered without losing synchronism?

Jun24 Q4b (past exam, 6 marks). *4b) Synchronous Machines [6 marks]* A round-rotor synchronous generator has a synchronous reactance of 1.8 per-unit. Its excitation system is controlled so that its terminal voltage is always 1 per-unit. Resistance and saturation may be neglected.

i) What excitation (in per unit) is required to produce the generator's rated volt-amp output at 0.9 leading power-factor? [3 marks]

ii) What is the rotor angle at this loading? [1 mark]

iii) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power, in per-unit, that can be delivered without losing synchronism? What is the rotor angle corresponding to this condition? [2 marks]

PQ 8.9 (practice questions). A 200 MVA generator has an unsaturated synchronous reactance of 1.8 per unit. If the excitation is set at a level which would give an open-circuit voltage of 1.5 per unit, and saturation and resistance can be neglected, what is the power in

megawatts that can be delivered when the machine's terminal voltage is 2% above its rated value and the rotor angle is not to exceed 70°?

PQ 8.11 (practice questions). A power station generator is rated at 250 MVA, 15 kV and has a star-connected stator winding. After manufacture, an open-circuit test is performed and a line voltage of 5.75 kV is measured with an excitation of 50 A. Under the same value of excitation, a current of 2050 A is measured with the short-circuit test. The resistance of each stator phase is 0.15 ohm. The generator is supplying its full-load rating at nominal voltage and a power factor of 0.85 lag. Ignoring the effects of magnetic saturation, calculate :

(a) The unsaturated synchronous reactance in per-unit
(b) The per-unit stator current magnitude and phase relative to the terminal voltage
(c) The induced voltage E_0 in per-unit showing both its magnitude and phase angle [Hint : draw the synchronous machine equivalent circuit and hence see that $E_0 = V + I(R + jX_S)$]
(d) The corresponding value of E_0 as a line voltage. [Hint : multiply the magnitude of the per-unit value by the base line voltage]
(e) The excitation required in amps.
(f) The rotor angle?

PQ 8.12 (practice questions). A 120 MVA, 50Hz, 3,000rpm, 16 kV power station generator gives the following test results :

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
50	4.4	600

The stator resistance is measured as 0.1 ohm. Magnetic saturation can be ignored.

(a) From the test results, calculate the values of the unsaturated synchronous reactance in per unit
(b) Calculate the stator current magnitude and phase in per-unit when the generator is supplying a load of 102 MW at a power factor 0.95 lag when the terminal voltage is at its rated

value. Use the terminal voltage as the reference phasor.
(c) Calculate the required excitation (in amps) for the situation in part (b). What is the ratio of this value of excitation current compared to the excitation current required for synchronization?

PQ 8.13 (practice questions). The stator winding of a synchronous motor is rated at 330 kVA. If the motor is supplying a mechanical load of 270 kW at an efficiency of 92%, what is the maximum reactive power it can provide for power factor correction?

PQ 8.14 (practice questions). A production line uses an induction motor to drive a 200 kW mechanical load. The induction machine has an efficiency of 90% and a power factor of 0.82. A 100 kVA synchronous motor is used to drive an 80 kW load nearby. Assume the synchronous motor has an efficiency of 100%. What is the highest input power factor of the two machines combined as a system?

Jun23 Q4b (past exam, 7 marks). *4b) Power Factor Correction [7 marks]* An industrial plant uses two identical induction motors, each driving a mechanical load of 200kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i) Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii) Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]

iii) Use the above results to determine the maximum power factor for the plant. [3 marks]

PQ 8.15 (practice questions). An industrial installation has an existing load of 2760 kVA at a power factor of 0.71 lag. A major extension is planned, requiring 1500 kW of mechanical load. It is decided to satisfy 800 kW of this demand using induction motors, and to purchase synchronous motors to supply the remainder. The parameters of the additional motors are as follows :

Induction Motors : Efficiency 89% : Power factor 0.75 lag
Synchronous Motors : Efficiency 93% : Total rating 1350 kVA

What is the maximum predicted power factor of the completed installation?

PQ 8.10 (practice questions). A synchronous generator has a rating of 150 MVA and a rated terminal voltage of 15 kV. It's stator is star-connected. It is given the standard tests after manufacture with the following results :

Open-Circuit Test : Slope of linear portion = 500 volts (line) per field amp.
Short-Circuit Test : Slope = 100 amps per field amp

The stator resistance is measured as 0.3 Ω / phase. Magnetic saturation can be neglected.

(a) Find the base current and base impedance. [Hint : choose the machine rated VA and voltage as the base power and base voltage.]
(b) What is the value of the unsaturated synchronous reactance *in per-unit* ? [Hint : choose any value of field current, say 1A and calculate the open-circuit voltage and short-circuit current corresponding to this. Find the synchronous impedance and hence the synchronous reactance from the stator resistance.]
(c) What value of field current would be required for synchronization? (Note that to synchronize the machine to the system, the open-circuit voltage E_0 is set to rated voltage.)
(d) At synchronization, what is the value of the open-circuit voltage E_0 in per-unit?
(e) Under a certain operating condition, the open-circuit voltage E_0 is calculated to be 1.65 pu. What field current (in amps) would be required to produce this?

Jun23 Q4c (past exam, 8 marks). *4c) Short-Answer Questions [8 marks]* Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) Explain the concept of power-factor correction. Why is power-factor correction useful in power systems? [3 marks]

ii) Carefully explain the differences and similarities between transformers and induction machines. [3 marks]

iii) Consider a squirrel-cage induction machine. If the rotor cage was made from a perfect conductor (with infinite conductivity) how would this affect the properties of the machine? [2 marks]

Jun24 Q4c (past exam, 9 marks). *4c) Short-Answer Questions [9 marks]* Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) We have discussed three types of electric power: real, reactive and apparent. Explain briefly what each one is and their practical significance. [3 marks]

ii) Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power. [3 marks]

iii) Carefully explain why the input current of an induction machine under no-load conditions is not zero. What are the physical effects which cause this? [3 marks]

Practice Questions, Ch8 short answers (j) and (k) (bonus). From “Some Short-Answer Questions” at the end of Chapter 8:

j) With the aid of sketch, explain what limits the reactive power capability of a synchronous machine.

k) Why is the power-factor of industrial plants usually lagging? Why are synchronous motors sometimes used in such plants?

New questions

N1. [3 marks] A 50 MVA synchronous compensator (P = 0) has $X_S = 1.5$ pu and is connected to a bus held at 1 pu. Neglect R.

a. What E_0 is needed for it to supply 0.8 pu reactive power to the bus? Is it over- or under-excited? [2]
b. What E_0 is needed for it to absorb 0.5 pu? [1]

N1 circuit: compensator E0 behind j1.5 pu on a 1 pu bus

N2. [4 marks] A 50 MVA, 11 kV star-connected generator has $X_S = 2.4\ \Omega$ per phase. Its excitation gives $E_0 = 14$ kV (line). It runs at $\delta = 25^\circ$ on an 11 kV bus. Neglect R.

a. Find the three-phase output power in MW using per-phase quantities, then check it with line quantities. [2]
b. Find the maximum power at this excitation, in MW and in per-unit. [2]

N2 circuit: E0 = 14 kV line behind j2.4 ohm on an 11 kV bus

N3. [6 marks] A generator with $X_S = 1.6$ pu delivers S = 0.8 pu at V = 1 pu. Neglect R.

a. Find E_0 and δ at 0.85 pf lagging. [2]

b. The real power is held constant, and the excitation is changed until the pf is 0.85 leading. Find the new E_0 and δ. [2]
c. Find P_{max} in each case, and comment on stability. [2]

N3 circuit: E0 behind j1.6 pu, V = 1 pu, S = 0.8 pu

N4. [5 marks] A generator ($X_S = 1.8$ pu, V = 1 pu) supplies a constant P = 0.6 pu. The operator lowers the excitation to absorb reactive power. Neglect R.

a. What is the minimum E_0 if δ must not exceed 75°? [2]
b. At that point, find the stator current phasor, the reactive power, and the pf. [3]

N4 circuit: E0 behind j1.8 pu, V = 1 pu, P = 0.6 pu

N5. [10 marks] A plant takes 600 kW at 0.75 pf lagging from a 415 V supply. A new 400 kVA, 415 V synchronous motor ($X_S = 1.0$ pu on its own rating, η = 95%) is installed to drive a 300 kW load. It is run fully over-excited to correct the pf. Neglect R in the phasor model, and take the motor's electrical input as the phasor power.

a. Find the motor's input power and its maximum reactive power. [2]
b. Find the plant's new pf. [2]
c. On the motor base, find the motor current phasor (in pu and in amps) and the required excitation E_0 (in pu and as a line voltage). [4]
d. The excitation is then reduced until the motor runs at unity pf. Find the new E_0 and the plant pf. [2]

N5 plant: 600 kW at 0.75 lag in parallel with a 400 kVA synchronous motor

6. Quick recall check

1. At what rotor angle does a round-rotor machine deliver maximum power, and what happens if you try to go past it?
2. A generator's output current is leading. Is it over- or under-excited, and does it supply or absorb Q?
3. Does the per-unit $P = \frac{VE_0}{X_S}\sin\delta$ give per-phase or three-phase power?
4. A 100 kVA synchronous motor drives 60 kW at 100% efficiency. What is the most leading reactive power it can supply?
5. On a V-curve at constant P, where is the stator current a minimum, and what is the pf there?

ELECTRICAL MACHINES · PAPER

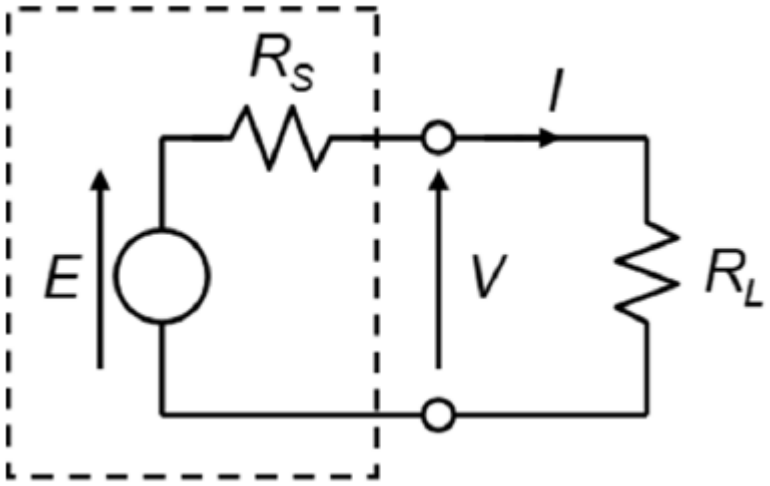
Jun 2023 exam: full worked solutions

Practice questions · Day 18 · Tue 10 Nov · pack MX1

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Battery Modelling [5 marks]

The equivalent circuit on the right is used to model a car battery.

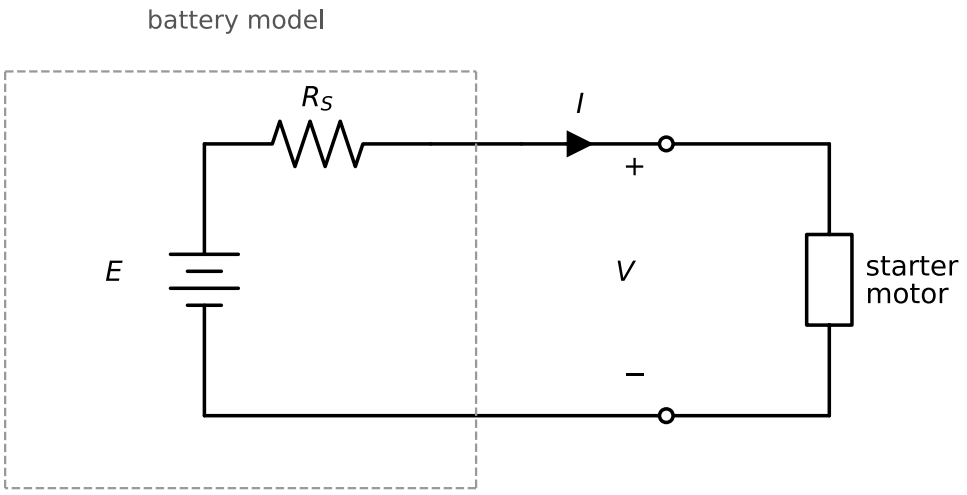


Jun23 Q1a figure: battery model, E in series with R_S, terminal voltage V, load R_L (from the exam paper)

i) The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]

ii) A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]

iii) Find the battery output voltage when a load drawing a current of 20A is connected to it. [1 mark]



open circuit: V = 13.8 V; at I = 300 A: V = 9.5 V

Q1a: battery modelled as EMF E in series with RS, feeding the starter motor; open circuit 13.8 V, 9.5 V at 300 A

MARKS: (i) 1. (ii) 1 for R_S, 1 for the loss, 1 for the motor power. (iii) 1.

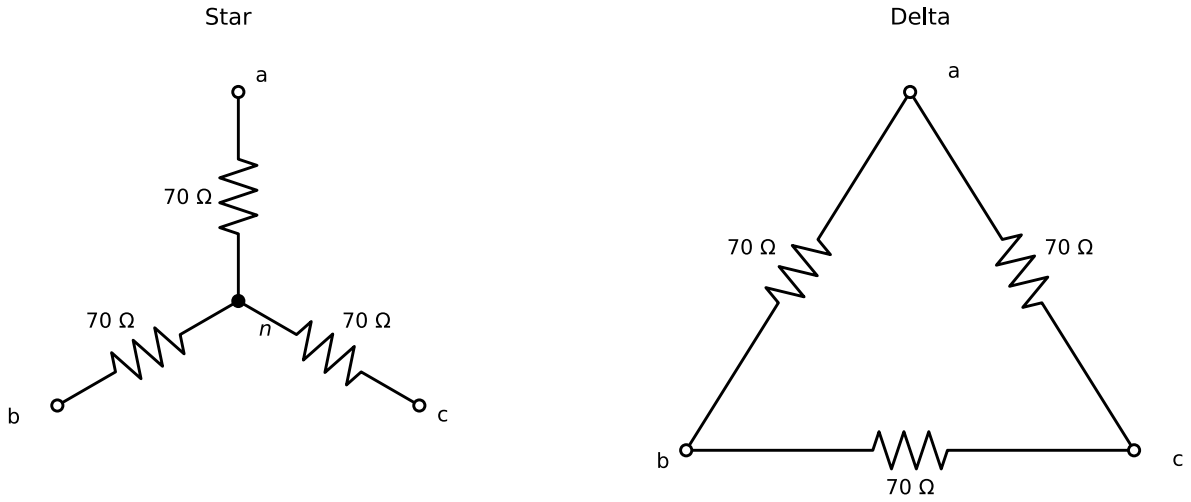
> **COMMON MISTAKE:** giving $EI = 4140$ W as the motor power. The power delivered to the load uses the **terminal** voltage, VI.

1b) Star-Delta Equivalent Circuits [6 marks]

Consider a three-phase load consisting of three 70 Ω resistors connected to a 460V, 3ph supply.

i) If the load is star-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

ii) If the load is delta-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]



line voltage 460 V between any two of a, b, c

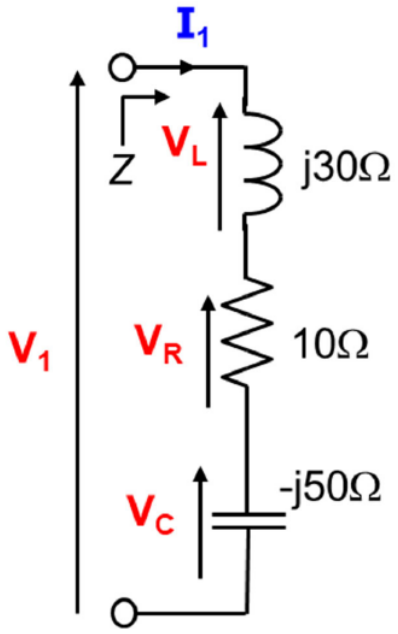
Q1b: three 70 ohm resistors in star and in delta on a 460 V line supply

MARKS: 1 each for phase current, line current and power, in each case (3 + 3).

> **COMMON MISTAKE:** using 460 V across each resistor in star (gives 3 times the true power), or forgetting $\sqrt{3}$ between phase and line current in delta.

1c) Phasor Calculations [9 marks]

Consider the single-phase series RLC load as shown in the figure on the right. The current $\mathbf{I}_1$ is known to be $13\angle+20^\circ$ A.



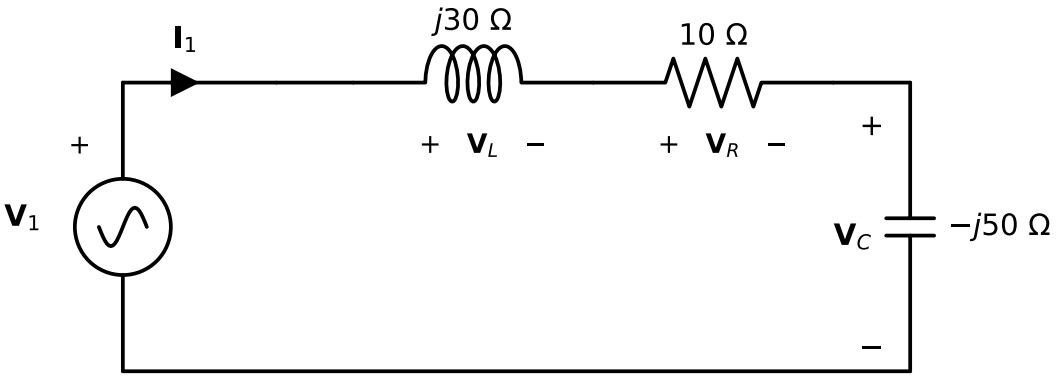
Jun23 Q1c figure: series j30 Ω, 10 Ω and –j50 Ω carrying I1, with V_L, V_R, V_C and V1 marked (from the exam paper)

i) Using the known current $\mathbf{I}_1$, find the voltage phasors across the resistance $\mathbf{V}_R$, inductance $\mathbf{V}_L$ and capacitance $\mathbf{V}_C$ in polar form using Ohm's law. [3 marks]

ii) Find the phase impedance Z of the load in polar form and hence the voltage phasor $\mathbf{V}_1$ in polar form using Ohm's law. [3 marks]

iii) The complex power S is given by $S = \mathbf{V}_1\mathbf{I}_1^*$, where * refers to the complex conjugate. Find the complex conjugate of the current $\mathbf{I}_1^*$, thus the complex power S and hence the total real power P and reactive power Q supplied from the power source. [3 marks]

$\mathbf{I}_1 = 13\angle 20^\circ$ A (element values from the paper's figure, as in pack M1)



Q1c: series j30 ohm, 10 ohm and -j50 ohm carrying I1 = 13 at 20 degrees A (values from the paper's figure, as in pack M1)

MARKS: (i) 1 per voltage. (ii) 1.5 for Z, 1.5 for $\mathbf{V}_1$. (iii) 1 for $\mathbf{I}_1^*$, 1 for S, 1 for P and Q with signs.

Discrepancies with the official sheet:

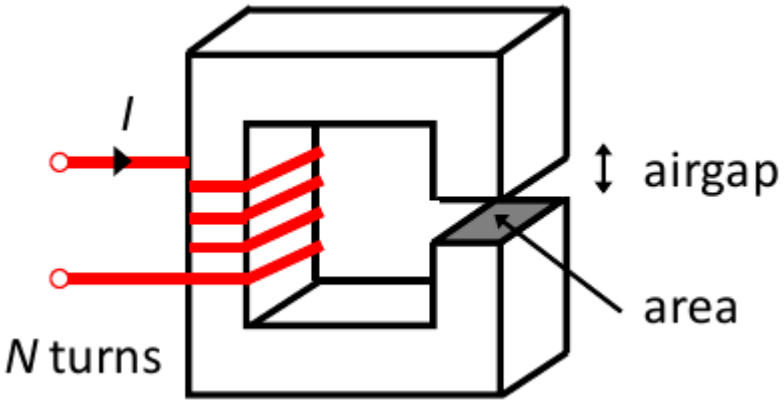
- The official sheet gives $\mathbf{V}_R = 130\angle 120^\circ$ V. This is a **typo**: a resistor's voltage is in phase with its current, so it is $130\angle 20^\circ$ V. The official $\mathbf{V}_1$ (which needs $\mathbf{V}_R$ at 20°) confirms this.
- The official Q is -3379 VAr. The exact value is -3380 VAr ($13^2 \times 20$). The official number comes from multiplying the rounded 290.7 V. Either earns the mark.

> **COMMON MISTAKE:** writing $\mathbf{V}_C$ at $+110^\circ$ or adding the reactances as $30 + 50$. A capacitor is $-jX_C$, so its voltage lags the current by 90°.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [9 marks]

Consider the iron c-core shown, which has a magnetic cross-sectional area of 20cm² and a mean magnetic path length of 50cm. The core has a winding with 32 turns and has a relative permeability of 3,000. The airgap length is 1.5mm.

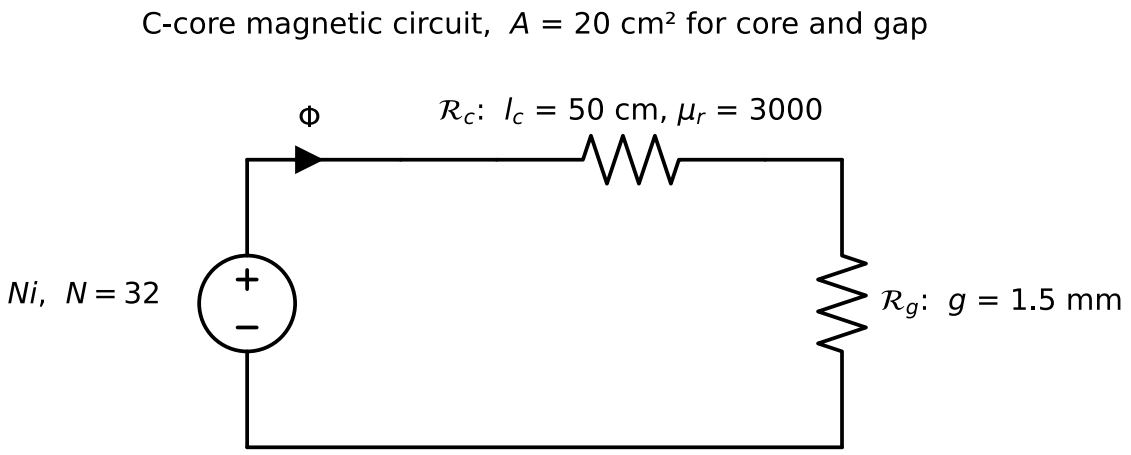


Jun23 Q2a figure: iron c-core with an N-turn winding carrying I, and an airgap (from the exam paper)

i) Calculate the reluctances of the airgap $\mathcal{R}_G$ and the core $\mathcal{R}_C$ and hence find the inductance of the coil. [4 marks]

ii) Calculate the maximum energy stored in the airgap only, if the maximum allowable peak core flux density is 1.8 T. [2 marks]

iii) The magnetic core is operating with a peak flux density of 1.5T at 60 Hz. It has an iron loss of 12W, of which 30% is hysteresis losses and 70% is eddy-current losses. Calculate the new hysteresis, eddy-current and total iron loss if the peak flux density is reduced to 1.2T, whilst the frequency is increased to 100Hz. [3 marks]



Q2a: magnetic circuit, MMF Ni with N = 32 driving flux through the core and gap reluctances in series

MARKS: (i) 1 each for $\mathcal{R}_G$ and $\mathcal{R}_C$, 2 for L (formula and value). (ii) 1 formula, 1 value. (iii) 1 for the split, 1 for each scaled loss (and the total).

COMMON MISTAKE: in (iii), scaling the whole 12 W by one law. In (ii), using $\frac{1}{2}LI^2$ (total energy) when the question asks for the gap only.

2b) Transformer Parameters [5 marks]

The following tests are performed on a single-phase 240V:32V, 2400VA transformer.

- Short-circuit test. Primary measurements are 27 V, 9.7 A, and 55 W.
- Open-circuit test. Primary measurements are 240 V, 1.3 A, and 45 W.

Find all the simplified equivalent circuit parameters.

Q2b test circuits: OC 240 V, 1.3 A, 45 W sees only the shunt branch; SC 27 V, 9.7 A, 55 W sees only the series branch

MARKS: 1 for each of R, X, R_C, X_M, plus 1 for stating they are referred to the primary.

COMMON MISTAKE: $X_M = V/I_{OC} = 184.6\ \Omega$. That is $|Z_M|$ of the parallel pair, not X_M. Separate P and Q first, or find the magnetising current $I_M = I_{OC}\sin\phi$.

2c) Transformer Analysis [6 marks]

Consider the transformer simplified equivalent circuit shown on the right. The values of the primary voltage $\mathbf{V}_1$ and current $\mathbf{I}_1$ phasors are known and it is desired to find the value of the secondary voltage $\mathbf{V}_2'$ phasor.

Jun23 Q2c figure: simplified transformer equivalent circuit, R_C parallel jX_M across V1, then jX1+jX2′ and R1+R2′ to V2′ (from the exam paper)

i) Using Kirchhoff's voltage law (KVL), write an expression for $\mathbf{V}_2'$ in terms of $\mathbf{V}_1$, $\mathbf{I}_2'$ and the relevant equivalent circuit parameters shown. [2 marks]

ii) Assuming the load is purely resistive with a value of R_L', use the voltage divider rule to write an expression for $\mathbf{V}_2'$ in terms of $\mathbf{V}_1$ and the relevant equivalent circuit parameters shown. [2 marks]

iii) For a purely resistive load, would you expect that $\mathbf{V}_2'$ will be leading, in phase or lagging with respect to $\mathbf{V}_1$? Carefully explain why. [2 marks]

Q2c: simplified transformer circuit, shunt RC parallel jXM across V1, series R1 + R2′ and j(X1 + X2′) to the load RL′

MARKS: (i) 2 (correct sign and both series terms). (ii) 2. (iii) 1 for "lagging", 1 for the reason chain (current lags because of X; load voltage in phase with current).

COMMON MISTAKE: in (iii), saying "lagging because it is a transformer" with no reason, or saying "in phase because the load is resistive". The load voltage is in phase with $\mathbf{I}_2'$, not with $\mathbf{V}_1$.

Question 3. AC and Induction Machines [20 marks]

3a) AC Machines [6 marks]

Consider a 50 Hz AC machine which has a rated speed of 560 rpm.

i) How many poles does it have? [1 mark]

ii) Is the AC machine a synchronous machine or an induction machine? Briefly explain why. [1 mark]

iii) At rated speed, what is the ideal (that is, maximum possible) efficiency of the machine? [2 marks]

iv) The machine is operating at one-quarter of rated torque. What is the operating speed in rpm? You can assume a linear torque versus speed curve. Find also the output power expressed as a percentage of rated output power. [2 marks]

MARKS: (i) 1. (ii) 1 (answer with reason). (iii) 1 for $\eta = 1 - s$, 1 for the value. (iv) 1 for 590 rpm, 1 for 26.34%.

> **COMMON MISTAKE:** in (iv), saying the power is 25%. The torque is a quarter, but the speed went up slightly, so the power is a little more than a quarter.

3b) Induction Machine Torque vs Speed [5 marks]

The equivalent-circuit parameters of a wound-rotor induction machine are listed below.

stator resistance R_1 = 0.4 Ω, rotor resistance R_2' = 0.5 Ω
stator leakage reactance X_1 = 1.3 Ω, rotor leakage reactance X_2' = 1.5 Ω
magnetising reactance X_M = 95 Ω, core loss resistance R_C = 270 Ω

i) Calculate the external resistance, R_{EXT}, required such that maximum torque occurs at standstill. [2 marks]

ii) On the axes below, draw a typical torque vs. speed characteristic curves for an induction machine – assume that this curve represents the above induction machine with zero external

resistance. In addition, draw the torque curves for external resistance equal to 0.5 and 1pu, where 1pu represents the R_{EXT} calculated in part i) above. [3 marks]

(The paper gives blank torque–speed axes, torque up and speed n across, starting at 0.)

Q3b: simplified circuit with 270 ohm parallel j95 ohm, 0.4 ohm, j1.3 ohm, j1.5 ohm and (R2′ + REXT′)/s

MARKS: (i) 1 for the s_{max} condition, 1 for the value. (ii) 1 for the correct $R_{EXT} = 0$ shape (zero at n_S, peak near n_S, lower starting torque), 1 for equal peaks, 1 for the 1 pu curve peaking at $n = 0$.

> **COMMON MISTAKE:** drawing the curves with different peak heights, or making them cross zero at different speeds. Adding rotor resistance only stretches the curve along the slip axis.

3c) Induction Machine Torque Calculation [9 marks]

A 415 V, 50 Hz, 1450 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):

stator resistance R_1 = 0.3 ohms, rotor resistance R_2' = 0.2 ohms
stator leakage reactance X_1 = 2.5 ohms, rotor leakage reactance X_2' = 2.5 ohms
magnetising reactance X_M = 50 ohms, core loss resistance R_C = 200 ohms

In your calculations you should use the simplified equivalent circuit.

i) Find the number of poles and hence the synchronous speed in rpm. [2 marks]

ii) Consider the machine operating at 1520 rpm. Is it operating as a motor or generator? Briefly indicate why? [1 mark]

iii) Consider the machine operating at rated voltage at its rated speed of 1450 rpm. What is the output torque? Hint: use the torque formula. [3 marks]

iv) Consider the machine operating at rated voltage at its rated speed of 1450 rpm. Ignoring the magnetising reactance and core loss resistance, find the input current magnitude and phase. [3 marks]

Q3c: simplified circuit with 200 ohm parallel j50 ohm, 0.3 ohm, j2.5 ohm, j2.5 ohm and R2′/s

MARKS: (i) 1 poles, 1 n_S. (ii) 1 with reason. (iii) 1 for s and ω_S, 1 for the formula with phase voltage, 1 for the value. (iv) 1 for the impedance, 1 for the magnitude, 1 for the angle.

COMMON MISTAKE: using the line voltage 415 V in the torque formula (gives 3 times the torque, 305 Nm), or dividing by the rotor speed ω instead of ω_S. $\tau = P_{AG}/\omega_S$ always.

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

A 150 MVA, 50Hz, 3,000rpm, 14 kV power station generator gives the following test results:

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
60	4.6	720

The stator resistance is measured at 0.3 ohm.

i) Find the value of synchronous reactance in ohms (not per-unit). [2 marks]

ii) The generator is delivering a real power of 100MW and reactive power of 50 MVAr at rated voltage. Calculate the magnitude and phase of the stator voltage and current in per-unit. [3 marks]

Q4a test circuits: OC 4.6 kV line and SC 720 A, both at a field current of 60 A, R = 0.3 ohm

MARKS: (i) 1 for $|Z_S|$ with phase voltage, 1 for X_S. (ii) 1 for $\mathbf{V}$, 1 for the magnitude of $\mathbf{I}$, 1 for the (negative) angle.

Discrepancy: the official sheet gives 0.7453 pu. The exact value is $\sqrt{0.6667^2 + 0.3333^2} = 0.74536$, which rounds to 0.7454. This is just truncation in the

official sheet.

> **COMMON MISTAKE:** a positive current angle (the OCR'd text version of the answer sheet drops the minus sign, but the PDF shows $-26.57°$), or using the line OC voltage directly ($4600/720 = 6.39\ \Omega$, a factor of $\sqrt{3}$ too big).

4b) Power Factor Correction [7 marks]

An industrial plant uses two identical induction motors, each driving a mechanical load of 200kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i) Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii) Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]

iii) Use the above results to determine the maximum power factor for the plant. [3 marks]

Q4b plant: the remaining induction motor and the 250 kVA synchronous motor in parallel on the supply

MARKS: (i) 1 each. (ii) 1 each. (iii) 1 for P total, 1 for net Q with the correct sign, 1 for pf with "lag".

Discrepancy: the official sheet gives 114.6 kVAr. The exact value is 114.53 kVAr. The official number comes from using the rounded 222.2 kW ($\sqrt{250^2 - 222.2^2} = 114.58$). The final pf is 0.9955 either way.

> **COMMON MISTAKE:** using the 200 kW output instead of the input power, adding the two reactive powers instead of subtracting, or treating 250 kVA as real power.

4c) Short-Answer Questions [8 marks]

Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) Explain the concept of power-factor correction. Why is power-factor correction useful in power systems? [3 marks]

ii) Carefully explain the differences and similarities between transformers and induction machines. [3 marks]

iii) Consider a squirrel-cage induction machine. If the rotor cage was made from a perfect conductor (with infinite conductivity) how would this affect the properties of the machine? [2 marks]

MARKS: (i) 1 what reactive power is and why it is a problem, 1 capacitor in parallel generates reactive power, 1 the benefit (lower current, losses, better use of rating). (ii) about 1.5 for similarities, 1.5 for differences (rotating short-circuited secondary, different rotor frequency, R_2'/s). (iii) 1 for "zero R_2', no rotor loss, higher efficiency", 1 for "vertical τ–n curve at n_S and zero starting torque".

> **COMMON MISTAKE:** in (iii), saying the machine would be "more efficient" and stopping there. The second mark needs the zero starting torque. In all of 4c, do not draw a phasor diagram or write a formula: the question forbids it.

Jun 2024 exam: full worked solutions

Practice questions · Day 21 · Fri 13 Nov · pack MX2

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Fill-In-The Blank Questions [4 marks]

i) A transformer has an input voltage of 240V and an output voltage of 24V. Based on this it has a turns ratio n of _____ and is a step _____ (choose down or up) transformer. Another transformer has an input voltage of 240V and an output voltage of 480V. Based on this it has a turns ratio n of _____ and is a step _____ (choose down or up) transformer.

ii) An electric motor has an input power of 1000W and a power loss of 300W. This results in an output power of _____W and an efficiency of _____%. Increasing the power loss for the same input power would cause the efficiency to _____ (choose decrease or increase). Increasing the input power for the same power loss would cause the efficiency to _____ (choose decrease or increase).

MARKS: 8 blanks at ½ mark each.

> **COMMON MISTAKE:** writing 1:10 for the first ratio. The course writes primary:secondary, so 240 V to 24 V is 10:1. The other mistake is answering “decrease” for the last blank. Think of it as the loss staying fixed while the input grows.

1b) Electrical and Mechanical Power [3 marks]

Consider a 3ph, 4 pole, 50 Hz induction motor. Under a certain operating condition, the motor speed is 1420 rpm, input voltage is 415 V, input current is 6.8 A, input electrical power is 4 kW and output torque is 23 Nm.

i) Find its input power factor (you can assume this is lagging). [1 mark]

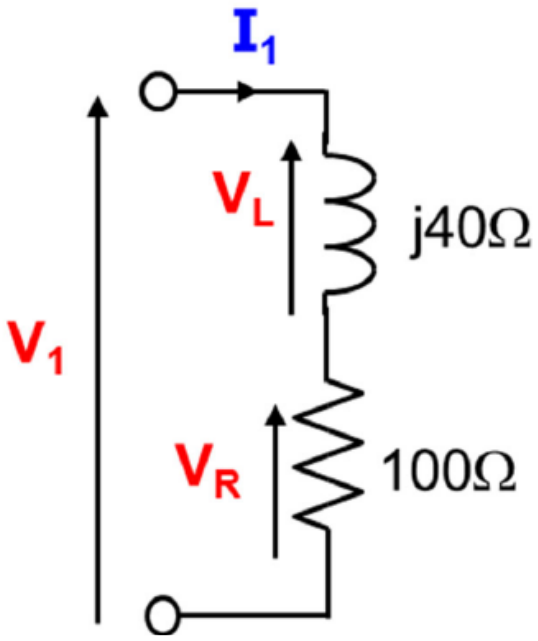
ii) Find its output power and hence efficiency. [2 marks]

MARKS: (i) 1 for $\sqrt{3}V_L I_L$ and pf. (ii) 1 for $P = \tau\omega$ with ω in rad/s, 1 for η.

COMMON MISTAKE: leaving out $\sqrt{3}$, which gives pf = 1.417. A pf above 1 is impossible, so this should trigger a re-check. The second classic error is $P = \tau \times n$ with n in rpm.

1c) AC Circuit Analysis [5 marks]

Consider the series resistor-inductor (RL) circuit shown on the right.

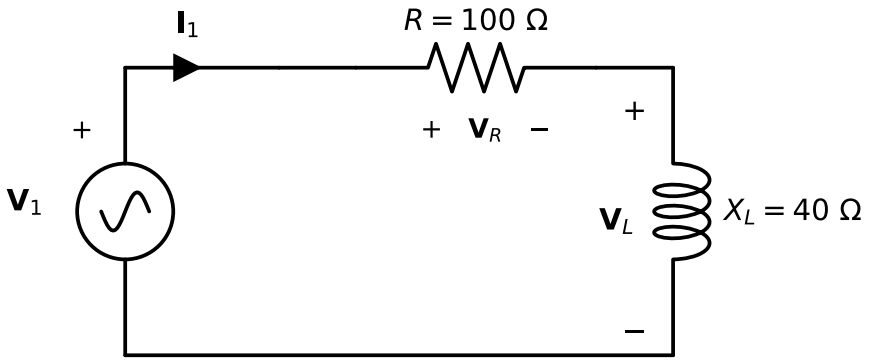


Jun24 Q1c figure: series j40 Ω and 100 Ω carrying I1, with V_L, V_R and V1 marked (from the exam paper)

i) The input current I_1 has a magnitude of 2A. Find the value of the real power and reactive power absorbed by the resistor. Find also the real and reactive power absorbed by the inductor. [3 marks]

ii) For a different value of input current I_1, it is known that the resistor voltage V_R has a magnitude of 50V and the inductor voltage V_L has a magnitude of 20V. Hence find the magnitude of the input voltage V_1. [2 marks]

series RL; values from the paper's figure



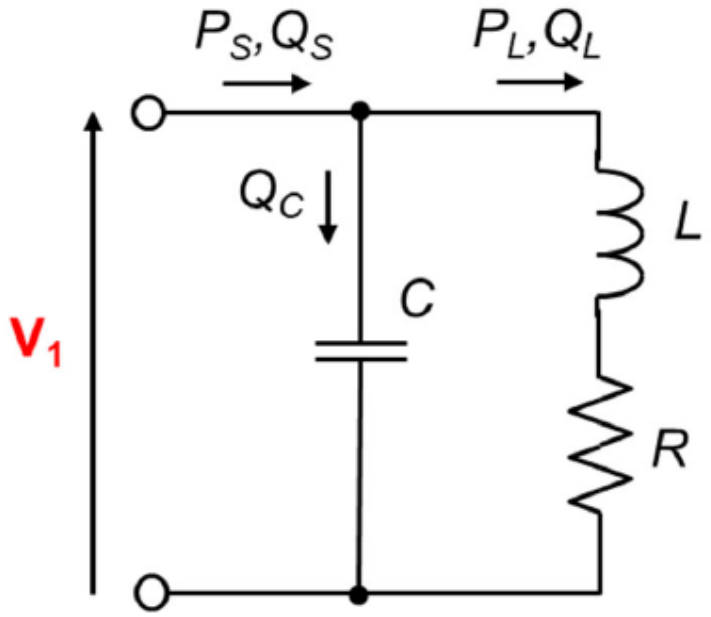
Q1c: series R = 100 ohm and XL = 40 ohm, as in the paper's figure

MARKS: (i) ½ mark for each of the four values, plus 1 for showing I^2R and I^2X (3). (ii) 1 for the perpendicular-phasor reasoning, 1 for 53.85 V.

COMMON MISTAKE: adding magnitudes, $50 + 20 = 70$ V. Also, in (i), using the source voltage instead of the current, or giving the inductor a non-zero P.

1d) Power Factor Correction [8 marks]

Consider the circuit shown on the right which contains a series RL load which is known to absorb a real power of 80W and a reactive power of 60VAr. A power-factor correction capacitor C is connected across the input supply.

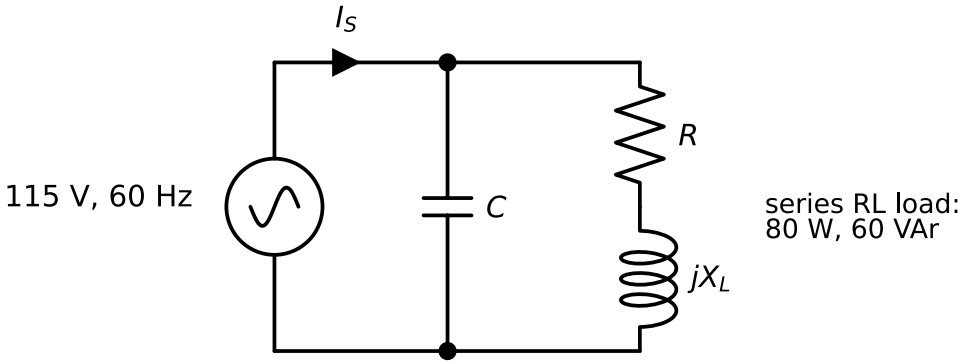


Jun24 Q1d figure: capacitor C across V1 in parallel with the series L and R load; P_S, Q_S at the supply, P_L, Q_L into the load, Q_C into C (from the exam paper)

i) Sketch the power triangle for the RL load and hence calculate its power factor. [2 marks]

ii) By adding the power factor correction capacitor C, it is desired to have a supply (input) power factor of 0.9 lag at the input to the circuit shown. Find the supply real power P_S and the value of supply reactive power Q_S which gives the desired power-factor. Hence find the value of the reactive power absorbed by the capacitor Q_C. [4 marks]

iii) For another desired value of power-factor, the capacitor is found to absorb -30VAr. What is the required value of capacitance (in µF) for the above capacitor if the input power supply is 115V, 60 Hz? [2 marks]



Q1d: capacitor C across the 115 V, 60 Hz supply in parallel with the series RL load (80 W, 60 VAr)

MARKS: (i) 1 for the labelled triangle, 1 for 0.8 lag. (ii) 1 for P_S, 1 for Q_S with a correct $\tan$, 1 for the Q balance, 1 for Q_C with its sign. (iii) 1 for the formula, 1 for the value.

COMMON MISTAKE: $Q_C = +21.25$ VAr (wrong sign). Also using −21.25 VAr in (iii), which gives 4.262 µF. Using 50 Hz gives 7.220 µF. The question says 60 Hz.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [7 marks]

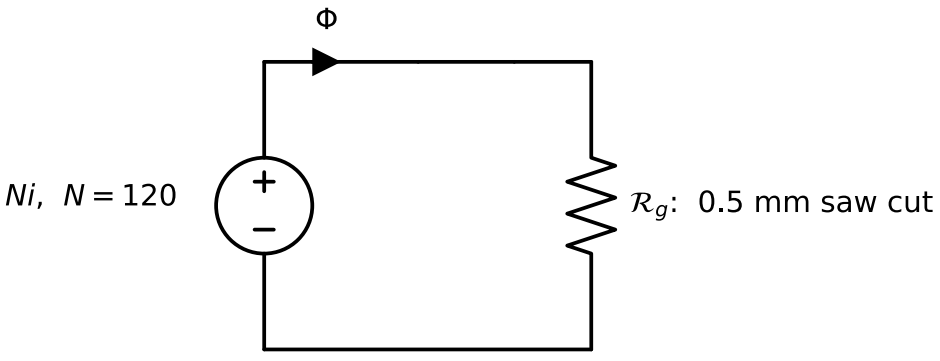
Consider a toroidal magnetic core which has a circular cross-section whose diameter is 2.5cm. The core has a winding of 120 turns and can be assumed to have an infinite permeability. A fine saw is used to cut a thin slot to produce an airgap of 0.5 mm in one side of the core.

i) Calculate the reluctance of the airgap $\mathcal{R}_g$ and hence the inductance of the winding. [3 marks]

ii) If the maximum allowable peak core flux density is 1.5T, calculate the maximum 50Hz voltage which can be applied to the winding. [2 marks]

iii) Using a flux density of 1.5T, calculate the force between the sides of the slot in the core. [2 marks]

toroid, $\mu = \infty$ (core reluctance zero), circular section 2.5 cm diameter



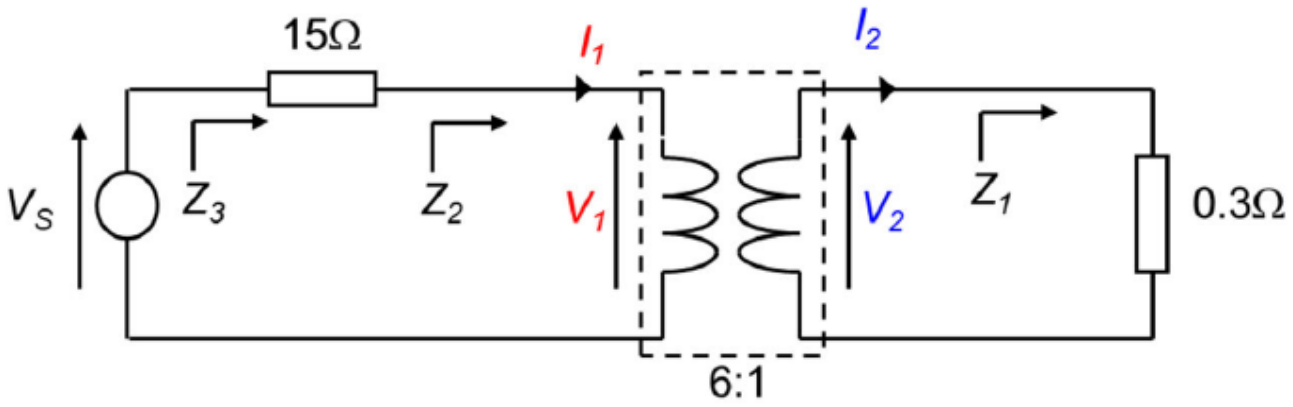
Q2a: magnetic circuit, MMF Ni with N = 120 across the gap reluctance only

MARKS: (i) 1 for the area, 1 for $\mathcal{R}_g$, 1 for L. (ii) 1 for the back-emf equation, 1 for the value. (iii) 1 for the formula, 1 for the value.

COMMON MISTAKE: using the diameter as the radius (A four times too big). Also writing $V = \hat{B}$-limited from L and a current, which needs the reluctance and goes wrong. In (iii), using $B = 1.5/\sqrt{2}$: the force formula uses the stated flux density, 1.5 T.

2b) Ideal Transformers [5 marks]

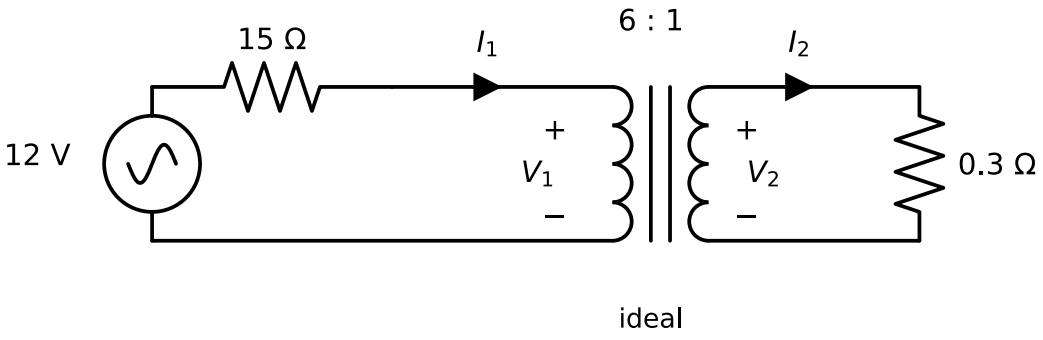
In the diagram on the right the transformer is ideal and has a turns ratio of 6:1.



Jun24 Q2b figure: source V_S, 15 Ω series resistor, 6:1 ideal transformer and 0.3 Ω load, with Z3, Z2, Z1, I1, I2, V1, V2 marked (from the exam paper)

i) Find the values of the impedances Z_1, Z_2 and Z_3 shown in the figure. [2 marks]

ii) Assuming the AC supply voltage V_S is 12V, find the magnitudes of the primary current I_1 and the actual (not referred) secondary current I_2. Find also the magnitudes of the primary voltage V_1 and the actual secondary voltage V_2. [3 marks]



ideal

values from the paper's figure

Q2b: 12 V source, 15 ohm series resistor, 6:1 ideal transformer and 0.3 ohm load

MARKS: (i) 1 for $Z_2 = n^2 Z_1$, 1 for Z_3. (ii) 1 for I_1, 1 for I_2, 1 for both voltages.

> **COMMON MISTAKE:** referring the wrong way $(0.3/36)$, or using $I_2 = I_1/n$. A step-down transformer gives a larger secondary current. Also $V_1 = 12$ V: the 15 Ω resistor drops most of the source voltage.

2c) Three-Phase Transformers [8 marks]

A three-phase transformer is rated at 16 kV/220 kV, 150 MVA. It has the following equivalent circuit parameters (referred to the primary winding where appropriate):

primary resistance R_1 = 0.08 ohms, secondary resistance R_2' = 0.06 ohms
primary leakage reactance X_1 = 0.14 ohms, secondary leakage reactance X_2' = 0.14 ohms
magnetising reactance X_M = 14 ohms, core-loss resistance R_C = 250 ohms.

i) Draw a simplified equivalent circuit for the transformer showing the values for each of the equivalent circuit parameters as impedances. [2 marks]

ii) Consider the transformer in the open-circuit condition with rated input voltage. Find the input current magnitude and (real) power. [4 marks]

iii) Consider the open-circuit condition as discussed above, if the input voltage was halved, by what factor (e.g. x0.25, x0.5, x1, x2 or x4) would you expect the following quantities to change by? [2 marks]

- Input current magnitude: _____

- Input real power: _____
- Input reactive power: _____
- Input power-factor: _____

Q2c: simplified circuit, 250 ohm parallel j14 ohm at the input, then 0.08 + 0.06 ohm and j(0.14 + 0.14) ohm

MARKS: (i) 1 for the shunt branch at the input, 1 for the combined series values. (ii) 1 for $V/\sqrt{3}$, 1 for the parallel Z, 1 for I, 1 for P. (iii) ½ mark for each factor.

> **COMMON MISTAKE:** using 16 kV as the phase voltage, which gives 1145 A and 3.072 MW. Also including the series 0.14 + j0.28 Ω in the OC calculation: on open circuit no current flows in it. And answering pf ×0.5: it does not change.

Question 3. Electric Machines [20 marks]

3a) Torque vs. Speed Characteristics [6 marks]

Consider the torque versus speed curves shown in the figure below.

i) For each curve, write the type of machine in the space below the curve and draw the resulting torque curve if the no-load/synchronous speed was reduced to half of its original value. [3 marks]

ii) Alongside each curve, indicate how you could halve the no-load/synchronous speed while ensuring safe machine operation. [3 marks]

Jun24 Q3a figure: three torque–speed curves, top to bottom a vertical line, a falling straight line and an induction-machine curve (the paper labels the torque axis T) (from the exam paper)

MARKS: (i) 1 per machine for the correct name and halved curve. (ii) 1 per machine. For the AC machines the mark needs both V and f.

COMMON MISTAKE: "halve f" alone for the AC machines. The question says "safe", and the mark is for keeping V/f constant. Another error is drawing the halved DC line through the origin, or with a different slope.

3b) Induction Machine Parameters [4 marks]

The following tests are performed on a 415V, 50Hz three-phase induction motor which is star-connected.

- Stator resistance measurement: 2 ohms per phase
- No-load test: input voltage 415V, input current 3.5A, input power 270W
- Blocked-rotor test: input voltage 80V, input current 5A, input power 390W

Find the following simplified equivalent circuit series parameters: R_1, R_2' and X.

Q3b test circuits: no-load 415 V, 3.5 A, 270 W; blocked rotor 80 V, 5 A, 390 W; R1 = 2 ohm

MARKS: 1 for per-phase values, 1 for R_1, 1 for R_2', 1 for X.

COMMON MISTAKE: not dividing 80 V by $\sqrt{3}$ (this gives $X = 15.13\ \Omega$). Also halving 2 Ω: it is already a per-phase stator value. And using the no-load test for the series parameters.

3c) Induction Machine Calculation [10 marks]

A 415 V, 50 Hz, 960 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):

stator resistance R_1 = 2.5 ohms, rotor resistance R_2' = 3 ohms
stator leakage reactance X_1 = 6 ohms, rotor leakage reactance X_2' = 6 ohms

The magnetising reactance and core loss resistance can be neglected.

i) The above induction machine is run as a generator at 1025 rpm. Find the synchronous speed, the value of slip s and hence the effective rotor resistance R_2'/s. [3 marks]

ii) Draw a single-phase simplified equivalent circuit for the machine when operating at 1025 rpm, showing the parameter values as impedances. Assuming that the machine is operating at rated voltage, find the magnitude and phase of the input phase current. Hence find the input power. You should assume that the phase angle of the phase voltage is zero. [5 marks]

iii) Using the results from part (ii) find the corresponding airgap power P_{AG} using the fact that in this case $I_2' = I_1$, and hence find the output torque τ. [2 marks]

Q3c: series circuit 2.5 ohm, j6 ohm, j6 ohm and a negative R2′/s, no shunt branch

MARKS: (i) 1 each for n_S, s, R_2'/s. (ii) 1 for the labelled circuit, 1 for $V_1 = 415/\sqrt{3}$, 1 for Z, 1 for I_1 with its angle, 1 for P_{IN} with its sign. (iii) 1 for P_{AG}, 1 for τ (using ω_S).

COMMON MISTAKE: dropping the minus signs ($s = +0.025$ gives a motor at 975 rpm, which is a different machine). Also using $X = 6\ \Omega$ instead of $X_1 + X_2' = 12\ \Omega$, or dividing P_{AG} by the rotor speed instead of ω_S. An angle near ±180° is correct here, not an error.

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

A 14 kV, 400 MVA synchronous generator is tested at a field current of 250 A. The open-circuit voltage was found to be 12 kV and the short-circuit current was 6.5 kA. The stator resistance was measured at 0.2 ohms per phase.

i) Find the value of the synchronous reactance in per-unit. [3 marks]

ii) The synchronous generator is delivering 200MW at a terminal voltage of 3% below rated voltage and a power-factor of 0.8 lagging. Find generator output voltage and current phasors in per-unit. [2 marks]

Q4a test circuits: OC 12 kV line and SC 6.5 kA, both at a field current of 250 A, R = 0.2 ohm

OC and SC characteristics with the Q4a test point at 250 A

MARKS: (i) 1 for $\sqrt{3}$ and $|Z_S|$, 1 for removing R, 1 for Z_B and the pu value. (ii) 1 for $\mathbf{V}$, 1 for $\mathbf{I}$ with the angle.

COMMON MISTAKE: forgetting $\sqrt{3}$ (gives 3.745 pu), or leaving R in (gives 2.175 pu). In (ii), using $|I| = 0.625$ pu, which ignores the 0.97 pu voltage. Answering in ohms also loses the mark: the question asks for pu.

4b) Synchronous Machines [6 marks]

A round-rotor synchronous generator has a synchronous reactance of 1.8 per-unit. Its excitation system is controlled so that its terminal voltage is always 1 per-unit. Resistance and saturation may be neglected.

i) What excitation (in per unit) is required to produce the generator's rated volt-amp output at 0.9 leading power-factor? [3 marks]

ii) What is the rotor angle at this loading? [1 mark]

iii) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power, in per-unit, that can be delivered without losing synchronism? What is the rotor angle corresponding to this condition? [2 marks]

Q4b: generator E0 behind j1.8 pu, V = 1 pu, current out at rated VA and 0.9 pf leading

MARKS: (i) 1 for $\mathbf{I}$ with the $+$ angle, 1 for $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$, 1 for 1.634 pu. (ii) 1 for δ. (iii) 1 for P_{max}, 1 for 90°.

COMMON MISTAKE: using -25.84° (lagging). That gives $E_0 = 2.410\angle 42.23^\circ$ pu and loses the whole part. Also recomputing E_0 in (iii): "excitation fixed" means keep 1.634 pu and set $\delta = 90^\circ$.

4c) Short-Answer Questions [9 marks]

Answer the following questions using words only (that is, without using any sketches, drawings or equations).

These model answers are built from the official answer sheet. The sheet says you do not need every point for full marks. Aim for one solid point per mark.

4c(i) [3 marks]. We have discussed three types of electric power: real, reactive and apparent. Explain briefly what each one is and their practical significance.

MARKS: 1 per power type, for its definition plus its practical significance.

> **COMMON MISTAKE:** writing $S = P + jQ$ or drawing a triangle. The question forbids equations and sketches. Another is saying reactive power "is wasted". It is not consumed. It oscillates, and its cost is the extra current.

4c(ii) [3 marks]. Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power.

MARKS: 1 for the copper loss mechanism and model, 1 for the iron loss mechanism and model, 1 for how each varies with output.

> **COMMON MISTAKE:** putting the core loss in series, or saying iron loss rises with load. Also forgetting that the question asks "how do these vary": give both trends explicitly.

4c(iii) [3 marks]. Carefully explain why the input current of an induction machine under no-load conditions is not zero. What are the physical effects which cause this?

MARKS: 1 for magnetising current and X_M with the physical reason, 1 for core loss and R_C, 1 for mechanical losses (friction and windage) and the small slip.

> **COMMON MISTAKE:** saying "the current is zero because there is no load torque". Also listing only the equivalent-circuit elements without the physical cause: the answer sheet asks for both.

ELECTRICAL MACHINES · MOCK

Machines mock paper

Practice questions · Day 23 · Sun 15 Nov · pack MX3

How to use this: this is a new paper, not a past paper. Sit it under exam conditions 2 days before the exam: 120 minutes, timer on, your printed notes and reference sheet only, calculator. Do not open the marking guide until time is up. Then mark yourself strictly with the guide and use the score table at the end to plan your last 24 hours.

Instructions

- Total duration: **120 minutes**.
- This examination is **open book**. Hardcopy printed and handwritten materials are allowed. No electronic devices apart from a calculator without remote communication.
- Attempt **ALL FOUR** questions. All questions carry equal marks (20 each, 80 in total). Part marks are given in brackets.
- Show your working. Method marks are given for correct equivalent circuits and correct substitutions, even when the final number is wrong.
- Phasors are rms. Unless stated otherwise, three-phase voltages are line-to-line and three-phase powers are totals.
- Answers should be expressed clearly and written legibly. These aspects of presentation will be taken into account in assessment.

Question	1a	1b	1c	2a	2b	2c	3a	3b	3c	4a	4b	4c	Total
Marks	5	6	9	7	5	8	6	5	9	5	7	8	**80**

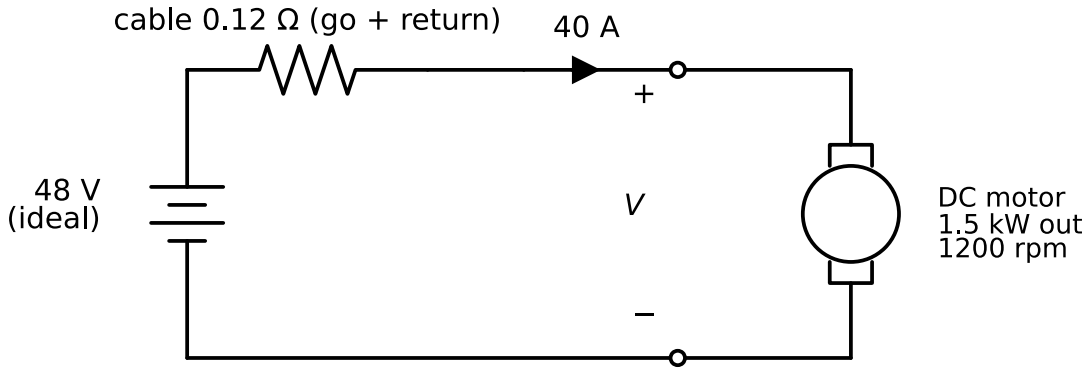
Time budget: 1.5 minutes per mark, so 30 minutes per question. Keep the last 5 minutes to check angles, signs and units.

Question 1 [20 marks]

AC and DC Power Circuit Analysis

1a) DC power and efficiency [5 marks]

A DC motor is supplied from an ideal 48 V DC source through a cable. The total resistance of the cable (go and return conductors together) is 0.12 Ω. The motor draws 40 A and delivers a mechanical output power of 1.5 kW at 1200 rpm.



Q1a: ideal 48 V source, 0.12 ohm total cable resistance, DC motor drawing 40 A

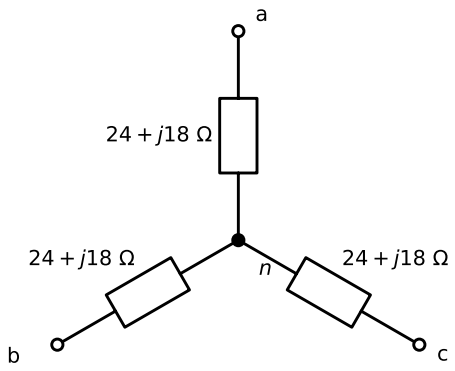
i. Find the voltage at the motor terminals and the power lost in the cable. [2 marks]

ii. Find the electrical input power to the motor, the motor efficiency and the motor output torque. [3 marks]

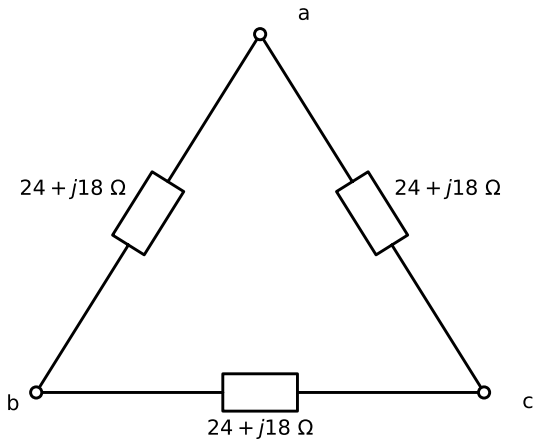
1b) Delta-connected load [6 marks]

A balanced three-phase load is connected in **delta** to a 415 V, 50 Hz, three-phase supply. Each phase of the load is a 24 Ω resistor in series with an inductive reactance of 18 Ω.

Star (part iii)



Delta (parts i, ii)



line voltage 415 V between any two of a, b, c

Q1b: three 24 + j18 ohm impedances in delta and then in star on a 415 V supply

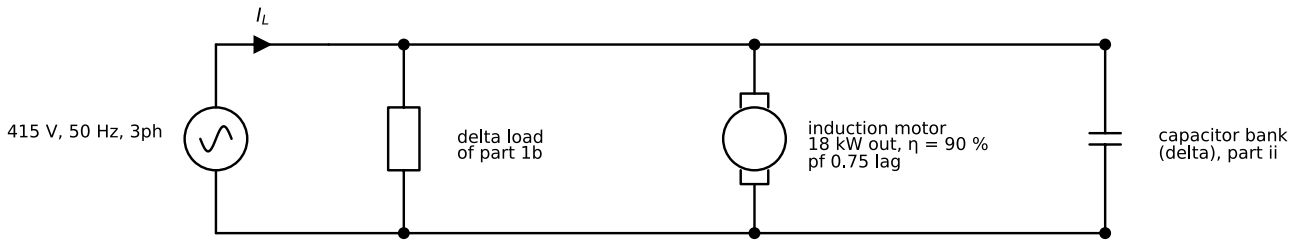
i. Find the phase current (magnitude and phase, with the phase voltage as reference), the line current magnitude and the power factor of the load. [3 marks]

ii. Find the total real power and the total reactive power absorbed by the load. [2 marks]

iii. The same three impedances are reconnected in **star** to the same supply. Find the new total real power. [1 mark]

1c) Power-factor correction [9 marks]

A small factory is supplied at 415 V, 50 Hz, three-phase. Its load consists of the delta-connected load from part 1b (connected in delta, as in parts (i) and (ii)) in parallel with a three-phase induction motor. The motor drives a mechanical load of 18 kW with an efficiency of 90 % and a power factor of 0.75 lagging.



Q1c: factory supply feeding the delta load, the induction motor and the delta capacitor bank in parallel

i. Find the total real power, the total reactive power, the power factor and the line current drawn from the supply. [3 marks]

ii. A bank of three capacitors is to be added in parallel with the factory to raise the supply power factor to 0.95 lagging. Find the total reactive power that the capacitor bank must absorb, and the new line current. [3 marks]

iii. The three capacitors are connected in **delta**. Find the required capacitance of each capacitor. [2 marks]

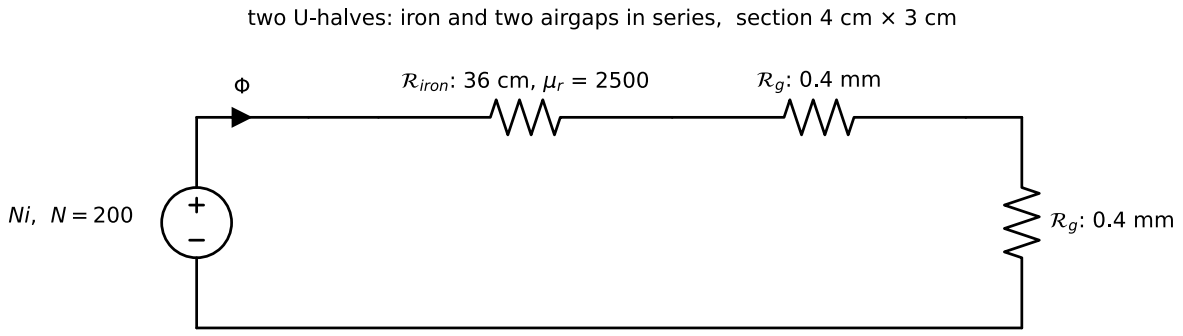
iv. How much reactive power would the capacitor bank need to absorb to give 0.95 **leading** instead? Give one reason why this is not done. [1 mark]

Question 2 [20 marks]

Magnetic Circuits and Transformers

2a) Electromagnetics [7 marks]

A magnetic core is made from two identical U-shaped halves placed with their ends facing each other, so that the flux crosses **two** airgaps in series, each of length 0.4 mm. The core has a rectangular cross-section of 4 cm × 3 cm everywhere, a mean magnetic path length in the iron of 36 cm (total for both halves), and a relative permeability of 2500. A coil of 200 turns is wound on one of the halves. Neglect fringing.



Q2a: magnetic circuit, MMF Ni with N = 200 driving flux through the iron and two 0.4 mm gaps in series

i. Calculate the reluctance of the iron, the reluctance of one airgap and the total reluctance, and hence the inductance of the coil. [3 marks]

ii. Find the DC coil current needed to produce a flux density of 1.2 T in the core, and the total energy stored in the two airgaps at this flux density. [2 marks]

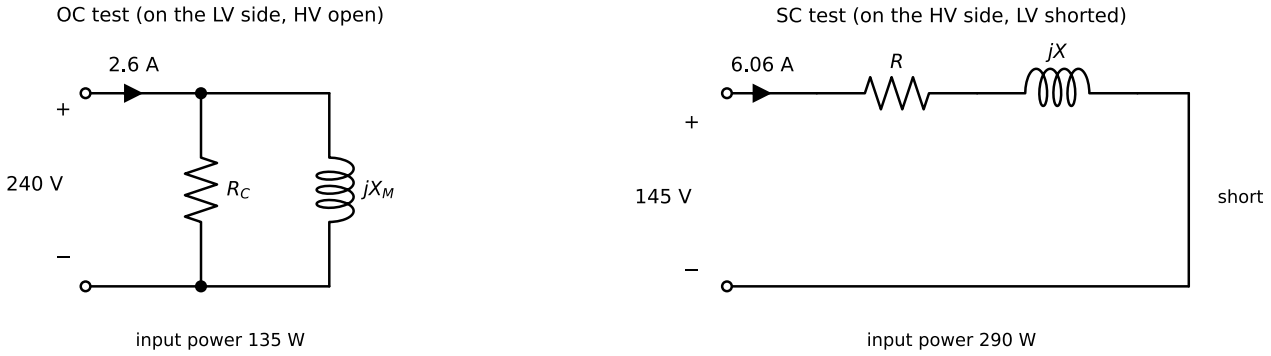
iii. At 1.2 T, find the total force pulling the two halves of the core together. [2 marks]

2b) Transformer parameters [5 marks]

The following tests are performed on a single-phase 3300 V : 240 V, 20 kVA, 50 Hz transformer.

- Open-circuit test, performed on the **low-voltage (240 V) side** with the HV side open: 240 V, 2.6 A, 135 W.
- Short-circuit test, performed on the **high-voltage (3300 V) side** with the LV side shorted: 145 V, 6.06 A, 290 W.

Find all the simplified equivalent circuit parameters (R, X, R_C and X_M), referred to the **high-voltage (primary)** side.



Q2b test circuits: OC on the LV side 240 V, 2.6 A, 135 W; SC on the HV side 145 V, 6.06 A, 290 W

2c) Transformer performance [8 marks]

The transformer of part 2b supplies its rated kVA at rated secondary voltage (240 V) to a load with a power factor of **0.8 leading**. Use the simplified equivalent circuit with the magnetising branch at the primary terminals, and take the referred secondary voltage as the phase reference.

Q2c: simplified circuit referred to the HV side with the 20 kVA, 0.8 leading load, V2′ as reference

i. Find the referred secondary current $\mathbf{I}_2'$ and the primary voltage $\mathbf{V}_1$ required, both in polar form. [3 marks]

ii. Find the voltage regulation in percent. Explain in one sentence why its sign is what it is. [2 marks]

iii. Find the efficiency at this load. [1 mark]

iv. Find the load (in kVA) at which this transformer has maximum efficiency, and the value of that maximum efficiency at 0.8 power factor. Assume rated voltage. [2 marks]

Question 3 [20 marks]

AC and Induction Machines

3a) AC machines [6 marks]

A three-phase AC machine has a nameplate showing 60 Hz and a rated speed of 1164 rpm.

i. How many poles does it have? Is it a synchronous machine or an induction machine? Briefly explain why. [2 marks]

ii. Find the rated slip and the frequency of the rotor currents at rated speed. [1 mark]

iii. At rated speed, what is the ideal (that is, maximum possible) efficiency of the machine? [1 mark]

iv. The machine is operating at one-half of rated torque. Assuming a linear torque versus speed curve, find the operating speed in rpm and the output power as a percentage of rated output power. [2 marks]

3b) Induction machine parameters [5 marks]

The following tests are performed on a 400 V, 50 Hz, three-phase induction motor whose stator is **delta-connected**. All voltages and currents below are **line** values as measured at the motor terminals, and all powers are three-phase totals.

- Stator resistance: 2.8 Ω per phase winding.
- No-load test: input voltage 400 V, input current 4.2 A, input power 320 W.
- Blocked-rotor test: input voltage 92 V, input current 10 A, input power 620 W.

Find the simplified equivalent circuit parameters per phase of the delta: R_2', X (the total leakage reactance $X_1 + X_2'$), R_C and X_M.

Q3b test circuits per phase of the delta: no-load 400 V, 4.2 A, 320 W; blocked rotor 92 V, 10 A, 620 W; R1 = 2.8 ohm

3c) Induction machine calculation [9 marks]

A 415 V, 50 Hz, six-pole, three-phase, **star-connected** squirrel-cage induction motor has the following equivalent circuit parameters per phase (referred to the stator where appropriate):

- stator resistance R_1 = 0.6 Ω, rotor resistance R_2' = 0.45 Ω
- stator leakage reactance X_1 = 1.2 Ω, rotor leakage reactance X_2' = 1.2 Ω
- magnetising reactance X_M = 40 Ω, core-loss resistance R_C = 300 Ω

Use the simplified equivalent circuit (magnetising branch across the input terminals). Friction and windage can be neglected. The motor runs at rated voltage at 970 rpm.

Q3c: 300 ohm parallel j40 ohm at the terminals, then 0.6 ohm, j1.2 ohm, j1.2 ohm and R2′/s

i. Find the synchronous speed, the slip and the effective rotor resistance R_2'/s. [2 marks]

ii. Find the referred rotor current $\mathbf{I}_2'$ and the input phase current $\mathbf{I}_1$ (magnitude and phase, with the phase voltage as reference), and the input power factor. [3 marks]

iii. Find the input power, the airgap power, the output torque and the efficiency. [3 marks]

iv. Find the slip and the speed in rpm at which maximum torque occurs. [1 mark]

Question 4 [20 marks]

Per-Unit and Synchronous Machines

4a) Synchronous machine tests [5 marks]

A 60 MVA, 13.8 kV, 50 Hz, three-phase, star-connected synchronous generator gives the following test results at a field current of 180 A:

- open-circuit line voltage: 9.2 kV
- short-circuit current: 1.25 kA

The stator resistance is 0.05 Ω per phase. Use rated values as the per-unit base.

Q4a test circuits: OC 9.2 kV line and SC 1.25 kA, both at a field current of 180 A, R = 0.05 ohm

OC and SC characteristics with the Q4a test point at 180 A

i. Find the base impedance and the base current. [1 mark]

ii. Find the synchronous reactance in ohms and in per-unit. [3 marks]

iii. What field current gives rated voltage on open circuit? Assume the open-circuit characteristic is linear. [1 mark]

4b) Generator at leading power factor [7 marks]

The generator of part 4a is connected to a large grid at rated terminal voltage. It delivers 45 MW at a power factor of **0.85 leading**. Neglect the stator resistance and saturation.

Q4b: generator E0 behind jXs at rated terminal voltage, delivering 45 MW at 0.85 pf leading

i. Find the terminal voltage and the stator current phasors in per-unit, with the terminal voltage as reference. Find also the stator current in kA. [2 marks]

ii. Find the excitation voltage E_0 in per-unit, the rotor angle, and the field current. [3 marks]

iii. With the field current held at this value, the operator tries to raise the real power output to 60 MW. Find the theoretical maximum power, in MW, that the generator can deliver without losing synchronism, and say whether 60 MW is possible. If it is not, find the minimum field current that would make 60 MW theoretically possible. [2 marks]

4c) Short-answer questions [8 marks]

Answer the following questions using words only (that is, without any sketches, drawings or equations).

i. A synchronous generator is connected to a large power grid. Explain what is meant by over-excited and under-excited operation, and explain how the field current and the mechanical input power each affect the real and reactive power delivered. [3 marks]

ii. The speed of an induction motor can be reduced either by adding resistance to the rotor circuit of a wound-rotor machine, or by using an inverter to reduce both the supply voltage and the frequency. Explain why the first method is inefficient and why the second is not. [3 marks]

iii. Explain why the per-unit system is useful when analysing a power system that contains transformers. [2 marks]

END OF QUESTIONS

Short-answer bank and final pass

Practice questions · Day 24 · Mon 16 Nov · pack MXF

1. The 20 calculations that come up every year

Q1 territory (Ch1–2): power, AC circuits, three-phase

C1. Motor power, torque, efficiency and power factor

SPOT IT: you are given a rating (kW), the speed in rpm, and an efficiency or input V, I and P. You are asked for torque, losses, efficiency or pf.

METHOD:

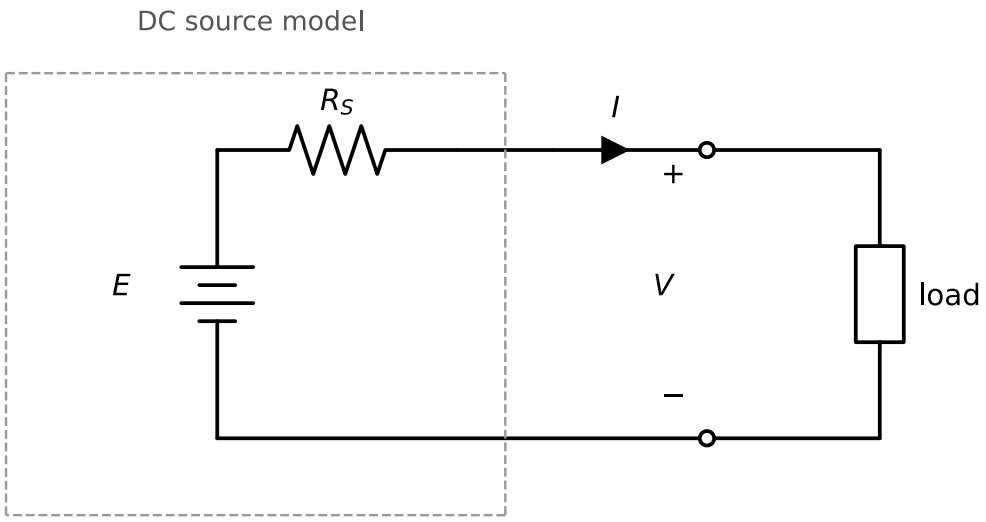
1. Convert the speed: $\omega = 2\pi n/60$.
2. Output power $P_{OUT} = \tau\omega$. The rating of a motor is its *output* power.
3. $P_{IN} = P_{OUT}/\eta$ and loss $= P_{IN} - P_{OUT}$.
4. Three-phase pf: $\cos\phi = P_{IN}/(\sqrt{3}V_L I_L)$.

KEY FORMULA: $\tau = \dfrac{P}{2\pi n/60}, \quad \eta = \dfrac{P_{OUT}}{P_{IN}}, \quad P = \sqrt{3}V_L I_L \cos\phi$

TRAP: using rpm instead of rad/s, or dividing the rating by efficiency to get the *output* power.

C2. DC source with internal resistance

SPOT IT: a battery or DC supply is given with an open-circuit voltage and a loaded voltage and current.



DC source model: EMF E behind RS, terminal voltage V = E - I RS

METHOD:

1. The open-circuit voltage is E, because no current flows and so there is no drop across R_S.
2. $R_S = (E - V)/I$ from the loaded reading.
3. Internal loss $= I^2 R_S$. Power to the load $= VI$.
4. New load: $V = E - IR_S$.

KEY FORMULA: $V = E - IR_S$

TRAP: forgetting that E is the *open-circuit* reading. Also, giving the power to the load as EI when it is VI.

C3. Star or delta resistive or impedance load

SPOT IT: "three identical X Ω loads on a V-volt 3ph supply, star then delta".

METHOD:

1. Star: $V_{ph} = V_L/\sqrt{3}$ and $I_L = I_{ph}$.
2. Delta: $V_{ph} = V_L$ and $I_L = \sqrt{3} I_{ph}$.
3. $I_{ph} = V_{ph}/|Z|$.
4. $P = 3I_{ph}^2 R$ (or $3V_{ph} I_{ph} \cos\phi$).

KEY FORMULA: delta power $= 3\times$ star power for the same impedances.

TRAP: putting V_L across a star phase. The quoted supply voltage is always the line voltage.

C4. Phasor circuit and complex power

SPOT IT: a series or parallel RLC circuit, with one phasor known (current or voltage). You are asked for the other phasors and for S, P and Q.

METHOD:

1. Write every element as an impedance: R, $+jX_L$, $-jX_C$.
2. Series: use the common current, so $V_k = IZ_k$ and $Z = \sum Z_k$. Parallel: use the common voltage, so $I_k = V/Z_k$ and $I = \sum I_k$.
3. Add the phasors in rectangular form and convert to polar at the end.
4. $S = VI^*$. Then $P = \operatorname{Re} S$ and $Q = \operatorname{Im} S$. A positive Q means the load absorbs reactive power, which is inductive.
5. For magnitudes only: in series RL, $|V| = \sqrt{V_R^2 + V_L^2}$, because the two phasors are at 90°.

KEY FORMULA: $S = VI^* = P + jQ, \quad Q = VI\sin(-\phi)$ where ϕ is the angle of I relative to V.

TRAP: adding magnitudes $(50 + 20)$ instead of phasors. Also, forgetting the conjugate, which flips the sign of Q.

C5. R and X from measured P, Q (or P, I, V)

SPOT IT: "a load draws I amps, absorbs P watts and Q VAr. Find series R and X, and L or C".

METHOD:

1. Series model: $R = P/I^2$ and $X = Q/I^2$.
2. If you only have V, I and P, use $|Z| = V/I$ and then $X = \sqrt{|Z|^2 - R^2}$.
3. Q > 0 means inductive, so $L = X/\omega$. Q < 0 means capacitive, so $C = 1/(\omega|X|)$.

KEY FORMULA: $R = P/I^2, \quad X = Q/I^2$

TRAP: using V^2/P, which is the *parallel* resistance, when the question asks for the *series* resistance.

C6. Power-factor correction with a capacitor

SPOT IT: a target supply pf is given. You are asked for Q_C or the capacitance.

METHOD:

1. Find the load P and Q, and draw the power triangle.
2. The capacitor does not change P, so $P_S = P_{load}$.
3. Find the target $Q_S = P_S \tan(\cos^{-1} \text{pf}_{target})$.
4. $Q_C = Q_S - Q_{load}$. This is negative, because the capacitor generates Q.
5. $C = |Q_C|/(\omega V^2)$. In three-phase, use the per-phase Q and the voltage across each capacitor.

KEY FORMULA: $|Q_C| = \omega C V^2$

TRAP: using the line voltage for star-connected capacitors. Also, quoting Q_C as positive when the question uses "absorbs" (the capacitor absorbs a negative Q).

Q2 territory (Ch3): magnetics and transformers

C7. Reluctance and inductance with an airgap

SPOT IT: a core with dimensions, μ_r, turns and a gap. You are asked for the reluctance(s) and L.

METHOD:

1. Convert everything to metres and m². The area of a circular section is $\pi d^2/4$.
2. $\mathcal{R}_g = l_g/(\mu_0 A)$ and $\mathcal{R}_c = l_c/(\mu_r \mu_0 A)$. If the core has infinite μ, then $\mathcal{R}_c = 0$.
3. Add them in series. One saw cut gives one gap. Two C-cores give two gaps.
4. $L = N^2/\mathcal{R}_{total}$.

KEY FORMULA: $\mathcal{R} = \dfrac{l}{\mu_r \mu_0 A}, \quad L = \dfrac{N^2}{\mathcal{R}}$

TRAP: cm² to m² is $\times 10^{-4}$, and mm to m is $\times 10^{-3}$. Also, the gap dominates even though it is tiny.

C8. Gap energy, force and DC current for a given B

SPOT IT: words like “energy stored in the airgap”, “force between the poles”, “hold the mass” or “DC current to give B”.

METHOD:

1. For DC current with infinite μ: $NI = Bl_g/\mu_0$. With finite μ: $NI = B(l_c/\mu_r\mu_0 + l_g/\mu_0)$.
2. Energy in the gap: $E = \frac{B^2}{2\mu_0} A l_g$.
3. Force: $F = \frac{B^2}{2\mu_0} A_{force}$. Two pole faces mean $A_{force} = 2A$.
4. To hold a mass, set $F = mg$ and solve for B.

KEY FORMULA: $E = \dfrac{B^2}{2\mu_0}\,\mathrm{Vol}, \quad F = \dfrac{B^2}{2\mu_0} A$

TRAP: missing the second gap or the second pole face, for example the paper in the 2024 test gives 2×0.07 mm of gap and 10 cm² of force area.

C9. Back-emf equation: turns, maximum voltage, B

SPOT IT: an AC winding with V and f, and either B_{max} or the turns given. You are asked for N, V or B.

METHOD:

1. $E = 4.44 N f \hat{B} A$, where E is rms and B is peak.
2. For N: $N = V/(4.44 f \hat{B}_{max} A)$, then round **up** to an integer, so that B stays below the limit.
3. Secondary turns: $N_2 = N_1 V_2 / V_1$, rounded to the nearest integer.
4. Wire resistance: $R = l/(\sigma \cdot \pi d^2/4)$.

KEY FORMULA: $V_{rms} = 4.44\, N f \hat{B} A$

TRAP: putting reluctance or μ_r in. They do not appear here: the voltage fixes the flux.

C10. Iron-loss scaling

SPOT IT: a total iron loss with a hysteresis and eddy split, then a new B and f.

METHOD:

1. Split the loss: P_H and P_E.
2. Use the course model: $P_H \propto B^2 f$ and $P_E \propto B^2 f^2$.
3. Scale each part separately, then add.
4. If only V and f change, then $B \propto V/f$.

KEY FORMULA: $P_{H2} = P_{H1}\left(\frac{B_2}{B_1}\right)^2 \frac{f_2}{f_1}, \quad P_{E2} = P_{E1}\left(\frac{B_2}{B_1}\right)^2 \left(\frac{f_2}{f_1}\right)^2$

TRAP: scaling the *total* with one law. The course formula sheet uses B^2 for hysteresis, not $B^{1.6}$.

C11. Ideal transformer and impedance referral

SPOT IT: an "ideal transformer, turns ratio a:1" with impedances on both sides.

Ideal transformer a:1 with a source impedance on the primary and a load Z on the secondary

METHOD:

1. $a = N_1/N_2$. Referring to the primary: $Z' = a^2 Z$.
2. Solve the single loop on the primary side.
3. Actual secondary values: $I_2 = aI_1$ and $V_2 = V_1/a$.

KEY FORMULA: $Z' = a^2 Z, \quad V_1/V_2 = a, \quad I_2/I_1 = a$

TRAP: using a instead of a^2, or referring in the wrong direction (to the low-voltage side, impedances get *smaller*).

C12. Transformer open-circuit and short-circuit tests

SPOT IT: "short-circuit test: V, I, P; open-circuit test: V, I, P". You are asked for the equivalent-circuit parameters.

Transformer tests: OC sees only RC parallel jXM, SC sees only the series R + jX

METHOD:

1. Three-phase: convert to per-phase first, with $V_1 = V_L/\sqrt{3}$ and $P_1 = P/3$.
2. Short-circuit test (series branch): $R = P_1/I_1^2$, $|Z| = V_1/I_1$, $X = \sqrt{|Z|^2 - R^2}$.
3. Open-circuit test (shunt branch): $R_C = V_1^2/P_1$, $Q_1 = \sqrt{(V_1 I_1)^2 - P_1^2}$, $X_M = V_1^2/Q_1$.

KEY FORMULA: short-circuit gives the series R and X. Open-circuit gives the parallel R_C and X_M.

TRAP: using series formulas on the open-circuit test, which gives a series R and X instead of the parallel R_C and X_M.

C13. Transformer on load: phasors, losses and efficiency

SPOT IT: a load in kW or MVA with a pf (or "generating Q"), at rated output voltage, using the simplified circuit referred to the primary.

Transformer simplified circuit referred to the primary: shunt branch at the input, series R1 + R2′ and j(X1 + X2′)

METHOD:

1. $V_2' = V_{L,rated}/\sqrt{3}\angle 0^\circ$ on the primary side.
2. $S = P/\text{pf}$ or $\sqrt{P^2 + Q^2}$. $|I_2'| = S/(\sqrt{3}V_L)$. The angle is $-\cos^{-1}\text{pf}$ for lagging and $+$ for leading (or when the load generates Q).
3. $V_1 = V_2' + I_2'(R + jX)$. The line voltage is $\sqrt{3}|V_1|$.
4. Losses: $P_{CU} = 3I_2'^2 R$ and $P_{FE} = 3V_1^2/R_C$. Then $\eta = P_{OUT}/(P_{OUT} + \text{losses})$, or $(P_{IN} - \text{losses})/P_{IN}$.
5. Open circuit: $I_1 = V_1/|R_C \parallel jX_M|$ and $P = 3V_1^2/R_C$.

KEY FORMULA: $V_1 = V_2' + I_2'(R + jX)$

TRAP: forgetting the factor of 3 on the losses. Also, forgetting $\sqrt{3}$ at the end when the question asks for the *line* voltage.

Q3 territory (Ch4–5 basics in part a, Ch7 induction machines)

C14. Poles, synchronous speed, ideal efficiency and a linear torque–speed curve

SPOT IT: "a 50 Hz AC machine with a rated speed of n rpm", or a fraction of rated torque.

METHOD:

1. $60f/n_R = p$ (pole-pairs). If it is not an integer, the machine is induction: round **down**, then poles $= 2p$ and $n_S = 60f/p$.
2. Ideal induction-machine efficiency $= n/n_S = 1 - s$.
3. Linear curve: the slip speed is proportional to torque. So $n = n_S - x(n_S - n_R)$, where $x = \tau/\tau_R$.
4. $P/P_R = x \cdot n/n_R$.

KEY FORMULA: $n_S = \dfrac{60f}{p} = \dfrac{120f}{\text{poles}}$

TRAP: p in the course formula is the number of pole-*pairs*. A 4-pole machine has $p = 2$.

C15. Induction-machine tests to parameters

SPOT IT: a DC resistance measurement, a no-load test and a blocked-rotor test on a star machine.

Induction machine tests: no-load sees RC parallel jXM, blocked rotor sees R1 + jX + R2′

METHOD:

1. Per phase: $V_1 = V_L/\sqrt{3}$ and $P_1 = P/3$.
2. Blocked rotor ($s = 1$, ignore the shunt branch): $R = P_1/I^2 = R_1 + R_2'$, so $R_2' = R - R_1$.

3. $X = \sqrt{(V_1/I)^2 - R^2} = X_1 + X_2'$. Split it equally if asked.
4. No load ($s \approx 0$): $R_C = V_1^2/P_1$ and $X_M = V_1^2/Q_1$.

KEY FORMULA: $R_2' = \dfrac{P_{BR}/3}{I_{BR}^2} - R_1$

TRAP: using the line voltage in $|Z|$. Also, forgetting to subtract R_1.

C16. Induction machine at a given speed (motor or generator)

SPOT IT: parameters, a speed, and "find I_1, P_IN, P_AG, τ". This is the Q3c staple worth 9–10 marks.

Induction machine simplified circuit: RC parallel jXM at the terminals, R1 + jX + R2′/s in series

METHOD:

1. Find n_S and $s = (n_S - n)/n_S$. If $n > n_S$, then $s < 0$ and the machine is a generator.
2. Compute R_2'/s, which is negative for a generator. Draw the circuit: $R_1 + jX + R_2'/s$, with the shunt branch across the terminals if it is not neglected.
3. Find $I_2' = V_1/(R_1 + R_2'/s + jX)$. Add the shunt current if it is included.
4. $P_{IN} = 3\,\mathrm{Re}(V_1 I_1^*)$ and $P_{AG} = 3I_2'^2 R_2'/s$.
5. $\tau = P_{AG}/\omega_S$. Check it with the torque formula. $P_M = (1-s)P_{AG}$.

KEY FORMULA:

$$\tau = \frac{3V_1^2\,(R_2'/s)}{\omega_S\left[(R_1 + R_2'/s)^2 + (X_1 + X_2')^2\right]}$$

TRAP: dividing by the *rotor* speed instead of the synchronous speed. Also, using the line voltage in place of V_1.

C17. Maximum torque and external rotor resistance

SPOT IT: "slip for maximum torque", "external resistance so maximum torque occurs at standstill", or sketching curves for added resistance.

METHOD:

1. $s_{\tau,max} = R_2'/|R_1 + jX|$.

2. Maximum torque at standstill means $s_{\tau,max} = 1$. So $R'_2 + R'_{EXT} = |R_1 + jX|$.
3. τ_{max} does not depend on R'_2. Adding resistance moves the peak towards standstill but keeps its height.

KEY FORMULA: $s_{\tau\max} = \dfrac{R'_2}{\sqrt{R_1^2 + X^2}}$

TRAP: using only X, or only X'_2, in the denominator. Also, drawing the higher-resistance curves with a higher peak.

Q4 territory (Ch8 and per-unit)

C18. Synchronous reactance from tests, and V, I in per-unit

SPOT IT: a table of excitation, open-circuit line voltage and short-circuit current, plus R. Then “delivering P (and Q or pf) at rated voltage”.

Synchronous machine OC and SC tests at the same field current

METHOD:

1. Use the same field current for both readings: $|Z_S| = (V_{OC,L}/\sqrt{3})/I_{SC}$.
2. $X_S = \sqrt{|Z_S|^2 - R^2}$.
3. $Z_B = V_B^2/S_B$ (line voltage, 3ph VA), so $X_S(\text{pu}) = X_S/Z_B$.
4. $V = |V|\angle 0^\circ$ pu (for example 0.97 for “3% below”). $S = P/\text{pf}$ or $\sqrt{P^2 + Q^2}$, divided by S_B.
5. $I = (S/V)\angle -\cos^{-1}\text{pf}$ when lagging or delivering Q.

KEY FORMULA: $Z_B = \dfrac{V_B^2}{S_B}, \quad I_B = \dfrac{S_B}{\sqrt{3}V_B}$

TRAP: forgetting $\sqrt{3}$ on the open-circuit *line* voltage. Also, dividing MW by MVA without first converting to S.

C19. Excitation, rotor angle and maximum power

SPOT IT: X_S in pu, $V = 1$ pu, a load at some pf. You are asked for E_0, δ and P_{max}.

Synchronous machine model: generator convention I out (top), motor convention Im in (bottom)

METHOD:

1. $I = (S/V)\angle \pm \cos^{-1}\text{pf}$, with $+$ for leading.
2. Generator: $E_0 = V + jX_S I$ (add R if it is given). Motor: $E_0 = V - jX_S I_{in}$.
3. $|E_0|$ is the excitation in pu, and $\delta = \angle E_0$.
4. $P = \frac{VE_0}{X_S}\sin\delta$, so $P_{max} = VE_0/X_S$ at $\delta = 90°$.

KEY FORMULA: $E_0 = V + jX_S I, \quad P = \dfrac{VE_0}{X_S}\sin\delta$

TRAP: using the wrong sign for the angle of a leading current. Leading output means the machine is under-excited, so E_0 is small and δ is large.

C20. Plant power factor with a synchronous motor

SPOT IT: induction motors (load kW, η, pf) plus a synchronous motor rated in kVA. You are asked for the best plant pf.

METHOD:

1. For each induction motor: $P = P_{mech}/\eta$ and $Q = P\tan(\cos^{-1}\text{pf})$, which is positive.
2. Synchronous motor: $P = P_{mech}/\eta$, and the most leading Q is $-\sqrt{S_{rated}^2 - P^2}$.
3. Add up all the P and all the Q.
4. $\text{pf} = \cos(\tan^{-1}(Q_{tot}/P_{tot}))$, and say lag or lead.

KEY FORMULA: $|Q|_{max} = \sqrt{S_{rated}^2 - P_E^2}$

TRAP: using the mechanical power in place of the electrical input power ($P_E = P_M/\eta$) in the capability formula.

2. Short-answer bank (Q4c style)

Past-paper short answers (official points)

S1. [Jun23 Q4c(i), 3 marks] Explain the concept of power-factor correction. Why is it useful in power systems?

S2. [Jun23 Q4c(ii), 3 marks] Carefully explain the differences and similarities between transformers and induction machines.

S3. [Jun23 Q4c(iii), 2 marks] A squirrel-cage rotor is made from a perfect conductor (infinite conductivity). How would this affect the machine?

S4. [Jun24 Q4c(i), 3 marks] Explain real, reactive and apparent power and their practical significance.

S5. [Jun24 Q4c(ii), 3 marks] Explain the main losses in a transformer, how they are modelled in the simplified equivalent circuit, and how they vary with output power.

S6. [Jun24 Q4c(iii), 3 marks] Why is the input current of an induction machine on no load not zero? What physical effects cause it?

S7. [2023 quiz Q7, 4 marks] How do you choose the number of primary turns for a given voltage and core? What happens with too few turns? With too many?

S8. [2024 test Q8, 4 marks] An ideal transformer drives a resistive load. What happens to the output voltage and current if the load resistance is doubled? How is a real transformer different?

S9. [Jun23 Q2c(iii), 2 marks] A transformer has a purely resistive load. Does the referred secondary voltage lead, lag or stay in phase with the primary voltage? Why?

S10. [Jun23 Q3a(ii), 1 mark] A 50 Hz AC machine has a rated speed of 560 rpm. Is it synchronous or induction? Why?

S11. [Jun24 Q3a(ii), 3 marks] How would you halve the no-load (synchronous) speed of a synchronous machine, a DC machine and an induction machine while keeping operation safe?

S12. [Jun24 Q2c(iii), 2 marks] A transformer is on open circuit and its input voltage is halved. By what factor do the input current, real power, reactive power and power factor change?

Likely new short answers

S13. [Ch7–8, 3 marks] Compare induction and synchronous machines.

S14. [Lecture 7.9.4–7.9.5, PQ short answer (f), 3 marks] Why is the voltage varied in proportion to frequency when an induction motor's speed is controlled?

S15. [PQ short answer (a), Lecture 7.4, 3 marks] How does the induction machine equivalent circuit differ from that of a short-circuited transformer, and why?

S16. [Ch3, Ch7, 3 marks] What is the effect of the airgap in a magnetic circuit, and in an induction machine?

S17. [Lecture 8.3, 3 marks] What are the advantages of the per-unit system?

S18. [Lecture 8.6.6–8.6.7, PQ short answer (k), 3 marks] Why is the power factor of industrial plants usually lagging, and why are synchronous motors sometimes used there?

S19. [Lecture 7.2.2, Jun24 Q3c, 3 marks] How can an induction machine operate as a generator? What are its limitations?

S20. [Lecture 7.1, 3 marks] Explain how an induction motor produces torque, and why it produces none at synchronous speed.

S21. [PQ short answer (b), Lecture 7.7, 3 marks] Explain the tests used to find the induction machine equivalent-circuit parameters.

S22. [PQ short answer (d), Lecture 7.9.1–7.9.3, 3 marks] Name the two types of induction-motor rotor and their main difference. What is the effect of extra rotor resistance?

S23. [PQ short answer (e), Lecture 7.8, 3 marks] Describe (in words) the torque–speed characteristic of an induction motor, naming the starting, rated and breakdown torques.

S24. [PQ short answer (g), Lecture 8.1–8.2, 3 marks] Using rotating magnetic fields, explain how a synchronous machine operates.

S25. [PQ short answer (h), Lecture 8.1.4–8.1.5, 2 marks] Why are salient-pole rotors used for low-speed generators and round rotors for high-speed generators?

S26. [PQ short answer (j), Lecture 8.6.5, 3 marks] What limits the reactive power capability of a synchronous machine?

S27. [Lecture 8.6.3–8.6.4, 3 marks] Explain the effect of the field current on a synchronous generator connected to a large power system.

S28. [PQ short answer (i), Lecture 8.1–8.4, 3 marks] Compare the synchronous machine equivalent circuit with that of a DC machine.

S29. [Ch3 iron loss, 2 marks] Why are transformer and machine cores laminated, and made from silicon steel?

S30. [PQ 9.1(d), Lecture 9.1, 2 marks] Why is electric power transmitted over long distances at high voltage?

S31. [Lecture 1.6.5, 2 marks] Why are large industrial machines three-phase rather than single-phase?

S32. [Lecture 7.1.1, 8.1, 2 marks] Why is a synchronous motor not self-starting on a fixed-frequency supply?